# What a Disaster Means: Interpretation and the Demand for Green Assets

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**Abstract**

How events are interpreted locally substantially shapes portfolio choices, fund flows, and asset prices. In the setting of extreme weather events, I use fine-tuned large language models on 1.5 million articles from U.S. news outlets to construct a new measure of local event interpretation that distinguishes attention to a disaster from coverage explicitly linking it to climate change. I combine this measure with geographically identified portfolios of household-serving advisers and mutual fund managers. Among disaster-affected counties, stronger climate attribution leads to substantial reallocation toward green ETFs in household portfolios. The relationship extends to ESG mutual-fund inflows and to nearby fund managers, who tilt their active trading toward greener firms as climate-linked coverage rises. Predetermined county and investor characteristics and the direct economic effects of the disaster are not driving the responses. Using changes in the ownership structure of the county's newspapers, I provide causal evidence that locally supplied climate interpretation narratives move households' and fund managers' portfolios independently of readers' prior beliefs and preferences. In a calibrated model where institutional investors respond to one of the many disasters occurring each year, an interquartile increase in local climate interpretation implies a relative appreciation of green stocks equivalent to up to 15% of the observed annual green return spread.

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# 1 Introduction

Many events that affect portfolio decisions and asset prices, such as elections, technological innovations, or natural disasters, are not directly about individual firms, and their implications for those firms can be ambiguous. Following the release of DeepSeek's cheaper AI model, for example, analysts and investors disagreed about whether it would reduce demand for Nvidia's chips, as fewer chips would be needed for each AI service, or increase demand by encouraging wider adoption of AI.[1] Such differences in interpretation may lead investors to make different portfolio decisions in response to the same event (Kandel and Pearson, 1995). Yet isolating the role of interpretation in portfolio choices, trading, and prices is difficult. Investors may also differ in whether and what they learn about an event, their exposure to its fundamental consequences, and how readily they assess its implications (Fedyk, 2021; Wagner et al., 2018).[2]

In this paper, I use extreme weather events to distinguish the role of interpretation in investors' portfolio decisions from that of information and attention. I examine whether households and mutual fund managers who experience salient events such as floods, hurricanes, or wildfires reallocate toward sustainable assets when these events are interpreted locally as evidence of climate change, compared to other investors exposed to the same disaster. I find that differences in how the same event is interpreted have substantial consequences for portfolio choices and capital allocation, beyond the information the event conveys or the attention it attracts.

Extreme weather events are particularly well suited to studying this question. First, they can affect investors across several counties simultaneously, allowing me to compare how the same event, with the same objective characteristics and implications for climate change, is interpreted in different places. Although climate change is already affecting many weather and climate extremes (IPCC, 2023), the extent to which a particular disaster can be attributed to it remains uncertain, leaving room for different interpretations. For instance, political polarization is a major source of disagreement over the same event (Druckman and McGrath, 2019; Hazlett and Mildenberger, 2020; Djourelova et al., 2024).[3]

Second, a single disaster provides limited information about long-run climate trends (Trenberth et al., 2015), and the green assets I study are not directly exposed to the disaster. Portfolio reallocations toward these assets, and differences in these reallocations across investors exposed to the same disaster, are therefore difficult to explain through fundamental information updating alone.

Finally, disasters are salient events known to affect portfolio decisions independently of their climate interpretation (Alok et al., 2020; Dessaint and Matray, 2017). Investors who do not respond to a disaster may have rationally seen no reason to respond, or may not have paid attention to it. I therefore test whether investors who do not shift toward sustainable assets nevertheless adjust their portfolios in response to the disaster. These features make differences in portfolio responses to the same disaster more plausibly attributable to heterogeneous interpretations than to differences in objective information or attention.

Empirically, I study portfolio reallocation by geographically identified investors. First, I compare portfolio changes for investors in the disaster zone with those of unaffected or more distant investors around the same disaster. Because the timing of a disaster is exogenous, this comparison captures the overall response to the event. Second, and more importantly, among investors with the same exposure to the same

[1] See FT Alphaville (2025) and Reuters (2025).

[2] Wagner et al. (2018) find that after the 2016 election, returns are consistent with investors incorporating tax consequences quickly and taking longer to work out trade consequences.

[3] After Hurricane Ian, the share of respondents associating the storm with climate change differed by 21 to 41 percentage points between liberals and conservatives, while liberal outlets were more likely to suggest such a link and conservative outlets to deny it (Djourelova et al., 2024).

or similar disasters and comparable levels of local coverage, I compare the changes of those in places that interpret the event differently.

I study the effect of climate interpretation in two settings. The first is household demand for environmentally labeled funds, which I observe through the portfolios of local investment advisers that serve primarily household accounts (Pan et al., 2025; Kostovetsky et al., 2026). I measure their green investment as the share of their exchange-traded fund (ETF) holdings in green ETFs, those labeled as environmentally friendly. These labeled products provide a direct mapping from observed investment decisions to demand for green assets. I extend this setting to flows into ESG (environmental, social, and governance) mutual funds. The second is active stock trading by managers of mutual funds without an ESG mandate, which allows me to relate portfolio choices more closely to individual decision-makers, their prior views, and their local information environments, and to examine how climate interpretation affects active trading in individual stocks. I measure managers' political views through their donations and locate them at the offices where they work. These offices sometimes differ from their funds' headquarters, allowing me to distinguish responses associated with managers' local information environments from those associated with the location of their funds' investors.

To capture the local interpretation environment in which investors are embedded, the narratives shared locally and the discussions they trigger, I measure how local newspapers cover each disaster and how they frame it. The corpus holds 1.5 million disaster articles from 527 outlets. I classify every paragraph of these articles with fine-tuned large language models. The classifier records whether a paragraph discusses a specific disaster and the link it draws with climate change, including how strongly it suggests connection and whether it negates it or expresses skepticism. I construct investors' exposure to this environment by matching their offices to the newspapers circulating in the same ZIP code or county, weighted by each newspaper's circulation share. Most of the measure's variation comes from how a specific outlet reads a specific disaster, and little from the outlet and its audience. The measure predicts portfolio responses beyond prior political views and climate concern.

Households move toward green assets when a disaster is interpreted locally as climate change. Comparing counties affected by floods and hurricanes, I find that a ten-percentage-point higher share of local news coverage explicitly linking the disaster to climate change is associated with a 19-basis-point larger increase in the share of green ETFs in their ETF holdings over two quarters, approximately 20% of the average pre-event green-ETF holdings. Disaster coverage alone raises the turnover of households' portfolios without a net tilt toward green ETFs. Households in high-interpretation counties tilt toward green ETFs without trading more.

More generally, this pattern extends to flows into ESG mutual funds. Among funds headquartered in affected states, net inflows into ESG funds relative to non-ESG funds are larger after disasters with high climate interpretation, by 8.5 percentage points of pre-disaster assets over two quarters, with no such difference in neighboring unaffected states.

Mutual fund managers respond to local interpretation as well. I study their trades across firms with different environmental scores, which rate firms' environmental performance and are a standard proxy for their exposure to climate risk (Engle et al., 2020; Pástor et al., 2022).[4] I validate this measure by showing that managers tilt toward greener stocks after local heat waves, a climate shock unrelated to the disasters I study (Alekseev et al., forthcoming). I find that a ten-percentage-point higher share of climate-linked coverage reaching managers within 50 miles of a disaster raises their active tilt toward greener firms by 69 basis points

[4]I use the Environmental Pillar score from LSEG, the part of the ESG rating related to climate change (Section 2.1).

in the quarter of the disaster, about as large as the response to a local heat wave severe enough to cause injuries or fatalities. The volume of disaster coverage alone does not move the tilt. A one-standard-deviation increase in interpretation moves about $49 million from the browner to the greener third of stocks for the average nearby fund, with $4.3 billion in assets.

The identifying assumption is that, conditional on disaster exposure, coverage, and the controls, investors in places where a disaster is interpreted more strongly as climate change would otherwise have adjusted their portfolios similarly to investors in places where it is interpreted less strongly as climate change. Two main challenges arise. First, places that interpret disasters more strongly as climate change may already be shifting toward green assets. Pre-disaster portfolio changes do not differ by interpretation, and the interpretation effect is absent or much smaller among investors farther from the disaster.

Second, investors in these places may respond more strongly to any salient shock, regardless of how it is interpreted, because of their prior beliefs or preferences. I account for investors' prior political views, climate beliefs, demographics and climate attention, allowing these characteristics to shape the response to the disaster. County fixed effects account for counties' usual portfolio changes around disasters. After earthquakes, which local coverage does not interpret as climate change, counties that interpret other disasters more strongly as climate change show no differential green tilt. In both settings, saturated fixed effects and detailed controls address the main alternative explanations, including non-belief motives such as misaligned incentives between intermediaries and their investors, mechanical portfolio adjustments induced by disaster-related damage and fund flows, and confounding factors such as common national narratives.

Finally, to separate the effect of climate interpretation narratives from the prior beliefs and preferences of their readers (Schwartzstein and Sunderam, 2021), I instrument the local interpretation of a disaster with the size of the chain that owned the county's newspapers. Chain size changes through sales and mergers of the outlets unrelated to the preferences of the county's readers (Ewens et al., 2022; Gao et al., 2020). Larger chains supply centrally produced copy and read local disasters less as climate change. Changes in chain size do not predict the county's politics, climate beliefs, or pre-event portfolio changes. The instrumented response of advisers is in line with the estimates above, and that of managers has the same sign and is about two and a half times as large.

I provide suggestive evidence that the mechanism behind the response is a revision of perceived climate harm. If so, interpretation effects should extend beyond green investments to decisions that protect against climate risks, which sustainability preferences do not govern, and to reported expectations of harm. I therefore run the same estimation approach on first-time purchases of flood insurance. A flood makes flood risk salient directly, so take-up rises in affected counties regardless of how the flood is interpreted. A wildfire does not directly signal greater flood risk, yet flood insurance take-up after wildfires rises with the climate interpretation of the event, suggesting that households generalize from a specific disaster to broader climate risks.[5] As corroborating evidence, I use the *Climate Change in the American Mind* surveys to show that respondents interviewed in the four weeks after a flood begins in their media market are more likely to report that global warming is harming Americans now or will within ten years than those interviewed in the four weeks before.

At the aggregate level, I find that the climate interpretation of national disaster coverage is associated with additional flows into ESG funds, beyond the attention a disaster receives.[6] The estimated three-month

[5] Any direct increase in flood risk after a wildfire depends on location and not on how local coverage reads the fire (Section 7.1).

[6] This is consistent with evidence that newsworthy disasters raise flows into green funds and the relative returns of sustainable assets (Marshall et al., 2021; Fiordelisi et al., 2023; Misev and Balles, 2024). I add to this evidence by separating how coverage interprets a disaster from the attention it receives.

inflow response is approximately $580 million larger at the 75th than at the 25th percentile of the national interpretation share. As a reference point, ESG funds in the sample took in 0.7% of assets in an average quarter, so this difference is close to a typical quarter's inflow.

Shifts in demand may also have meaningful price implications, with shifts in institutional ESG demand accounting for 83% of the average annual return on a portfolio that buys ESG stocks and sells other stocks (van der Beck, 2026). I assess the economic significance of climate interpretation with a counterfactual in which disasters throughout a year collectively activate the institutional investor base. Each investor responds once, as the local households represented in my adviser sample do. Holding disaster attention fixed, an interquartile increase in climate interpretation implies roughly 34 basis points of one-time ESG-stock appreciation relative to other stocks. The calculation uses estimates of how much demand moves prices[7] and assumes comparable portfolio demand and market accommodation. This is equivalent in magnitude to 18% of the average annual return contribution attributed to institutional ESG demand and 15% of the observed annual return spread.

This paper contributes to research on disagreement, political views and financial responses to common events, which documents partisan differences in investment beliefs and capital allocation (Meeuwis et al., 2022; Kempf et al., 2023; Cookson et al., 2020). These studies proxy investors' worldviews with their political ideology. I study the effect of local narratives and interpretations beyond political views. My constructed measure captures geographic differences and changes over time in the explanations linking experience to climate change, as well as shared interpretations of specific events that more stable, cross-sectional differences in political views alone cannot capture.

The paper also relates to research linking climate attention to trading and asset returns (Choi et al., 2020; Pástor et al., 2022; Ardia et al., 2023).[8] I contribute by identifying local portfolio responses to disaster experience and examining how these responses depend on whether local narratives attribute the event to climate change. I therefore distinguish the attention an event receives from its attribution to climate change and trace how geographically dispersed investors translate disaster experience and local information into portfolio adjustments. Disaster coverage is associated with greater gross turnover, while climate attribution is associated with reallocation toward green assets.

I further contribute to research on institutional investors' responses to climate-related shocks. Prior work documents reduced exposure to disaster-zone stocks, greater support for environmental proposals, and portfolio responses to climate-related belief shocks (Alok et al., 2020; Dessaint and Matray, 2017; Fich and Xu, 2025; Alekseev et al., forthcoming).[9] These studies compare investors with different exposure to the event. I also compare investors who have similar exposure to the same disaster and comparable levels of local coverage but vary in how strongly their local narratives attribute the disaster to climate change. Stronger

[7]Recent estimates put the micro multiplier near 1 (Gabaix and Koijen, 2023). Following van der Beck (2026), I calibrate the multiplier for the ESG portfolio to approximately 1.1 using his December 2022 inputs.

[8]Choi et al. (2020) associate unusually warm local temperatures with increased attention to climate change, retail selling of carbon-intensive stocks, and their relative underperformance. Pástor et al. (2022) link green assets' realized outperformance to unexpectedly strengthening climate concerns. Ardia et al. (2023) construct a daily newspaper-based climate-concern index and show that green stocks outperform brown stocks on days when climate concerns rise unexpectedly. Their daily stock-return regressions also associate green outperformance with shocks to concerns about specific physical risks, including hurricanes and floods.

[9]Alok et al. (2020) find that funds headquartered within 100 miles of a disaster-zone firm reduce their portfolio weight in that firm more than distant funds, with evidence consistent with salience bias. Alekseev et al. (forthcoming) use local extreme heat and changes in shareholder disclosures to identify climate-belief shocks and connect them to industry reallocations, like buying automobile and semiconductor stocks and selling real estate. Fich and Xu (2025) find increased support for environmental proposals after hurricanes and increased holdings in firms where their votes could help environmental proposals pass, providing evidence consistent with both managers' own preferences and career concerns about investor flows.

attribution is associated with active reallocation toward greener assets.

In addition, the paper relates to research linking investors' beliefs and preferences to observed portfolio holdings (Riedl and Smeets, 2017; Giglio et al., 2025; Anderson and Robinson, 2026). These studies link survey data on beliefs and preferences to cross-sectional heterogeneity in green ownership. My evidence is complementary. I examine portfolio adjustments around particular disaster experiences instead, using pre-event allocations to distinguish investment responses from pre-existing differences in holdings.[10] Investors rarely adjust their portfolios: changes in beliefs explain little about whether they trade, even though they help explain the direction and magnitude of trading conditional on adjustment (Giglio et al., 2021). The reallocations I observe suggest that disaster experience and its local interpretation can overcome this inertia.

Finally, the paper is related to a broader literature on how experience, political views, and media shape climate attitudes (Deryugina, 2013; Ash et al., 2023; Djourelova et al., 2024). These studies measure the response in stated beliefs and attitudes. I show that the local interpretation of disaster experience also shapes consequential financial decisions, through active portfolio reallocations toward green assets. Complementary survey evidence also examines how responses to disaster experience vary with political views and local media coverage, distinguishing perceived climate risks from beliefs about human causation and policy preferences. These patterns suggest that risk perceptions can respond to experience among conservatives despite persistent political differences in climate attitudes.[11] Read alongside the portfolio responses, they point to the role of perceived risk in prompting green investment adjustments.

Section 2 describes the data, institutional setting, measurement of climate interpretation, and the local information environment and determinants of interpretation. Section 3 examines how local climate interpretation affects household demand for green ETFs through investment advisers, extends the analysis to ESG mutual fund flows, and addresses the main alternative explanations. Section 4 turns to active portfolio rebalancing by mutual fund managers, examining whether local climate interpretation also shapes their trading decisions. Section 5 exploits changes in newspaper ownership to isolate variation in the climate narratives supplied by local outlets, independently of readers' pre-existing beliefs and preferences, and examines the resulting portfolio responses of both advisers and mutual fund managers. Section 6 studies how national disaster coverage and climate interpretation affect aggregate ESG fund flows and asset prices, and quantifies the potential price effects of climate interpretation when a year's disasters collectively activate institutional investors. Section 7 discusses the mechanisms underlying these responses, distinguishing changes in beliefs from shifts in preferences. Section 8 concludes.

## 2 Data and Measurement

This section describes the data (Section 2.1) and the climate interpretation measure constructed to capture the local information environment (Section 2.2). I draw on four types of data: the portfolio holdings of households and of mutual fund managers, records of climate disasters, and local newspaper coverage of each disaster.

[10] Closest to my paper, Anderson and Robinson (2026) study climate-belief revisions following wildfires and global warming attention, and link it to cross-sectional differences in allocations to fossil-fuel-exclusion mutual funds within Sweden's premium pension system, measured about three and a half years after the wildfires. Instead, I compare active portfolio changes from pre-event allocations, tracing the timing and subsequent path of portfolio adjustments over shorter horizons around disasters.

[11] This pattern is consistent with Deryugina's (2013) finding that abnormally warm weather makes conservatives more likely to report that the effects of global warming have already begun. Similarly, Myers et al. (2013) find that personal experience raises certainty that global warming is happening, mainly among respondents initially less engaged with the issue.

## 2.1 Data

**Households: local investment advisers.** To represent household portfolio decisions, I use the portfolios reported by local U.S. registered investment advisers (RIAs) that predominantly serve household accounts. Advisers file Form ADV with the SEC, where they report their client composition, assets under management, and office locations. I combine quarterly Form ADV filings with raw Form 13F holdings to obtain the location and the clientele of each 13F filer. Following Pan et al. (2025), I define an adviser as serving predominantly household accounts when individuals and high-net-worth individuals make up a strict majority of its clients, and as local when all its offices are in one county. The final sample contains 4,040 advisers in 490 counties, covering all 50 states, the District of Columbia, and Puerto Rico (Table 1). FactSet classifies most of these advisers as investment advisers (66.4 percent) or as private banking and wealth-management firms (30.6 percent).

TABLE 1: REGISTERED INVESTMENT ADVISER SAMPLE CHARACTERISTICS

| Statistic | All registered advisers | Household-serving, single-county |
|---|---|---|
| Registered investment advisers | 8,489 | 4,040 |
| Adviser-quarters | 222,832 | 87,853 |
| U.S. counties represented | 1,439 | 490 |
| States and U.S. jurisdictions represented | 52 | 52 |
| Median regulatory assets under management ($ millions) | 593.8 | 426.1 |
| Median client accounts | 439 | 681 |
| Median advisory offices | 10.0 | 6.0 |
| Advisers ever holding any ETF (%) | 87.1 | 97.1 |
| Advisers ever holding a green ETF (%) | 36.3 | 44.0 |

*Notes.* This table reports characteristics of the registered investment advisers in the sample. The universe comprises investment advisers with a direct Form ADV filing or identified as relying advisers on Schedule R; exempt reporting advisers are excluded. Household-serving advisers are those whose individual and high-net-worth individual clients make up a strict majority of their clients, and single-county advisers have all their offices in one county. Regulatory assets under management, client accounts, and offices are measured at each adviser's latest observation.

Advisers that serve households hold a large share of ETFs, and an even larger share of green ETFs. Advisers whose clients are mostly high-net-worth individuals held 47 percent of their reported 13F assets in ETFs by 2023, against less than 7 percent for pensions and endowments.[12] Registered advisers held 41 percent of the $10.3 trillion U.S. ETF market at the end of 2024 and 45 percent of its green segment, against 20 and 18 percent for every other Form 13F institution combined.[13] Of the $38 billion of green ETFs held by all registered advisers at the end of 2024, local household advisers alone held about $12 billion. The final sample of local household-serving RIAs holds 7 percent of the regulatory assets reported by all registered advisers, while holding 21 percent of their ETF dollars and 31 percent of their green ETF dollars (Appendix Table C.2). Almost all of these advisers have ever held an ETF, and almost half have held a green ETF.

The local and household-account restrictions keep the part of an adviser's 13F portfolio that most likely reflects its clients' choices, especially for green ETFs. ETFs are held largely by households, directly or through their advisers. In 2024, 17 million U.S. households, 13 percent of the total, owned ETFs. These

[12] Source: Moussawi et al. (2025).

[13] Appendix Table C.1 shows ETF adoption, portfolio allocation, and market shares for registered advisers and for the household-serving group.

households are relatively wealthy, with a median income of $140,000 and median financial assets of $500,000, against a national median household income of about $87,000 in 2025. Since 58 percent of U.S. families hold equity directly or indirectly, more than one in five investing households owns ETFs.[14]

I combine 13F filers into stable adviser-level portfolio units[15] and assign each unit to a county. I restrict the sample to advisers that never change county, so that within-county portfolio changes are not mechanically driven by relocations. Then, I pool ETF holdings across local advisers to measure the allocation of the combined ETF wealth of the households they serve. Under the assumption that these households reside in their advisers' county, pooling holdings gives each household's ETF investment its corresponding weight in the county portfolio. Advisers serving more households or managing larger household investments therefore contribute more to the aggregate. The county green share is

$$g_{ct} = \frac{\sum_{a \in c} \sum_f V_{aft} G_f}{\sum_{a \in c} V_{at}^{ETF}} = \sum_{a \in c} \frac{V_{at}^{ETF}}{\sum_{b \in c} V_{bt}^{ETF}} g_{at}, \tag{1}$$

where $G_f$ identifies green ETFs, $V_{aft}$ denotes adviser $a$'s holdings in ETF $f$, $V_{at}^{ETF} = \sum_f V_{aft}$, and $g_{at}$ is the adviser's green ETF share.[16]

I define an ETF as green if its name in CRSP contains at least one of a list of climate and transition keywords, or if Morningstar assigns it one of a selected set of climate and clean-technology themes. ESG labels influence investor demand beyond differences in underlying holdings (Baker et al., 2024). Even as investors differ in which assets they think are green, buying a product labelled as environmentally friendly is a direct way to act on a climate shock. I exclude sectors whose climate mandate is ambiguous[17] and funds with only a social mandate. The rule classifies 227 funds as green over 2009–2024. Since green ETFs became common in adviser portfolios only around 2015, the main analysis covers 2015–2024 (Section 3).[18] At the end of 2024, about 180 green ETFs were active, with a market value of about $94 billion, or approximately 1 percent of the U.S. ETF market.

To ensure that an adviser's county is also where its clients live, I manually review the names of all retained advisers. I exclude internet-based advisers and advisers whose offices are all in one county but whose clientele is not local. To validate the local definition, I verify that the share of an adviser's stock portfolio invested in firms headquartered in its own state exceeds the share obtained by randomly reassigning adviser locations. I compute the green share within an adviser's ETF holdings, so that changes in mutual fund holdings reported on the same Form 13F do not enter the household measure.[19]

[14]Sources: Investment Company Institute (2025) for the ETF-owning households, U.S. Census Bureau (2026) for the national median household income, and Board of Governors of the Federal Reserve System (2023) for the share of families holding equity.

[15]The FactSet parent rollup and the S34 manager identity define the candidate aggregations. I split a parent into separate portfolios only when they belong to distinct advisers and are reported separately throughout. This keeps a portfolio's holdings unchanged when it is reported under a different filer identity.

[16]Adviser-level regressions weighted by pre-disaster shares of county ETF holdings approximate this aggregate response and yield very close estimates, as reported in Appendix Table C.3. They capture changes in green shares within advisers but omit the contribution of changes in advisers' relative ETF holdings; Appendix B.4 derives the conditions for equivalence. Equal-adviser-weighted regressions instead measure the average adviser's response, giving advisers equal importance regardless of the number of households they serve or the ETF wealth they manage. Although this differs from my target, I report these estimates in the Appendix (Appendix Table C.3, Panel B): the results remain statistically significant, while their magnitudes change with the different estimand.

[17]Metals and mining, resource extraction, upstream and midstream energy, commodities, agriculture and food, nuclear, water, and batteries.

[18]Green ETF market value grew from $2 billion in 2009 to a peak of $106 billion in 2021.

[19]Some advisers also manage mutual funds, and their 13F filings then include the funds' holdings. This overlap matters little for ETF positions. In the same quarter, mutual funds in S12 hold only 1.23 percent of ETF market value and 2.11 percent of green ETF market value, and only 0.59 percent of S12 funds hold any green ETF.

**Mutual fund managers.** I combine fund holdings from the Thomson Reuters S12 database, fund characteristics and manager names from Morningstar Direct, manager profiles from Revelio Lab., and the filings of the managing firms with the SEC to build a panel of fund portfolios located where their managers work. Since mutual fund investors cannot be located, I study the portfolio managers who actively rebalance the funds, whose location I observe. I locate each fund at the offices of its sub-adviser, or of its adviser when it has none, from Form N-SAR until 2017 and Form N-CEN thereafter, following Alekseev et al. (forthcoming), and from Morningstar portfolio-manager records. This way, I am able to locate the managing office for 76,652 fund-quarters. This office often differs from the fund's headquarters (Section 4.3). I also match 1,161 funds to 2,348 named managers by combining Morningstar Direct and Revelio, which allows me to measure the managers' political views from their political donations.

I construct fund-quarter holdings from S12 and compute changes only between consecutive quarterly reports. I also restrict the sample to actively managed U.S. equity funds.[20] Appendix Figure A.3 maps the county-weighted geography of fund management.

I classify a mutual fund as having an ESG mandate if its name includes a sustainability term or if Morningstar classifies it as having an intentional ESG mandate. I exclude funds with an ESG mandate from the analysis of managers (Section 4), because their portfolios follow climate considerations and incentives unrelated to local disasters (Alekseev et al., forthcoming).

**Climate disaster episodes.** I build disaster data from the NOAA Storm Events Database, which records storms and other weather events at the county level. I group related county records into disaster episodes, so that a disaster that affects several counties counts as a single event while each county keeps its own exposure.[21] I assign each episode a dominant hazard type and measure its severity from inflation-adjusted property and crop losses, fatalities, and injuries. I combine these measures into an index of the episode's aggregate impact.[22] For each sample, I retain the episodes occurring in the period of analysis that are in the top 15 percent of this index within their hazard type, and exclude episodes that had another local episode in their pre-period. Appendix Table A.1 reports the characteristics of the episodes by hazard type, and Figures C.1 to C.2 map them by county.

**Local news.** To measure how disasters are discussed locally, I collect newspaper articles from ProQuest TDM Studio. I query articles whose text includes climate and global-warming vocabulary, as well as extreme-weather and disaster vocabulary, yielding 1,485,896 disaster articles from 527 outlets. Restricting to articles that cover a disaster over 2015–2024, the corpus holds 782,714 articles, of which 724,969 come from local outlets and 57,745 from national ones. I combine the articles with local newspaper circulation data from the Alliance for Audited Media and Editor & Publisher, complemented by device-measured readership from Comscore. The circulation data give ZIP-level newspaper market shares, which I use to aggregate outlet-level measures to the ZIP and county level. Using these weights, I construct a sample of 433 outlets reaching 2,134 counties.

[20] I exclude funds whose Thomson Reuters objective is international, municipal bonds, bond and preferred, or balanced (Alok et al., 2020; Kacperczyk et al., 2008), passive funds and exchange-traded funds, identified by CRSP flags and fund names (Appel et al., 2016; Sammon and Shim, 2023; Alok et al., 2020), and funds with less than $5 million in assets or fewer than ten stocks (Alok et al., 2020).

[21] I audit every multi-event episode for coherence and split those that fail into sub-episodes, so that each episode approximates the footprint of one physical shock.

[22] The construction of this index builds on a Pareto approximation of the tail of disaster losses (Barro and Jin, 2011).

**Additional data.** To construct the instrument for local interpretation (Section 5), I combine newspaper ownership data from Ewing (2026) with the ownership panels of the Northwestern Medill Local News Initiative. I date ownership changes to the quarter using SEC Form 8-K filings and reports of newspaper and company acquisitions.

To measure the environmental profile of the stocks the funds trade, I use the Environmental Pillar score from Refinitiv (LSEG, 2024), the environmental dimension of the ESG rating, covering resource utilization, emissions, and innovation. I take the firms' GICS sectors and industries from Compustat and the location of their headquarters from the business addresses in their SEC filings.

The remaining sources enter as controls, such as characteristics of ZIP codes, counties, advisers, mutual funds, and disasters, or serve to construct placebo events and other newsworthy events that coincide with disasters.

## 2.2 Local Information and Interpretation Environment

Climate beliefs are shaped by local discourse, through family and friends, news from different sources, and social media. Thirty-nine percent of Americans hear about global warming in the media at least once a month, and 17 percent hear people they know talk about it as often (Leiserowitz et al., 2026). Local news reaches people through the same channels. Seventy-three percent of Americans get local news from friends, family, and neighbors, 64 percent from local television, half from local social media groups or forums, and a third directly from local daily newspapers, down from 43 percent in 2018 (Pew Research Center, 2024a,b).

Although only a third of residents read a local daily directly, newspapers shape much of what circulates locally. First, they originate most of the stories that set the agenda of other media,[23] and even small outlets can spark wider discussion (King et al., 2017). Second, their articles reach readers beyond print, through websites, news aggregators, and social media.[24] Finally, local newspapers also reach investors. Retail and institutional investors rely on geographically proximate information (Engelberg and Parsons, 2011; Baik et al., 2010; Bernile et al., 2015), and local newspapers are "a key channel through which institutional investors acquire geographically scattered information" (Kang and Nam, 2025).

To construct a measure of local climate interpretation that captures the narratives shared locally and the discussions they trigger, I quantify how much local newspapers cover each disaster and how they frame it. To do so, I classify every paragraph of the disaster articles with a hierarchical teacher–student pipeline of large language models. I first define the classes and hand-label about 300 paragraphs, which I use to tune the prompts of a teacher model, GPT-5.5. The teacher then labels a training set, on which I train two local student models, DeBERTa and Qwen. DeBERTa classifies the full corpus and passes uncertain cases to Qwen.

The classifier first detects whether a paragraph discusses a specific disaster event, identified in time and place. If it does, the classifier records whether the paragraph links the event to climate change and how strongly, and its stance on climate change. The stance can be affirming, denying, neutral, lukewarm (accepting global warming but minimizing its importance), or reporting contested opinions, and the classifier also records whether the stance is the reporter's own. In addition, I identify other common topics in each

[23] Pew Research Center, *How News Happens*, 2010, https://www.pewresearch.org/journalism/2010/01/11/how-news-happens/.

[24] Of the 546 local newspapers in the sample, at least 468 have an active website and 293 a Twitter account. Within a county, their print market shares track their web market shares (correlation 0.78; Appendix Figure D.1), and their Twitter audiences track their print circulation (rank correlation 0.84; Appendix Figure D.2). News aggregators matter most for the audiences of small and local outlets (Calzada and Gil, 2020; Athey et al., 2021).

paragraph, the most frequent being damages and the human toll (damage to homes and victims), and whether the paragraph tells the event episodically, through the experience of a specific affected person, or thematically, through facts and aggregates (Iyengar, 1991).

Against my own hand labels on held-out paragraphs, the two student models reach an $F_1$ score of 0.87 and 0.88 on the climate link, above general-purpose models such as gpt-4.1-mini (0.85) and gpt-5.4-mini (0.77).[25] The classifier substantially outperforms the keyword rule in detecting paragraphs on a specific event among paragraphs mentioning disasters in general, and keywords cannot read the intensity and stance of the attribution (Appendix Tables A.2 and A.3). Appendix Table A.4 shows two examples. The first, from the coverage of Hurricane Helene, links the storm to the burning of fossil fuels without ever mentioning "climate change," so a keyword search would miss it, while the classifier labels it as an affirming climate link. The second, from the coverage of Hurricane Katrina, tells the human toll through one person's story and makes no link to climate change.

The classifier yields two measures, the first one being local disaster coverage. Let $o$ index outlets, $c$ counties, and $e$ disaster episodes, and let $w_{co}$ be outlet $o$'s circulation share in county $c$, normalized to sum to one within the county, and $A_{oe}$ the number of articles on episode $e$ that outlet $o$ publishes in the four weeks from the episode's start. Local disaster coverage of episode $e$ in county $c$ is the circulation-weighted number of articles,

$$A_{ce} = \sum_{o} w_{co} A_{oe}. \tag{2}$$

When coverage enters the regressions, I use $\log(1 + A_{ce})$ to allow for diminishing returns to additional coverage.

The second measure is local climate interpretation, the extent to which local coverage of a disaster interprets the event as a result of climate change. Let $C_{oe}$ be the number of outlet $o$'s articles on episode $e$ that link the event to climate change, and $m_{oe} = C_{oe}/A_{oe}$ the outlet's interpretation ratio, defined for outlets with $A_{oe} > 0$. I aggregate these ratios to the county with the same circulation weights, renormalized over the outlets that cover the episode

$$M_{ce} = \frac{\sum_o w_{co}\, \mathbf{1}\{A_{oe} > 0\}\, m_{oe}}{\sum_o w_{co}\, \mathbf{1}\{A_{oe} > 0\}}. \tag{3}$$

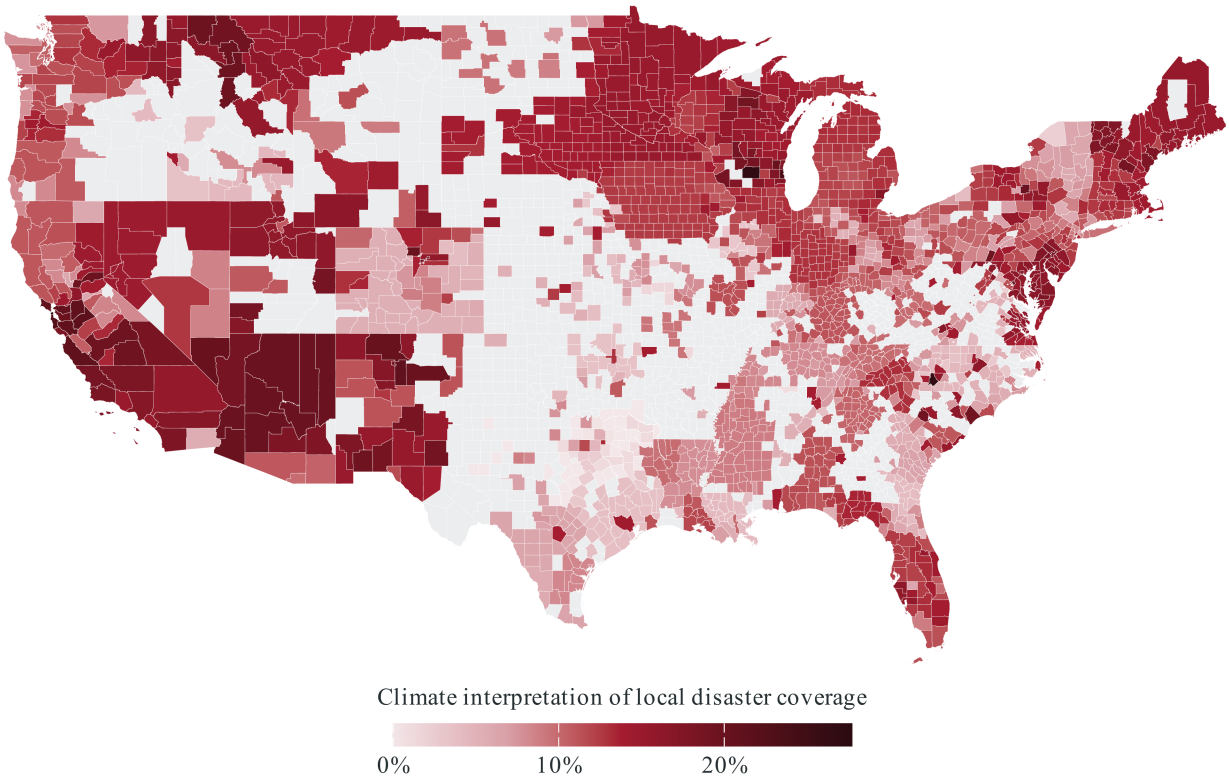
$M_{ce}$ is the share of climate-linked articles in the coverage of the average local outlet that covers the episode, with outlets weighted by their readership in the county.[26] The measure averages 0.172 across county–episode observations, with a standard deviation of 0.157, so an increase of 10 percentage points corresponds to 0.64 standard deviations. In a county served by a single outlet, such an increase means one more climate-linked article in every ten articles on the episode. In a county served by two outlets with equal circulation, it means either one more climate-linked article in ten in both outlets, or two more in ten in one outlet and no change in the other.

Figure 1 shows the average local climate interpretation of disaster coverage in each county. Interpretation varies widely across the country: it is highest along the West Coast and in the Southwest, and lowest in Texas and much of the South. It also rises over time, from 6.5 percent of a county's local disaster coverage before

[25] Appendix Figure A.1 shows the pipeline. Appendix Tables A.2 and A.3 report the performance of each model on every classification task, and Appendix A details the training and validation procedures.

[26] As a robustness check, I also use the pooled share $\sum_o w_{co} C_{oe} / \sum_o w_{co} A_{oe}$, which weights outlets by both their readership and the number of articles they publish on the episode.

FIGURE 1: LOCAL CLIMATE INTERPRETATION OF DISASTER COVERAGE




*Notes.* This figure plots the mean local climate interpretation of disaster coverage in each county (Equation (3)), over the disasters covered by its local outlets. Grey counties have no local coverage of disasters in the sample.

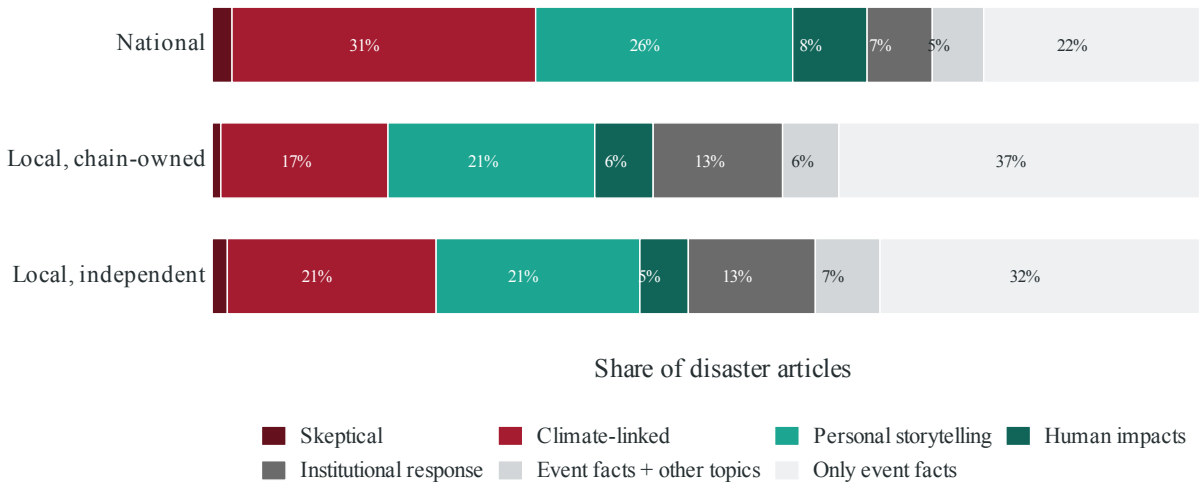
2015 to 17.9 percent from 2015 on.

To separate the role of the outlet from that of the disaster, I decompose the variance of coverage and of climate interpretation across outlet–episode pairs into an outlet component, an episode component, and their match, as in the literature on matched employer–employee data (Abowd et al., 1999; Kline et al., 2020). The outlet component captures stable differences across outlets, such as the political views and green attitudes of their audience and owner. The episode component captures the characteristics of the disaster, such as its year, hazard type, and damages. The match captures how a given outlet covers a given disaster beyond what the outlet and the disaster predict on their own. Appendix Figure D.3 shows that coverage depends mostly on the episode, which accounts for 55.4 percent of its variance against 19.5 percent for the outlet, mainly through the hazard type.

Climate interpretation instead lies mostly in the match, at 54.1 percent of its variance, against 40.7 percent for the episode and 5.5 percent for the outlet. The year and hazard type of the event explain part of the episode component, and over 90 percent of that part comes from the year, because the interpretation of each hazard type changes over time. Stable outlet and audience differences, together with common event characteristics, including time, still leave more than half of the variation unexplained. A substantial part of the variation therefore concerns how a specific outlet interprets a specific disaster, while serving a given audience at that time.

Part of the match reflects how articles reach an outlet. Many local newspapers belong to chains, groups of newspapers under the same owner, and a chain often writes an article once and prints it in several of its newspapers. Among the outlet-event pair characteristics I observe, chain copies account for 92 percent of the variation in the match that these characteristics explain. I leverage this variation to provide causal evidence on the effect of local interpretation in Section 5.

FIGURE 2: COMPOSITION OF DISASTER COVERAGE BY OUTLET OWNERSHIP




*Notes.* This figure plots the composition of disaster coverage for national outlets, chain-owned local outlets and independent local outlets. Shares are calculated among 424,888 disaster articles: 41,016 national, 353,748 chain-owned local and 30,124 independent local. A local outlet is chain-owned when its owner holds two or more newspaper titles in the quarter of publication, and independent when its owner holds one. Categories are mutually exclusive. Skeptical treatment takes precedence over climate linkage, and climate linkage over the narrative and contextual categories. An article is assigned to Only event facts when every relevant block reports the specific event. It is assigned to Event facts + other topics when at least one relevant block is not about that event. This block either discusses hazards as a class, a trend or a projection, or covers green products, sustainability or climate regulation.

The way a disaster is represented in the news therefore depends on the type of outlet that covers it. Figure 2 shows the composition of disaster coverage for national outlets, chain-owned local outlets, and independent local outlets. The key difference is in the share of disaster articles linked to climate change, which is 30.8 percent for national outlets, 16.9 percent for chain-owned local outlets, and 21.1 percent for independent local outlets. Across all disaster articles, chain-owned local outlets link disasters to climate change 4.2 percentage points less often than independent ones, and the gap remains statistically significant within the same quarter (Appendix Table D.1).

# 3 Households and Fund Investors: Green ETFs and ESG Fund Flows

I find that a higher share of local news coverage reading the extreme weather event as climate change is associated with a significantly larger increase in the green-ETF share of household portfolios over the following two quarters. The observed increase in sustainability comes on the extensive margin. Disaster coverage alone moves portfolio turnover without a net tilt toward sustainable funds (Section 3.4).

To show this, I begin with a difference-in-differences design that compares affected counties with clean comparison counties around the same episode and, among affected counties of the same hazard in the same year, counties whose newspapers read the disaster differently. The estimate is stable, conditional on detailed controls for the county's politics, climate beliefs, demographics and climate attention before the event, and with county fixed effects. I then present and rule out the alternative explanations by progressively saturating the specification, and by introducing a within-episode design.

## 3.1 Empirical Strategy

My objective is to estimate how the local interpretation of a disaster affects the green share of household portfolios. The ideal experiment would compare the response of investors in otherwise identical counties, with the same level of exposure to the same disaster at the same time, and differing only in how the local discourse (proxied by newspapers) has interpreted the disaster.

I approximate it by exploiting two sources of variation. First, because the timing of a disaster is exogenous, comparing the change in the green share of affected counties with that of unaffected counties around the same episode identifies the response to the disaster at average interpretation. Second, among the affected counties, I compare those with similar observable characteristics, exposed to the same type of disaster in the same year, differing in the climate-interpreted share of their local newspapers' coverage. In practice, I implement this comparison by estimating the following local-projection regression at each horizon $h$ from the extreme weather event onset:

$$\Delta_h q_{ie} = \gamma_h M_{ie} + \theta_h D_{ie} \cdot M_{ie} + D_{ie} \cdot \delta^h_{k(e),y(e)} + X'_{it}\left(\Gamma_h + D_{ie} \cdot \Lambda_h\right) + \alpha^h_{s(i)} + \eta^h_e + \varepsilon_{ieh}. \tag{4}$$

The unit $i$ is a county or an adviser, observed around episode $e$ starting in quarter $t(e)$, at horizon $h$.

Treatment is repeated and non-absorbing. A county can be affected by several episodes over the sample, and the comparison counties of one episode can be the affected counties of another. Therefore, I construct the sample of treatment and control units around each episode with a clean-window design. Treated units are those in the disaster zone of the episode with no earlier local disaster in the two quarters before the start, and control units are clean comparison counties[27] not affected by that episode or by any other major disaster

[27] More precisely, clean comparisons are defined relative to each episode and each horizon. For horizon $h$, control county-quarters

within a window.[28]

The outcome is the change in the share of a county's exchange-traded fund holdings in environmentally focused funds, $q_{it}$, aggregated to the county of the adviser's office.[29] I define the cumulative portfolio change from the quarter before the onset as $\Delta_h q_{ie} \equiv q_{i,t+h} - q_{i,t-1}$.

To isolate active rebalancing from valuation drift, I construct the portfolio weight change at fixed pre-event price. For each episode starting in quarter $t$ and each horizon $h$, I rebuild the county portfolio using quantities observed at $t + h$, valuing every ETF at the pre-event price vector $P_{j,t-1}$,

$$q_{i,t+h|t-1} = \frac{\sum_j G_j P_{j,t-1} Q_{ij,t+h}}{\sum_j P_{j,t-1} Q_{ij,t+h}}, \tag{5}$$

and $\Delta_h q^{FP}_{ie} \equiv q_{i,t+h|t-1} - q_{i,t-1|t-1}$.[30] The pre-event-price measure isolates active reallocation, while the total measure captures the net observed green exposure. When both are positive and close in magnitude, the observed increase in relative exposure is neither an artifact of revaluation nor a counterbalancing of negative drift to maintain the desired exposure. It is an active change to a new desired allocation. The main text reports the two-quarter cumulative pre-event-price portfolio change, rescaled to basis points, and Appendix Table E.1 adds the current-price change.[31]

$D_{ie}$ is an indicator function and takes value 1 when $i$ is in the disaster zone of episode $e$, while $M_{ie}$ is the climate-interpreted share of local coverage of the episode.[32] The coefficients of interest are $\theta_h$, the differential effect of interpretation in affected relative to unaffected units, and $\gamma_h + \theta_h$, the total interpretation effect among affected units.

$\delta^h_{k(e),y(e)}$ are affected × hazard × year fixed effects, where $k(e)$ and $y(e)$ are the hazard type and year of episode $e$, capturing the average response of affected counties in each hazard type and year. $\alpha^h_{s(i)}$ are state fixed effects, where $s(i)$ is the state of unit $i$. $\eta^h_e$ are episode fixed effects, absorbing the level of each disaster. The episode fixed effects compare units in the disaster zone with the unaffected units of the same disaster, while the affected × hazard × year fixed effects compare the interpretation effect among affected units within the same year and the same hazard type.

$X_{it}$ is a vector of county controls that may be correlated with both local interpretation and the portfolio response. It always includes disaster coverage, among both affected and unaffected units, to control for news attention to the disaster, with additional controls depending on the analysis. The additional controls hold fixed what could make a county both receive a climate reading of the disaster and buy green assets in any case. They are the county's Democratic vote share at the last presidential election and the share of its residents worried about global warming, its income, college share, share of white residents, total population

have no local disaster in the stabilization window before the reference quarter and none between it and $t+h$. Treated county-episodes have no other local disaster through horizon $h$. Episodes starting in the same quarter are kept, each county–episode weighted by one over the number of episodes of its county in that quarter.

[28] Following Dube et al. (2025), the first comparison is a local-projection difference-in-differences with clean controls. Because episode fixed effects absorb everything common to an episode, the effect of $D_{ie}$ is estimated within each episode. The regression is a stacked version of their quarter-level event study estimator in which each episode forms its own stack, as in Cengiz et al. (2019).

[29] As described in Section 2, holdings are pooled across the county's advisers before taking the share, and adviser-level regressions are weighted by pre-disaster shares of county ETF holdings.

[30] This follows the same logic as Calvet et al. (2009), adapted to fit the local projection estimation framework.

[31] For ease of exposition, most tables and figures in the main text report the pre-event-price portfolio change. The Appendix shows, for each of these tables and figures, the version at both pre-event and current prices side by side (Tables E.1 and G.1, Figures E.1 and G.1). Magnitudes are consistent throughout.

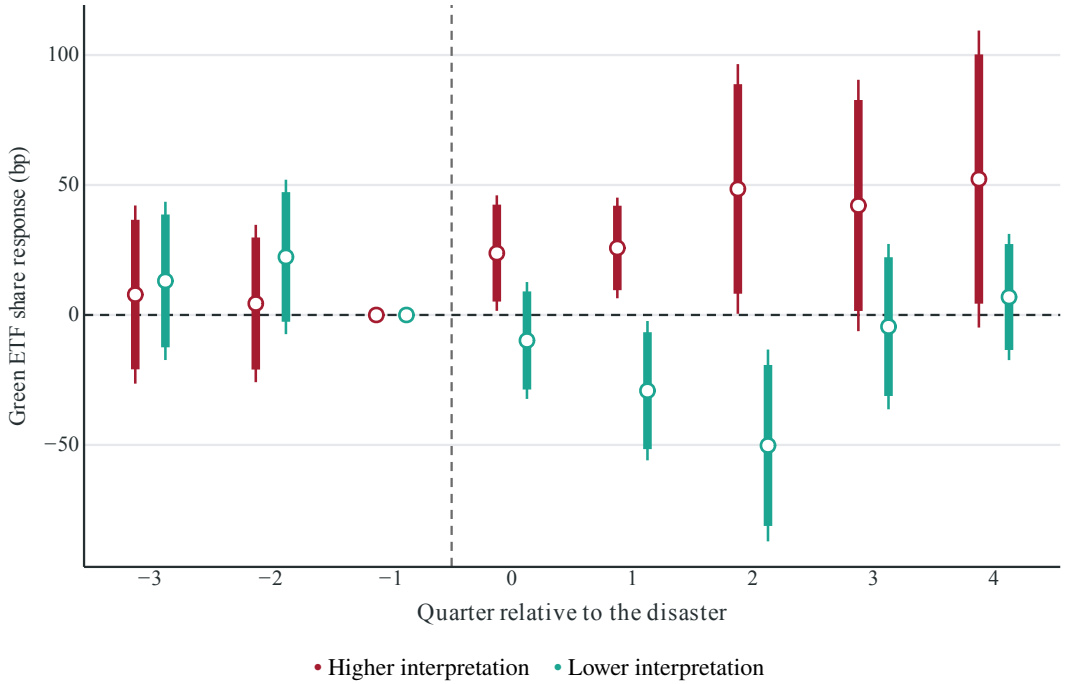
[32] $M_{ie}$ is missing for county-episodes with no local coverage, so the interaction is estimated on covered county-episodes.

and population density, the size of its advised portfolio in the previous quarter, the share of its local coverage devoted to climate policy and green topics in the quarter before the episode, and its own damage, fatalities and injuries. Each control enters alone ($\Gamma_h$) and interacted with the affected indicator ($\Lambda_h$), so $\theta_h$ is net of how the green share of counties with these characteristics changes around any episode and of how they respond to a disaster.[33] Section 3.3 discusses the specific confounders each set of controls rules out. Standard errors are clustered by county.

## 3.2 Results

**Interpretation Effect on Green Household Demand.** The full results are presented in Table 2. I find that a one-standard-deviation increase in local climate interpretation of the disaster raises the pre-event-price green share of affected counties by 30 basis points relative to the unaffected counties of the episode (Column 4). The estimate for observed prices (Appendix Table E.1) is within one basis point of the pre-event-price estimate, so price drift does not drive the increase in either direction.

FIGURE 3: DISASTER RESPONSE OF COUNTY GREEN ETF SHARES BY LOCAL INTERPRETATION, BY QUARTER




*Notes.* This figure plots the conditional disaster responses of the county green ETF share for counties with local interpretation above and below the median, by quarter relative to the flood, evaluated at a common reference. Changes are relative to quarter −1, normalized to zero. The regressions include calendar-quarter and disaster-episode fixed effects and Affected × year terms. Controls include lagged assets, prior politics, demographics, continuous interpretation and attention. Thin and thick bars are 95 and 90 percent confidence intervals, with two-way clustering by county and disaster episode. The joint test that the gap between the two groups is zero in quarters −4 to −2 gives $p = 0.463$. Appendix Figure E.1 shows the same estimates at pre-event and current prices.

Of two counties exposed to the same disaster, the one whose local coverage reads the event as climate change ten points more often, thirty percent of articles in place of twenty, holds 19 more basis points of green share at pre-event prices two quarters later. For the median county's advisers, that is about $0.66 million

[33] Damage, fatalities and injuries are zero outside the affected counties, so they enter once. Coverage of climate policy and green topics enters alone.

TABLE 2: COUNTY GREEN-ETF PORTFOLIO RESPONSE

| *Dependent variable:* Green-ETF share response (basis points, pre-event prices) | | | | | |
|---|---|---|---|---|---|
| | (1) | (2) | (3) | (4) | (5) |
| Affected × Interpretation | 25.48** | 25.86** | 22.46* | 30.31** | 22.76* |
| | (10.13) | (12.33) | (12.18) | (12.58) | (12.44) |
| Affected × Attention | .86 | 4.91 | 3.64 | 1.69 | −9.36 |
| | (10.93) | (10.92) | (10.30) | (10.06) | (10.01) |
| Total Interpretation effect among affected counties | 26.70** | 27.08** | 23.47* | 31.32** | 22.73* |
| | (10.45) | (12.44) | (12.26) | (12.53) | (12.31) |
| Observations | 105,662 | 105,662 | 105,662 | 105,662 | 105,662 |
| $R^2$ | .028 | .029 | .031 | .032 | .102 |
| Affected county-episodes | 244 | 244 | 244 | 244 | 244 |
| Episode and quarter FE | ✓ | ✓ | ✓ | ✓ | ✓ |
| State FE | ✓ | ✓ | ✓ | ✓ | |
| County FE | | | | | ✓ |
| Affected × hazard × year FE | | ✓ | ✓ | ✓ | ✓ |
| Financial and demographic controls | | | ✓ | ✓ | ✓ |
| County damages, fatalities and injuries | | | ✓ | ✓ | ✓ |
| Political and belief controls | | | | ✓ | ✓ |
| Lagged climate attention | | | | ✓ | ✓ |
| Affected × controls | | | | ✓ | ✓ |

*Notes.* This table reports estimates of the county green-ETF share response to disasters. The outcome is the average county green-ETF share over the quarter of the episode and the two following quarters minus its average over the two quarters before, in basis points, at pre-event prices, held at their level in the quarter before the episode. Appendix Table E.1 reports the same estimates at pre-event and current prices. The sample covers high-salience flood, hurricane and coastal episodes. Affected is one for counties in the episode footprint. Interpretation is the climate-interpreted share of local coverage of the episode and Attention the log volume of that coverage, both over the first four weeks of the episode and standardized; one standard deviation of Interpretation is 15.7 percentage points. The total Interpretation effect among affected counties is the sum of the Interpretation coefficient and its Affected interaction, with its standard error from their joint covariance. Controls are defined in Section 3.1. Standard errors, in parentheses, are clustered by county. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

more in green ETFs.

The estimate is similar with interpretation measured over the calendar quarter and at the adviser level (Appendix Tables E.2 and E.4). It does not rest on the comparison counties. The slope among affected counties alone is comparable, while the same contrast among unaffected counties is indistinguishable from zero (Appendix Figure E.2).

**Extensive Margin.** Per standard deviation of interpretation in affected counties, the share of advisers in the county holding a green ETF rises about 6 percentage points, and as much among advisers reporting from the same county across the two quarters, so the increase comes from advisers already there. That is more than four times the ordinary movement of this share after a disaster (Appendix Table E.5).

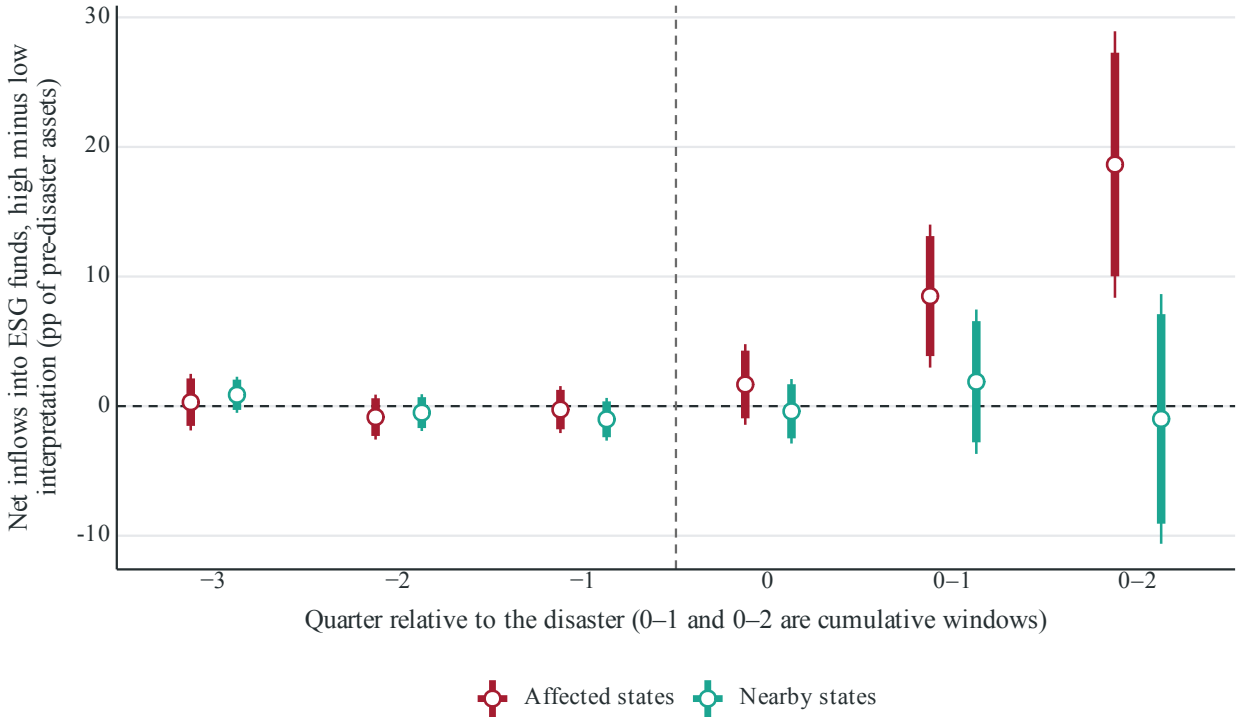
**Extension to ESG Mutual-Fund Flows.** To establish external validity beyond green ETFs, I extend the analysis to ESG mutual-fund flows.[34] Disaster exposure is assigned at the fund's headquarters state. The assumption is that headquarters location proxies for clientele exposure. With the same assignment, Marshall et al. (2021) find higher net inflows into green mutual funds relative to matched nongreen funds when disasters occur in the funds' headquarters states. In Figure 4 I compare relative net ESG inflows, measured against net inflows into non-ESG funds, across disasters with high and low climate interpretation, holding disaster coverage in the headquarters state and other covariates constant.

Fund fixed effects absorb persistent differences in fund flows. ESG-status-by-quarter fixed effects absorb broad national trends in ESG investment by allowing ESG and non-ESG fund flows to vary separately each quarter. Hazard-by-exposure-group fixed effects account for average differences across disaster types and geographic exposure.

Among funds headquartered in disaster-affected states, relative net ESG inflows are larger following high-interpretation disasters, by 8.5 percentage points of pre-disaster assets over two quarters, and by more over three (Figure 4). The corresponding differences in neighboring unaffected states are indistinguishable from zero.

[34]ESG funds are identified by a dated green or sustainable fund-name classification, a Morningstar intentional-ESG designation effective before the disaster, or a Morningstar climate-theme flag. The comparison group comprises confirmed non-ESG mutual funds. ETFs, low-carbon-only funds and funds with ambiguous classifications are excluded.

FIGURE 4: CLIMATE INTERPRETATION AND FLOWS INTO ESG MUTUAL FUNDS




*Notes.* This figure plots the High-minus-Low difference in relative net inflows into ESG mutual funds, measured against non-ESG funds, separately for funds headquartered in disaster-affected and neighboring unaffected states. High and Low interpretation are defined using the median interpretation among disaster-affected state observations, with the same cutoff applied to both groups. The outcome is net dollar flow divided by pre-disaster assets, in percentage points. Pre-event points use individual quarterly flows. Post-event points accumulate flows from quarter 0, with the same asset denominator. Each horizon is estimated separately on fund–disaster observations. High interpretation and its interaction with ESG status enter separately for affected and neighboring states, together with ESG status × Affected. Attention and its ESG interaction also enter separately for the two groups of states. Fixed effects are fund, ESG status × event-calendar quarter, and hazard × group of states. Controls are log disaster damage and its interactions with ESG status and with the group of states, and log pre-disaster assets. Thin and thick bars are 95% and 90% confidence intervals, clustered by disaster episode.

## 3.3 Identification

Equation (4) nests two designs. The first identifies the response to the disaster, through the experience of the event, its salience and the attention it receives, by comparing affected units with the clean comparison counties of the same episode. The response is therefore identified under no anticipation and parallel trends between affected and comparison counties.[35]

The second measures the interpretation effect among the affected counties by comparing those exposed to the same type of disaster in the same year, with similar observable characteristics and damage, whose local interpretation of the disaster differs. For the purpose of identifying $\theta_h$, the design is equivalent[36] to a

[35]No anticipation means that the green share of a county does not move before an episode in response to it. Parallel trends means that, absent the episode and given the controls, the green share of affected and comparison counties would have followed the same path. With repeated disasters, this holds for each episode a county is exposed to (Dube et al., 2025). The clean window of two quarters keeps earlier local episodes out of the pre-period of affected counties and any change in exposure out of the comparison counties through the horizon.

[36]The unaffected counties enter $\theta_h$ only through these components, and the total effect among affected counties, $\gamma_h + \theta_h$, through their own slope, $\gamma_h$. In this sense $\theta_h$ is a triple difference, across interpretation, between affected and unaffected counties, around

regression on the affected counties alone in which the fixed effects, the controls and the baseline interpretation response $\gamma_h$ are held at their estimates from the entire sample.

The identifying assumption is that, conditional on these controls, the climate reading of a county's coverage of a disaster is unrelated to how its green share would have changed under a different reading. Namely, affected counties discussing the disaster more as climate change would otherwise have changed their green share similarly to those discussing it less as climate change. A bias would arise if the readers who demand climate interpretation of a disaster would have bought green assets after it without that interpretation. Appendix Figure E.2 shows the event-study path with the coefficients on leads and lags. The leads show no pre-trend.

By exploiting progressively more fine-grained variation, the columns of Table 2 restrict the level at which a confound would have to operate. The episode fixed effect absorbs the disaster itself, its size, its national coverage, everything common to the affected and unaffected counties of each specific episode, including anything else in its quarter of onset. Because each horizon is estimated separately, the episode fixed effect also absorbs common shocks in the following quarters, such as national events or news. Furthermore, the state fixed effects absorb the average change in the green share of the counties in each state. On top of this, column 2 controls for the average response of affected counties in each hazard type and year, so that the interpretation effect does not come from comparing fundamentally different disasters, or from underlying trends in climate salience or market trends in sustainable products picked up by comparing disasters in different years.

Column 3 adds financial and demographic controls, including the size of the advised portfolio in the previous quarter, and the county's own damage, fatalities and injuries to check that the interpretation effect is not driven by how strongly a specific county was affected by the event.[37] Column 4 adds the county's prior Democratic vote share, its share worried about global warming and its coverage of climate policy and green topics in the quarter before onset, which absorbs the change in the green share that goes with how much its newspapers already write about climate. It also interacts the controls with the affected indicator, except that coverage, so that the interpretation effect is estimated net of the disaster response of richer, more Democratic and more worried counties.

Column 5 replaces state with county fixed effects. On the stacked panel a county appears in every episode it is exposed to, almost always as a nearby or distant comparison county,[38] so the county effect absorbs its trend in the green share and its usual change around large disasters, wherever they occur. The interpretation effect is then read from how an affected county's change after its own disaster departs from that usual change, compared with the other affected counties in the same hazard and year. The interpretation effect survives county fixed effects. If the estimate only reflected counties that react strongly to every disaster, it would have vanished once each county is compared with its own usual response.

Nor is interpretation a stable trait of local outlets. A fixed outlet component explains only 5.5 percent of its variance, and most of it reflects how a specific outlet reads a specific disaster (Section 2.2). This suggests that the effect is tied to the interpretation of the current event.

Remaining confounds must operate within a county and among affected counties of the same hazard type

the event.

[37] Damage could itself be the confounder. A county more strongly impacted may be less likely to buy green or may be forced to liquidate assets, and local news in a severely affected county may turn to fatalities, losses and emergency information, at the expense of climate interpretation articles, so that both lower interpretation and lower rebalancing toward green appear in the most affected counties.

[38] Even a county that is affected by some episode is affected in only 0.4 percent of the episodes it enters, and is a comparison county in the rest.

and year. For example, an affected county whose newspapers read a flood as climate change would have to raise its green share around the event, by more than the other counties affected by similarly impactful floods that year, for reasons other than interpretation-driven climate concern, and those reasons would have to be correlated with how its newspapers read the flood.

Section 3.4 addresses the remaining identification concerns. The first is that the interpretation effect picks up predetermined county beliefs, attitudes or trends unrelated to disasters, which I address with placebo tests and with the newspaper-ownership design of Section 5, where the interpretation that local outlets supply shifts with ownership changes. The second is that it reflects comparisons across disasters, which the within-episode design of Section 3.4 replaces with a comparison of advisers at the same distance from the same disaster in the same state. The third is that it mostly captures attention to the disaster, which I address by controlling for coverage and by showing that turnover rises with coverage but does not respond to interpretation. Section 3.4 also discusses threats to reading the response as a demand shock for sustainability.

### 3.4 Alternative Explanations

**Confounders of Interpretation.** The interpretation effect could capture something else that happened in the county in the same quarter, such as climate policy events and their coverage. Appendix Table F.1 adds the share of local coverage devoted to climate policy and green topics, in the same quarter and in the quarter before. It adds them as controls and interacted with the affected indicator, with interpretation and with their product, allowing other forms of climate attention to compete against the interpretation of the disaster in explaining the portfolio response. The climate-attention interactions are all indistinguishable from zero, and the interpretation effect, estimated against them, remains unchanged.

The interpretation effect could reflect attitudes the county already held, with the disaster as the occasion on which they are acted upon. If residents more concerned about climate change buy green assets whenever a salient local event occurs, and if coverage in those same counties is also more climate-interpreted, the correlation between interpretation and green reallocation would be spurious. The rows of Appendix Table F.1 that add the county's share of residents worried about global warming, its perceived scientific consensus, its Democratic vote share and its college share suggest that this is not the case.

If local climate interpretation mainly captured a general propensity of climate-concerned counties to buy green assets after any salient local shock, the same pattern would appear after earthquakes, which no coverage reads as climate change. There is no differential green tilt after earthquakes in counties whose coverage of other disasters was more climate-interpreted (Appendix Table F.2). Finally, Section 5 uses changes in the ownership of the county's newspapers, which shift the climate-interpreted share of their disaster coverage, to isolate the interpretation that local outlets supply.

Interpretation could proxy for attention to the disaster. More heavily affected places may be both covered more and interpreted more. Affected counties most likely pay attention to the disaster in any case. The main specification controls for its damage, and every specification controls for coverage.

If interpretation only measured attention, advisers in high-interpretation counties would trade more after the disaster, as advisers do where coverage is heavier. I examine how the gross turnover[39] of advisers' books after the disaster changes with local disaster coverage and with interpretation, in affected and unaffected counties (Appendix Table F.3). Turnover of the whole reported book rises with coverage in both affected and unaffected counties, by 0.107 and 0.081 log points per standard deviation. It does not rise with interpretation

[39]Following Calvet et al. (2009), turnover is the value of absolute quantity changes over the quarter at start-of-quarter prices, divided by the starting reported book value, aggregated to the county and measured as log(1 + turnover in basis points).

in affected counties. Advisers in high-interpretation counties tilt toward green ETFs without trading more than advisers in low-interpretation counties. Interpretation is therefore not a proxy for attention to the disaster.[40] The newspaper-ownership design of Section 5, which shifts the interpretation that local outlets supply while holding coverage fixed, separates the two further. Section 7 separates them in household insurance demand.

**Threats to Event Link and Specificity.** A concern with the county estimates in Table 2 is how much of the interpretation effect comes from comparing different disasters, and how much from differences in interpretation within one disaster.[41] The fixed effects and controls of Section 3.1 compute the interpretation effect only across disasters of the same type, in the same year, with comparable damage. Such disasters may still differ in the national attention and the national interpretation they receive, in the state where they occur and its politics and policies, and in the local economy of the affected counties, and these can correlate with the local interpretation. The within-episode design removes these differences. It compares advisers at the same distance from the same disaster in the same state.

To compare portfolio responses to the same disaster, I use advisers in affected counties and nearby advisers within 50 miles of the affected area, and I measure local interpretation at the adviser's office ZIP code. I estimate

$$\Delta_h q_{ie} = \theta_h M_{ie} + X'_{ie}\gamma_h + \alpha^h_{e,s(i)} + \lambda^h_{e,b(i,e)} + \mu^h_{c(i)} + u_{ie,h}, \qquad h \in \{0, 1\}, \tag{6}$$

where $\Delta_h q_{ie}$ is the change in green-ETF share from the previous quarter to the disaster quarter or the following quarter, measured in basis points at pre-event prices. $M_{ie}$ measures interpretation in ten-percentage-point units, with a common coefficient across affected and nearby advisers. $X_{ie}$ contains the same four groups of controls as the main design, namely attention, financial, demographic and political, and their group interactions.[42] Interpretation and attention are measured at the adviser's ZIP code and the financial controls at the adviser level. Standard errors are clustered by disaster episode and ZIP.

Episode by state fixed effects absorb the disaster itself, with its footprint, impact and national coverage, and everything common to the state, such as differential trends in political or climate views, in the insurance market and in state policy. Episode by band fixed effects allow affected advisers and nearby advisers 0–25 and 25–50 miles from the affected area to have different average responses to each disaster. With county fixed effects, the comparison is relative to the county's average portfolio response across episodes in which its advisers are affected or nearby.

The identifying assumption is that, conditional on the controls, advisers in the same state at the same distance from the same disaster would otherwise have changed their green share by the same amount. A ten-percentage-point increase in interpretation is associated with a 7-basis-point increase in green-ETF share in the disaster quarter and a 12-basis-point increase by the following quarter, both relative to the previous quarter (Appendix Table F.4, column 3).

**Threats to Revealed Demand Shock.** The response could reflect the advisers' views and not their clients'. I restrict the sample to local advisers serving predominantly household accounts, but most of

[40] The event-study plots dividing counties at the median of interpretation (Figure 3) show that low-interpretation counties tend to reduce their green share in the quarters after the disaster, suggesting, if anything, that they are reacting by accommodating the demand.

[41] NOAA splits one storm system into separate episodes at state and forecast-office boundaries and by hazard type, so many episodes have a single affected county with observed advisers, and a separate intercept for the affected counties of each episode is too demanding at the county level.

[42] See Appendix Table F.4.

their assets are managed on a discretionary basis, so the adviser decides the trades. I therefore compare county voting, which proxies for clients' political views, with advisers' political views measured from staff donations, including both in the regression. If advisers' own views drive the response, their political leanings should help explain it. The response is stronger in less Democratic counties but does not vary significantly with advisers' own political views (Appendix Table F.7).

Less Democratic advisers are less likely to hold any green ETF to begin with. However, conditional on holding a green ETF, their own views do not significantly affect the disaster response and the interpretation effect.[43] This evidence suggests that the green ETF response to the local interpretation reflects the views of the households the advisers serve, and not the advisers' own, consistent with what Pan et al. (2025) find.

The effect could be mechanical. Disasters may generate liquidity needs, withdrawals, or precautionary adjustments that change ETF shares even without any active move toward green assets. In particular, advisers may cut ETFs first to meet their liquidity needs because ETFs tend to be particularly liquid and easy to trade, and may be held as a liquidity buffer. If, among ETFs, the non-green ones are liquidated more aggressively, the ETF denominator can fall and green ETF shares can rise even without a genuine tilt toward green funds. Instead, I find that green ETF assets rise alongside non-green and total ETF assets, while total observed 13F holdings show no statistically significant change (Appendix Table F.9). ETF and whole-portfolio flows show no systematic contraction in counties and episodes with higher climate interpretation (Appendix Table F.8). The interpretation effect also holds among advisers in nearby counties not directly affected by the disaster (Appendix Table F.5). These findings do not support liquidation as the explanation for the increase in the green ETF share.

Advisers may also act on their own incentives. One is to encourage clients to buy their green products, or to buy green ETFs and then accommodate that demand by selling the underlying stocks. About 2% of the local advisers in my sample sponsor or manage the ETFs in their holdings. I find that when advisers purchase green ETFs, they also tend to purchase the underlying constituent stocks directly (Appendix Table F.10). This pattern does not support systematic offsetting sales within the observed portfolios. Another is to use a disaster as an occasion to rebalance client portfolios and justify their fees. I control for advisers' prior ETF and whole-portfolio turnover. The interpretation response does not vary significantly with a one-standard-deviation increase in either measure (Appendix Table F.11).

Affected advisers may buy green ETFs because they expect higher returns (Giglio et al., 2025). Since a single local disaster event provides little information about either climate risk trends or green firms' cash flows, the return expectation motive requires anticipating that many investors will buy these products as a result of the event[44] and push up their prices. These expectations would be shared by affected and unaffected advisers. The common component of this response is therefore absorbed by the episode fixed effects.

For return expectations to explain the stronger response of affected and nearby advisers, and the interpretation effect among them, affected advisers in higher-interpretation places would need to expect a stronger reaction from others than unaffected advisers do. They would also need to expect a stronger reaction than affected advisers in places where the disaster was interpreted less as a climate event. Expectations about other investors would follow the national coverage of the event, so I add national coverage and national interpretation, alone and interacted with local interpretation, to the county and adviser regressions of Section 3

[43] Only smaller advisers show an effect of the firm's own leaning on the average disaster response, and none on the interpretation response.

[44] Morris and Shin (2002) show how public information can influence behavior through expectations of others' actions. Schmidt-Engelbertz and Vasudevan (2025) document speculation motivated by beliefs about other investors, even when it conflicts with investors' own valuations.

(Appendix Table F.12). The local interpretation response among affected counties and advisers remains positive after accounting for national news. If anything, national attention strengthens this response and widens the gap between affected places with high and low local interpretation,[45] while national interpretation does not significantly change it.

Finally, if advisers instead bought green ETFs on momentum, chasing past returns, the response would be larger after green ETFs had done well. The response does not differ after strong and weak years for green ETFs (Appendix Table I.7).

# 4 Mutual-Fund Managers and Active Rebalancing

The second setting I study is actively managed mutual funds. In this context, I can observe the investment teams that make the decisions and measure the political views of the managers themselves, as well as the local information environment they are embedded in. This allows me to separate the decisions of delegated portfolio managers and to study the effect among professional institutional investors.

I find that a higher share of local news coverage reading the disaster as climate change raises the shift of nearby mutual fund managers' portfolios toward stocks of environmentally friendly firms in the quarter of the event.

## 4.1 Empirical Strategy

The hypothesis is that managers are influenced by the narratives surrounding a disaster that occurred near where they work, and it affects their portfolio decisions, particularly toward more sustainable assets when local coverage interprets the disaster as climate change. To show this, I compare the shift toward stocks of more relative to less environmentally friendly firms in the quarter of the event across funds with offices within 50 miles of the affected area whose local newspapers read the event differently. In each quarter, I link a fund to the nearest disaster and to every disaster that affects one of its offices.[46] For fund $f$ and episode $e$, I estimate the following regression

$$\Delta_h q_{fe} = \theta_h^n D_{fe}^n M_{fe} + \sum_{b \neq n} \theta_h^b D_{fe}^b M_{fe} + X_{fe}' \Gamma_h + \sum_g D_{fe}^g Z_{fe}' \Lambda_h^g + \delta_{g(f,e),e}^h + \mu_{c(f,e)}^h + \kappa_{s(f)}^h + \varepsilon_{feh}. \quad (7)$$

The outcome, $\Delta_h q_{fe}$, is the green tilt of the fund's trading, the change in its weight in stocks in the top tercile of the score minus the change in its weight in stocks in the bottom tercile, over the quarter of the disaster, in basis points of the fund's scored stock portfolio. I measure each stock's environmental profile with its Environmental Pillar score (Section 2.1), after validating this measure with local heat waves (Section 4.3). As for households (Equation 5), I compute the weight changes holding constant prices at the beginning of the quarter.

$M_{fe}$ is the share of local coverage that interprets the disaster as climate change where the managers are (Equation 3), and $D_{fe}^b$ indicates the distance band of the fund's managers. The bands are the affected

[45] The strengthening is statistically significant for counties and book-weighted advisers, and remains positive but is not statistically significant with equal adviser weights.

[46] As disasters I consider floods, hurricanes, storms, and wildfires. I exclude a fund from the analysis of a disaster when one of its offices was affected by another disaster in the same or the previous quarter. I also exclude it when it has an office within 100 miles whose county had no local coverage of the disaster, since local interpretation is not defined there, and when stocks with an Environmental Pillar score make up less than half of its stock portfolio before the disaster. The latter excludes few observations, and the median fund's stock portfolio before the disaster is almost fully scored.

counties (affected managers), within 50 miles of them (nearby managers, $b = n$), between 50 and 100 miles, and more than 100 miles away (the comparison group).[47] Each band has its own slope, $\theta_h^b$. The coefficient of interest is $\theta_h^n$, the total effect of interpretation among nearby managers.

$g(f, e)$ is the fund's exposure group in the disaster. The groups are affected managers, managers within 100 miles, and the comparison group. $\delta^h_{g(f,e),e}$ are exposure-group-by-disaster fixed effects, which capture each group's average response to each disaster, including its damage, and give the comparison group its own level in each disaster. $\mu^h_{c(f,e)}$ are fixed effects for the county from which the fund is managed. $\kappa^h_{s(f)}$ are fund-style fixed effects, so that each fund is compared with funds of the same investment style. The strongest specification also includes fund fixed effects, so that each fund's change in green tilt is measured against its own average quarterly change across the disasters in the sample.

$X_{fe}$ contains the size, age and turnover of the fund before the disaster and an indicator for nearby managers, which gives them a level common to all disasters. $Z_{fe}$ contains, for the fund's offices in each exposure group $g$, the volume of local coverage and, depending on the column, county and political controls, as well as damage for the offices in the affected counties.[48] I also control for how the county's outlets interpret the other large disasters they cover, so that the comparison holds fixed the county's usual reading of climate events. I cluster standard errors by disaster.

**Identification.** The effect of interpretation is identified by comparing the response of managers at the same distance from the same disaster, managing funds of the same investment style, whose local coverage interprets it more or less as climate change, relative to how funds managed from their county usually respond to a disaster. The identifying assumption is that, conditional on the controls, the climate reading of a manager's local coverage is unrelated to how the fund's green tilt would have changed under a different reading. Namely, managers whose coverage reads the disaster more as climate change would otherwise have changed their green tilt similarly to managers at the same distance whose coverage reads it less as climate change. A bias would arise if managers whose newspapers read the disaster as climate change would have tilted toward greener stocks after it without that reading.

Isolating the effect of local interpretation on managers has two main challenges. First, how local outlets interpret a disaster may be endogenous and reflect stable characteristics of the place and of the managers who live there, and not the event itself. Namely, counties whose outlets often write about climate change, or with a given political leaning, may also host managers who respond differently to any disaster. Second, the disaster itself can change the stocks the managers hold and the flows into their funds, whatever its coverage.

The first challenge is addressed by the fixed effects for the county from which the fund is managed, $\mu^h_{c(f,e)}$, which absorb how funds managed from that county respond to disasters in general. The second challenge is addressed by the exposure-group-by-disaster fixed effects, $\delta^h_{g(f,e),e}$, which absorb the movements of greener and browner stocks in the quarter of the disaster that are common to managers within the same distance band from it.

The mutual fund manager design combines the advantages of each of the two main designs I run for

[47] Most funds are managed from a single county. For funds with offices in several bands, each indicator is the fraction of the fund's offices in the band, with equal weight on each office, and $M_{fe}$ averages $M$ over these offices. The results are robust to keeping only the funds with all their offices in one county, and to placing the fund at all its offices or only at the office closest to the disaster (Appendix Table G.4).

[48] Political controls are the county's Democratic two-party vote share and the managers' political score, from the political donations of the named portfolio managers or, without them, of the management company. Appendix Table G.3 shows that the results are robust to including funds whose managers and management companies have no matched donations.

counties and advisers. As in the within-episode design (Equation 6), it compares only managers at the same distance from the same disaster. As in the main design (Equation 4), it still includes a clean (unaffected and far) comparison group, allowing me to control for how funds managed from each county (and funds of the same style and characteristics) usually respond to a disaster.[49] The comparison group starts more than 100 miles away because the effects of a disaster spill over beyond the affected area.

To assess the identifying assumption, Figure 5 plots the effect quarter by quarter. In the three quarters before the disaster, the estimates are close to zero and jointly not significant, so there is no evidence of pre-trends.

## 4.2 Results

Table 3 reports the interpretation coefficient for nearby managers, $\theta_h^n$. Panel A shows that nearby managers tilt their portfolios toward greener stocks when local coverage interprets the disaster as climate change. A ten percentage point higher interpretation share raises their green tilt by 69 basis points at beginning-of-quarter prices in the quarter of the disaster (Column 2).[50] The tilt is driven by an increase in the weight of greener stocks as well as a reduction in the weight of browner stocks (Appendix Table G.5). The estimates are similar with political controls (Column 3) and with fund fixed effects (Column 4).[51] In contrast, the volume of local coverage does not move the tilt, with the coefficient on attention small and never significant.

A one standard deviation increase in the interpretation share raises the tilt by 1.1 percentage points, about a tenth of the gap in the tilt between a conventional fund and a fund with an ESG mandate. In dollar terms, the average nearby fund, with $4.3 billion in assets, moves about $49 million from bottom-tercile to top-tercile stocks. Summed over the nearby funds of the average disaster, the shift reaches about $1.4 billion.

The effect is concentrated near the disaster. Appendix Table G.7 reports, from the same regressions, the effect for affected managers and for managers more than 100 miles away. It is as large for affected managers as for nearby managers, while among managers more than 100 miles away a ten percentage point higher interpretation share raises the tilt by 4 basis points, about a twentieth of the effect among nearby managers.

One concern with the results of Panel A is that they could be driven by differences in the fundamentals of the stocks held by managers whose local coverage interprets the disaster differently. A disaster can move the fundamentals and prices of the firms it affects, and it can affect greener and browner firms differently. For example, funds whose local coverage interprets the disaster as climate change may hold more of the greener stocks, which could be directly affected by the policy response or the news about climate risk that the disaster brings.

To address this concern, I use more granular data at the level of the fund, stock and disaster, with the change in the fund's weight in the stock as the outcome. I can then include fixed effects for each stock around each disaster, separately for each fund style, which absorb everything common to the funds of that

[49] For the purpose of identifying the interpretation effect, the design is equivalent to a regression on the exposed managers alone, as in the within-episode design for advisers, in which the county fixed effects, the fund-style fixed effects and the fund controls are held at their estimates from the entire sample. Managers more than 100 miles away enter the interpretation effect only through these components. Restricting the sample to managers near the disaster would leave those components to be estimated only from the few disasters in which each county is affected or nearby. Because a county is far from most disasters, the comparison group pins down how funds managed from each county usually respond.

[50] The results are robust to measuring each stock's environmental profile with its continuous Environmental Pillar score (Appendix Table G.4).

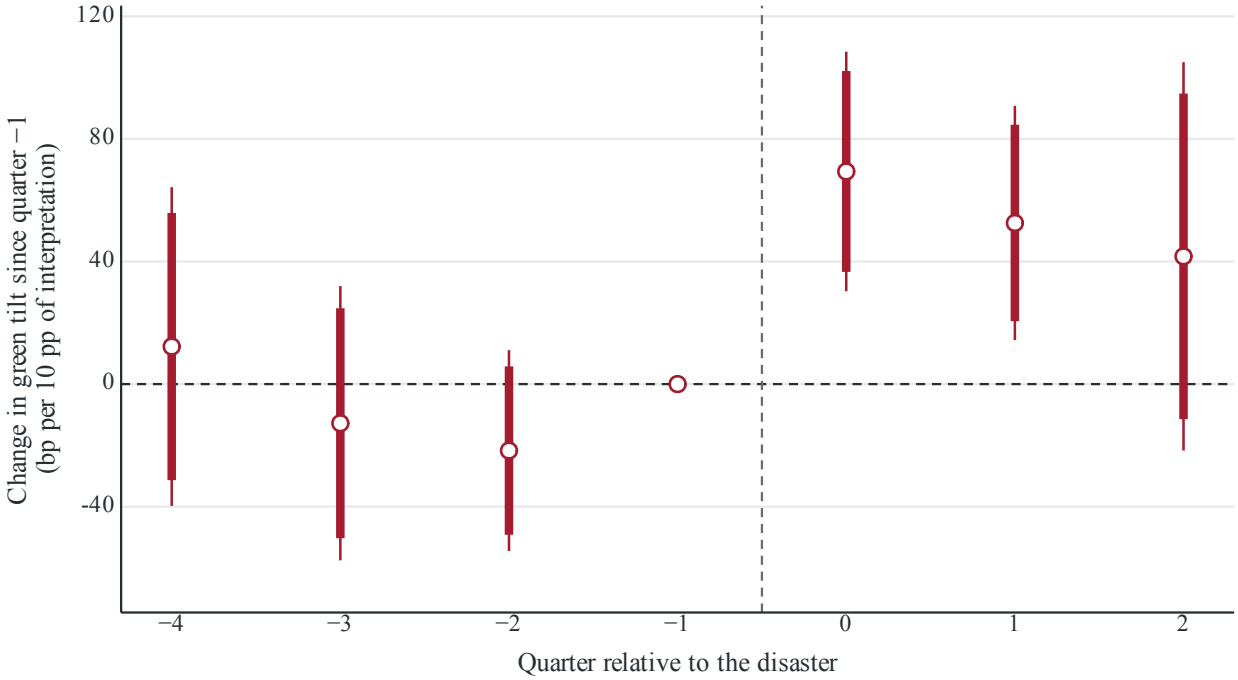
[51] The estimates are smaller but remain significant when I replace the exposure-group-by-disaster and county fixed effects with episode fixed effects and fixed effects for each combination of exposure group, hazard and calendar quarter (Appendix Table G.3, Columns 5 and 10).

TABLE 3: MUTUAL FUND PORTFOLIO RESPONSE

| *Dependent variable:* response in basis points (pre-event prices) | | | | |
|---|---|---|---|---|
| | (1) | (2) | (3) | (4) |
| *Panel A. Fund level: change in the fund's green tilt* | | | | |
| Interpretation | 69.34*** | 69.38*** | 66.25*** | 67.58*** |
| | (20.11) | (19.87) | (18.99) | (19.86) |
| Attention | −2.05 | −.85 | −1.09 | −1.95 |
| | (2.84) | (3.12) | (3.09) | (3.36) |
| Observations (fund-episodes) | 37,047 | 37,047 | 37,047 | 36,964 |
| Nearby fund-episodes | 1,712 | 1,712 | 1,712 | 1,710 |
| $R^2$ | .031 | .032 | .033 | .093 |
| Exposure × episode FE | ✓ | ✓ | ✓ | ✓ |
| County FE | ✓ | ✓ | ✓ | ✓ |
| Fund style FE | ✓ | ✓ | ✓ | ✓ |
| Fund FE | | | | ✓ |
| Fund controls | ✓ | ✓ | ✓ | ✓ |
| Damage and coverage controls | ✓ | ✓ | ✓ | ✓ |
| Exposure × county controls | | ✓ | ✓ | ✓ |
| Exposure × political controls | | | ✓ | ✓ |
| *Panel B. Stock level: change in the fund's weight in a stock, top minus bottom tercile* | | | | |
| Interpretation | | 3.57*** | 3.56*** | |
| | | (1.28) | (1.28) | |
| Attention | | .04 | .03 | |
| | | (.27) | (.28) | |
| Observations (fund-episode-stocks) | | 5,426,327 | 5,426,327 | |
| Fund-episodes | | 53,028 | 53,028 | |
| $R^2$ | | .577 | .577 | |
| Fund × episode FE | | ✓ | ✓ | |
| Fund style × stock × episode FE | | ✓ | ✓ | |
| Exposure × episode FE | | ✓ | ✓ | |
| Damage and coverage controls | | ✓ | ✓ | |
| Exposure × political controls | | | ✓ | |

*Notes.* This table reports estimates of equation (7). In Panel A the dependent variable is the change in the fund's green tilt over the episode quarter, the change in its weight in stocks in the top tercile of the Environmental Pillar score minus the change in its weight in bottom-tercile stocks; in Panel B it is the change in the fund's weight in each stock, and the rows report the response of top-tercile relative to bottom-tercile stocks. Appendix Table G.1 reports the same estimates at pre-event and current prices. Rows report the effects among managers within 50 miles of the episode. Interpretation is per 10 percentage points and Attention per 10 log points. Fund controls are lagged log total net assets, log age and turnover; county controls are the county's interpretation of its other episodes, log fatalities, log population, log population density, log income, college share, unemployment rate and urban share; political controls are the county Democratic two-party vote share and the managers' political score. County and political controls enter separately for each band. Standard errors, in parentheses, are clustered by episode. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

FIGURE 5: INTERPRETATION EFFECT AMONG NEARBY MANAGERS, BY QUARTER




*Notes.* This figure plots the effect of local climate interpretation on the green tilt of nearby managers by quarter relative to the disaster, per 10 percentage points. The outcome is the change in the tilt since the end of quarter −1, in the specification of Table 3, Column 2. Thin and thick bars are 95 and 90 percent confidence intervals, with standard errors clustered at the disaster level. Appendix Figure G.1 shows the same estimates at pre-event and current prices.

style that hold the stock, such as its return and fundamentals, the location of the firm and its own exposure to the disaster, national coverage of the firm and the shock to its industry. Since I compare funds of the same style that hold the same stock around the same disaster, the stock's cash-flow news from the disaster, the part agreed by everyone, cannot explain the effect.

In addition, I include fixed effects for each fund around each disaster, which absorb everything common to the fund's stocks, such as the attention and interpretation of its local coverage, its distance from the disaster, the damage in its county, its flows and its average trading, as well as the fund and county controls of Panel A. Interpretation, attention, damage and the exposure-group-by-disaster fixed effects therefore enter interacted with indicators for the top and bottom terciles of the Environmental Pillar score, as do the political controls in Column 3. The estimate therefore compares how managers whose local coverage interprets the same disaster more or less as climate change reweight the same stocks, relative to the other stocks in their portfolios and to funds of the same style that hold the same stocks.

Panel B shows the results. A ten percentage point higher interpretation share raises the weight of nearby managers in top-tercile relative to bottom-tercile stocks by 4 basis points (Column 2). The estimates are similar with political controls (Column 3) and at current prices (Appendix Table G.1).

## 4.3 Alternative Explanations

**Investors or Managers.** Differences in portfolio responses may also reflect the pressure of the funds' investors. This is especially relevant for smaller funds, whose clientele is often local, so that a disaster can be experienced and interpreted both by the manager and, sometimes, by a significant part of the clientele.

Managers whose clients are more concerned about climate change may face stronger incentives to demonstrate that their funds have responded to a disaster. It is much easier to show investors that the fund acted on climate change risk by showing a higher Environmental Pillar score than through adjustments based on the firms' exposure to physical risk, which are harder to explain and appear less transparent. Conversely, managers facing less pressure may still adjust their portfolios in response to perceived climate risks, but along different dimensions. The differences in the observed environmental tilt may then reflect client pressure and the margins along which managers choose to respond, regardless of their interpretation of the disaster.

To distinguish managers from their clients, I exploit differences between the locations of fund headquarters and managers' offices. Studies of mutual funds and local disasters typically use headquarters as a proxy for manager location (e.g., Fich and Xu, 2025). Yet, headquarters is also a natural proxy for investor location, since individual investors favor locally managed funds (Bailey et al., 2011). Instead, I here use more granular data on managers' locations and treat headquarters as the approximate location of the investors. While managers' offices are usually in the same county as headquarters, the locations differ in 35 percent of cases and are more than 100 miles apart in 30 percent. For sub-advised funds, whose portfolios are run by an outside firm, the locations are more than 100 miles apart in 63 percent of cases. If client pressure drives the environmental tilt, the response should follow the climate interpretation near headquarters and not near managers' offices.

Appendix Table G.11 exploits this separation for funds with known headquarters, using the main specification. I estimate the effect of managers' local interpretation separately for managers near a disaster whose funds' headquarters are within or beyond 100 miles of the disaster. I also estimate the effect of the interpretation near headquarters for funds whose headquarters are within 50 miles of the disaster but whose managers are more than 100 miles away. Nearby managers respond equally strongly whether their headquarters are near or far from the disaster. Conversely, funds with nearby headquarters but distant managers exhibit a smaller, statistically insignificant response. The contrast is stronger when including all fund-episodes, and stronger still among below-median-size funds, whose investors are more likely to live near headquarters (Appendix Table G.12). These results suggest that the response depends more on the location of managers than on that of their funds' investors.

**Measure of Green Stocks.** A related concern is that managers respond along dimensions of climate risk that the Environmental Pillar score does not capture. Managers may disagree on how green a firm is (the ESG ratings of the same firm from different providers correlate only between 0.38 and 0.71, Berg et al., 2022), as well as on how to respond to a higher perceived climate risk, for example by tilting toward sustainable firms or away from firms exposed to certain types of physical or transition risk. The choice of measure biases the interpretation effect only if the part of the response that the score misses is correlated with the interpretation of the disaster. In any case, using the approach and (extended) data of Alekseev et al. (forthcoming), I take local heat waves as the climate shock, since they raise managers' climate concerns, and check that the response goes in the direction of the score. A fund with all its offices exposed to such a heat wave raises its green tilt by 65 basis points at beginning-of-quarter prices (Appendix Table G.2).[52]

**Sale of Firms in the Disaster Area.** The estimated tilt could reflect managers selling firms located

[52]In an ideal experiment, I would observe each climate shock as the manager perceives it and let the portfolio response reveal which stocks managers treat as green. Such shocks cannot be isolated. A disaster bundles the damage it causes, the policy responses it triggers and the news it brings about climate risk. Comparing managers at different distances from it still requires that the response to the bundle is the sum of the responses to its elements. Any more sophisticated approach would rest on the same assumption. I therefore use a sustainability rating, which investors chase (Hartzmark and Sussman, 2019) and fund managers trade to obtain (Gantchev et al., 2024; Ceccarelli et al., 2024). The validation exercise is then an indirect approximate version of the ideal experiment.

in the disaster area because of expected losses or concerns about investors' exposure, if these firms are disproportionately brown. I decompose the tilt of Table 3, Column 2, into reallocations across firms at different distances from the disaster and reallocations among firms at similar distances (Appendix Table G.9). The former component is small and insignificant, while the latter is nearly identical to the main estimate. Consistently, the stock-level effect outside the disaster area is unchanged, while within the affected area the effect is larger, at 19 basis points (Appendix Table G.10). Managers thus do not sell local firms as a block, but reweight toward the greener ones.

**Fund Flows or Liquidity Shocks.** The effect could also capture fund flows and not the managers' own choices. Funds managed in the affected counties can face outflows from local investors (Tubaldi, 2021), which raises two concerns. First, outflows may correlate with climate interpretation through disaster severity or other local characteristics.[53] Second, flow-induced sales may disproportionately affect certain stocks (Coval and Stafford, 2007) and generate an environmental tilt even after accounting for proportional portfolio adjustments.

To address the first concern, I measure active trades net of proportional adjustments to flows, following Sammon and Shim (2023). A ten percentage point higher interpretation share raises the net active purchases of nearby managers in top-tercile relative to bottom-tercile stocks by 30 basis points of assets before the disaster (Appendix Table G.8, Column 2). To address the second, I have estimated the effect on managers outside the affected counties, and kept the affected managers in a band of their own. Nearby managers do not face the outflows of local investors in the affected counties (unless their fund is headquartered there). Nearby managers respond equally strongly whether their headquarters are near or far from the disaster (Appendix Table G.11), so forced sales cannot explain their tilt.

# 5 Separating Interpretation Narratives from Prior Beliefs and Preferences

The previous household, adviser and fund-manager designs compare units whose local coverage gives different interpretations to the same type of disaster, holding fixed the climate and political views of the county as well as those of advisers and fund managers. Event-study leads, placebo tests and additional fixed effects ruled out confounders such as a trend in green investment and a general propensity of climate-concerned counties to buy green assets even after salient but non-climate shocks. So far, I have estimated the response to local interpretation as it emerges in equilibrium from local discussions and narratives, regardless of whether the county's outlets supply them or cater to their readers' predispositions.

Next, I move to the role of locally supplied interpretation models and narratives (Schwartzstein and Sunderam, 2021), by testing whether the local interpretation moves portfolios when it is driven by changes in the narrative that the outlets supply, independently of the prior beliefs or preferences of their readers. I do so by adapting an instrumental variable approach that uses media ownership as a shifter of content (Martin and McCrain, 2019; Garz and Rickardsson, 2023; Ananyev and Volkova, 2024; Ewens et al., 2022; Kostovetsky et al., 2026). In particular, I instrument the interpretation of the disaster in local news with the size of the chain that owned the county's newspapers. Larger newspaper chains are more detached from the counties they serve and cater less to local demand, supplying centrally produced copy.[54]

[53] For instance, in counties where the disaster was more destructive, climate narratives may be less common, and liquidity needs might raise outflows. As a result, the interpretation effect would pick up differences in outflows.

[54] Outside the United States, newspapers held by one owner share that owner's slant (Garz and Rickardsson, 2023), and acquisitions shift slant in a direction that depends on the acquirer (Ananyev and Volkova, 2024); in American local broadcasting, ownership changes shift content (Martin and McCrain, 2019).

In practice, the instrument exploits changes in the ownership of the newspapers serving a county between two disasters of the same type. For each county, I compare the disasters of the same type that affected it directly or occurred within 50 miles, in particular those that occurred before with those that occurred after the chain size of its newspapers changed.

**Specification and estimation.** The sample contains the counties directly affected or within 50 miles and with positive local coverage of the event. I define the chain size of newspaper $o$ in calendar quarter $t$, $ChainSize_{ot}$, as the number of newspapers in the sample held by the outlet's owner as recorded prior to the event, so that an independent newspaper has a chain size of one. The county instrument is the mean of the logarithm of chain size[55] across the chained newspapers serving the county, weighted by their circulation.

$$Z_{c,t(e)} = \frac{\sum_{o \in C_{c,t(e)}} w^D_{oc} \log ChainSize_{o,t(e)}}{\sum_{o \in C_{c,t(e)}} w^D_{oc}}, \tag{8}$$

where $C_{ct}$ is the set of chained newspapers serving county $c$ and $w^D_{oc}$ is the fixed circulation weight. The first-stage equation is

$$M_{ce} = \tilde{\pi} Z_{c,t(e)} + \tilde{\beta}_A A_{ce} + X'_{ce}\tilde{\gamma} + \tilde{\mu}_c + \tilde{\tau}_{t(e)} + \tilde{\lambda}_{h(e)\times b(c,e)} + u_{ce}, \tag{9}$$

where $Z_{c,t(e)}$ is the county instrument of equation (8), $A_{ce}$ is local disaster coverage in logs, $X_{ce}$ collects the logs of one plus own-county fatalities and damage, the county Democratic two-party vote share at the previous presidential election and, in columns 2 and 4 of Table 4, the audience overlap of the county's newspapers with the event footprint,[56] $\tilde{\mu}_c$ are county fixed effects, $\tilde{\tau}_{t(e)}$ calendar-quarter fixed effects, and $\tilde{\lambda}_{h\times b}$ hazard-by-exposure-band fixed effects, where the two exposure bands are the directly affected counties and the counties within 50 miles; the comparison is among counties at the same exposure to disasters of one hazard family.[57] I estimate

$$Y_{ue} = \beta_M M_{ce} + \beta_A A_{ce} + X'_{ce}\gamma + \mu_c + \tau_{t(e)} + \lambda_{h(e)\times b(c,e)} + \varepsilon_{ue}, \tag{10}$$

where $u$ is a county or an adviser in county $c$ and $M_{ce}$ is instrumented with $Z_{c,t(e)}$. Adviser regressions weight each adviser by its share of the county-episode ETF book in the quarter before the episode. Standard errors are clustered by county.

**Results.** Table 4 reports two-stage least squares estimates of the instrumental variable design. Panel A reports the first stage at the outlet and county level, Panel B the county response of Table 2, Panel C the

[55] I use the natural logarithm to measure proportional changes in chain size, since the same change in the number of newspapers is more meaningful and impactful for a small chain than for a large one.

[56] Separating attention from interpretation requires holding fixed the incentive of the county's newspapers to cover the event. Beyond disaster coverage itself, I control for newspaper-audience overlap with the event footprint outside the county, inspired by the newspaper-market congruence designs of Snyder and Strömberg (2010) and Myers (2026). Consider three nearby counties, A, B, and C. A disaster affects only A. People in B mostly read newspaper X, which has 90 percent of its readers in A. People in C mostly read newspaper Y, which has no readers in A. X therefore has more incentive than Y to cover the disaster, so people in B should receive more coverage than people in C, even though neither county is affected. Instrumenting coverage jointly with this measure gives similar interpretation estimates from a weakly identified system; that version is in the Appendix (Appendix Table H.8).

[57] For mutual fund managers, as in Section 4, I focus on managers whose offices are outside the affected counties but within 50 miles of the disaster footprint, to address outflows from local investors. With a single exposure band, the band fixed effects reduce to hazard fixed effects. The county is the county of the manager's offices, and the instrument is averaged over those offices.

adviser response of Table C.3 and Panel D the mutual fund manager response of Table 3. Columns 1 and 2 use pre-event prices and columns 3 and 4 current prices; all columns control for coverage volume, own losses and the county's Democratic vote share, and columns 2 and 4 add the audience overlap of the county's newspapers with the disaster footprint. The coefficients remain stable in magnitude across columns. As expected, the use of 2SLS reduces statistical power, but for counties and advisers the weak-instrument-robust confidence sets exclude zero at the 10% level and the instrumented response is of the same magnitude as the local-projection estimates of Table 2.[58]

For mutual fund managers whose offices are all within 50 miles of the disaster, a 10-percentage-point increase in the supplied climate interpretation narrative raises their green tilt by 177 basis points at pre-event prices, about two and a half times the ordinary least squares estimate of Section 4 (69 basis points, Table 3). The 90% set excludes zero in columns 1, 2 and 4 and includes it in column 3.

**Relevance and exclusion.** The climate-interpreted share of disaster coverage differs persistently across owners, and newspaper chain size predicts that share at the outlet level; under a larger prior owner, local disaster coverage has a lower climate-interpreted share (Table H.1). Chain size is measured at the newspaper–quarter level; Panel A of Table 4 shows that its relation to the interpretation share extends to the event-specific share of the county's coverage.

The identifying assumption is that, conditional on the fixed effects, coverage volume and own losses, a change in the chain size of a county's newspapers is unrelated to any change in how its investors respond to the disaster, other than through the climate interpretation of the coverage they receive. Calendar-quarter fixed effects absorb everything common to the counties exposed in the same quarter, such as a rise in chain size or a fall in green shares across the country. Hazard-by-exposure-band fixed effects restrict the comparison to counties exposed to the same hazard type in the same exposure band: affected counties are compared with affected counties and nearby counties with nearby counties.

County fixed effects absorb permanent differences between counties with larger and smaller chains, such as market size, politics, climate beliefs and the usual reading of their newspapers. They also absorb a steady trend in a county's green share, since it adds the same amount to each of the county's changes around its disasters. What remains is the county's log chain size at each disaster minus its average over the county's disasters.[59]

The remaining concern is that changes in chain size systematically occur in counties that, even without such shocks, would have bought fewer green assets and whose newspapers would have interpreted disasters less as climate change. How likely this selection bias is depends on the nature of the change.

Two kinds of ownership shock account for most of the changes in chain size. The first is an upstream shock, a change at the level of the owner: the owner expands or contracts through a merger or an acquisition by or of other owners, or through purchases and sales of other newspapers, as in the 2019 merger of Gannett and GateHouse.[60] Owners take these decisions at the chain level, for reasons that follow from the economics

[58]The interpretation share is a classifier output and carries measurement error that can bias its coefficient (Ash and Hansen, 2023; Hausman, 2001); since ownership changes are unrelated to the classifier's errors on a given article, the instrument removes this bias as well.

[59]For example, the Hartford Courant, which holds 97 percent of the circulation of Hartford County, Connecticut, was owned by Tribune Publishing, a chain of 18 newspapers, at the time of the flood of 28 June 2020 and by Alden Global Capital, a chain of 56, at the time of the flood of 1 September 2021, both within 50 miles of the county.

[60]Table H.2 in the Appendix analyses this merger on its own: after it, the interpretation share and the green-ETF response of affected and nearby counties are both lower where the county's pre-merger Gannett circulation share is higher.

of the newspaper industry, such as declining advertising revenue, and are plausibly unrelated to the climate concern of the county's readers.[61] Upstream shocks are the most common.

The second is a direct change in ownership, in which the county's newspaper itself passes from one owner to another, as in the 2023 sale of the San Diego Union-Tribune by NantMedia to Alden Global Capital. Although acquisitions and sell-offs also follow the economics of the newspaper industry, a chain buying an independent newspaper is the only case where one could argue that the owner chooses the county's market. Removing these purchases leaves the county and adviser results similar (Appendix Table H.5). Changes in chain size do not predict the county's latest presidential vote share, its climate beliefs measured before the sample, its portfolio change before the event, or the losses, distance and coverage of the disaster (Table H.4).

The final concern is that disasters that attract greater attention may also lead newspapers to discuss their climate implications more extensively, so the estimated interpretation effect could reflect how much attention the event receives. A change in ownership may also change how much the county's newspapers cover the disaster, so I instrument interpretation while controlling for the volume of local coverage. At the outlet level, chain size leaves disaster coverage unmoved (Table H.1).

# 6 Interpretation, Aggregate Flows and Relative Prices of Green Stocks

At the aggregate level, disasters have been found to raise flows into green ETFs and the relative returns of sustainable assets (Fiordelisi et al., 2023; Misev and Balles, 2024). To complement this evidence, I check that national news moves ESG fund flows in the same direction. Each month I measure national disaster attention as the number of disaster articles in national newspapers, and national interpretation as the share of those articles that read the disaster as climate change. An attention surprise is the prediction error of an autoregression of the monthly series fitted on the previous three years, in standard-deviation units, as in Ardia et al. (2023). I ask whether a surprise raises flows into ESG-labelled funds more than flows into other funds, and whether it does so more when national interpretation is high. I use my sample of ESG-labelled and other US domestic-equity mutual funds and ETFs over 2015–2024.[62]

Figure 6 shows the responses over the surprise month and the following two months. At the average level of national interpretation, the estimated three-month response of flows into ESG-labelled funds relative to other funds is about 29 basis points of fund assets, or about $288 million. The response grows with national interpretation. At the 25th percentile of national interpretation the response is close to zero. At the 75th percentile it is positive, and about 58 basis points of assets larger, a difference of about $580 million.[63] As a reference point, ESG-labelled funds in the sample took in 0.7% of assets in an average quarter, so one high-interpretation surprise adds close to a typical quarter's inflow over the following three months.

In a counterfactual exercise, I compute the additional ESG demand and the relative appreciation of green stocks that a stronger climate interpretation of disasters would generate. Van der Beck (2026) shows that ESG flows and, more generally, institutional ESG demand are important drivers of green-minus-nongreen stock returns. The exercise extends the response of affected advisers to investors outside the sample and

[61] Acquisitions and consolidation in the industry follow declining advertising revenue, the liquidity needs of legacy owners, and cost cutting in roll-ups (Ewens et al., 2022; Gao et al., 2020).

[62] The specification, adapted from Hartzmark and Sussman (2019), compares ESG-labelled funds with other funds within the same investment category and calendar month through investment-category-by-month fixed effects. It controls for fund characteristics and weights observations by log lagged AUM.

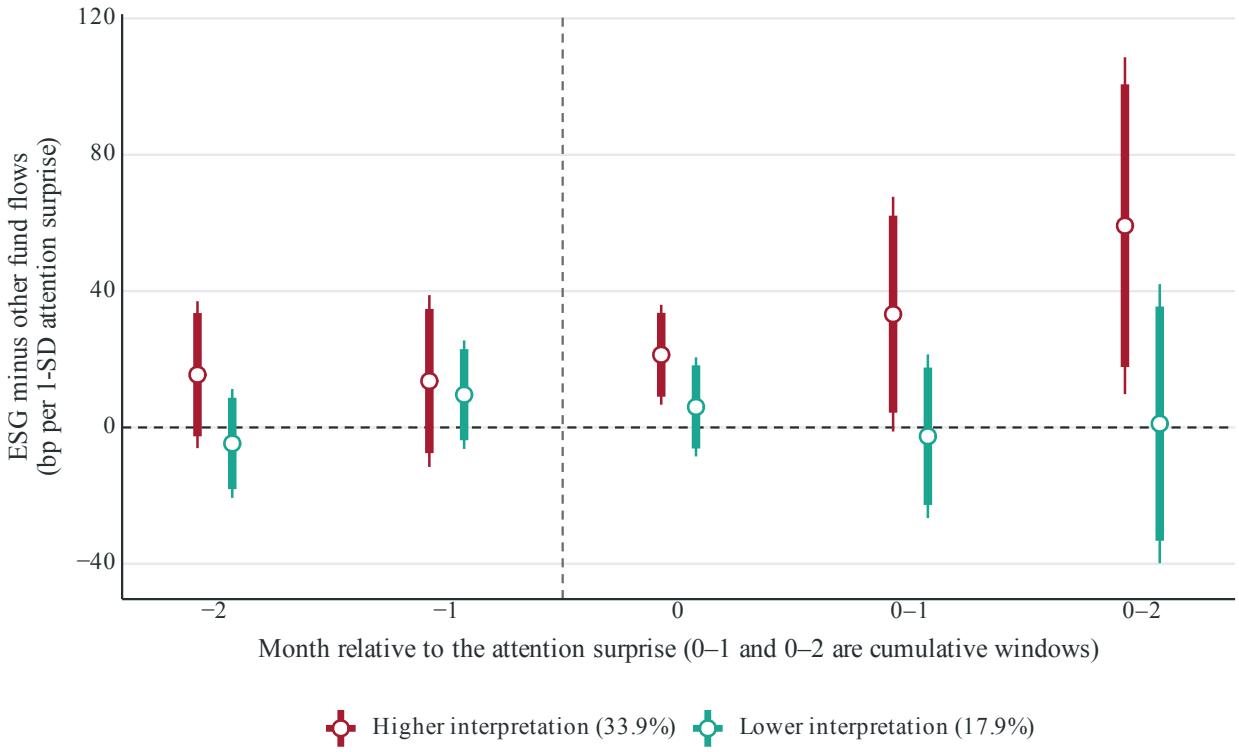
[63] Following Hartzmark and Sussman (2019), I convert the estimated relative flow rates into dollars by multiplying each month's response by each ESG fund's assets in the previous month, summing across funds and the three response months, and averaging across the eligible starting months.

TABLE 4: PORTFOLIO RESPONSE TO INSTRUMENTED INTERPRETATION

| | Portfolio response (basis points) | | | |
|---|---|---|---|---|
| | Pre-event prices | | Current prices | |
| | (1) | (2) | (3) | (4) |
| *Panel A. First stage* | | | | |
| Outlet level | −2.29*** | −2.29*** | −2.29*** | −2.29*** |
| | (.50) | (.50) | (.50) | (.50) |
| County level | −8.81*** | −8.82*** | −8.81*** | −8.82*** |
| | (2.89) | (2.89) | (2.89) | (2.89) |
| Newspaper-quarters | 6,711 | 6,711 | 6,711 | 6,711 |
| County-events | 694 | 694 | 694 | 694 |
| First-stage $F$, outlet level | 20.85 | 20.85 | 20.85 | 20.85 |
| First-stage $F$, county level | 9.32 | 9.33 | 9.32 | 9.33 |
| *Panel B. County ETF holdings* | | | | |
| Interpretation (IV) | 22.75 | 22.47 | 24.96* | 24.60* |
| | (14.30) | (14.21) | (14.69) | (14.59) |
| Observations | 694 | 694 | 694 | 694 |
| Anderson–Rubin $p$ | .064 | .067 | .051 | .054 |
| Anderson–Rubin 90% set | [2.5, 61.9] | [2.3, 61.2] | [3.9, 64.7] | [3.6, 63.9] |
| *Panel C. Adviser ETF holdings* | | | | |
| Interpretation (IV) | 19.34* | 19.18* | 22.54** | 22.38** |
| | (10.51) | (10.53) | (11.30) | (11.31) |
| Observations | 4,342 | 4,342 | 4,342 | 4,342 |
| County-events | 660 | 660 | 660 | 660 |
| First-stage $F$ | 9.43 | 9.44 | 9.43 | 9.44 |
| Anderson–Rubin $p$ | .029 | .032 | .025 | .027 |
| Anderson–Rubin 90% set | [4.7, 48.4] | [4.5, 48.2] | [6.3, 52.8] | [6.0, 52.5] |
| *Panel D. Mutual fund managers* | | | | |
| Interpretation (IV) | 176.56** | 209.87** | 107.27 | 164.27* |
| | (84.17) | (97.94) | (82.25) | (97.44) |
| Observations | 2,052 | 2,052 | 2,052 | 2,052 |
| County-events | 136 | 136 | 136 | 136 |
| First-stage $F$ | 11.47 | 11.13 | 11.47 | 11.13 |
| Anderson–Rubin $p$ | .062 | .043 | .190 | .073 |
| Anderson–Rubin 90% set | [26.5, 350.6] | [47.1, 429.3] | [−31.0, 287.6] | [14.5, 400.3] |
| Coverage, county fatalities and damages | ✓ | ✓ | ✓ | ✓ |
| Democratic vote share | ✓ | ✓ | ✓ | ✓ |
| Audience reach | | ✓ | | ✓ |
| Fund size, age and turnover (Panel D) | ✓ | ✓ | ✓ | ✓ |
| County, quarter and hazard × distance-band FE | ✓ | ✓ | ✓ | ✓ |

*Notes.* This table reports two-stage least squares estimates that use circulation-weighted log newspaper-chain size as the instrument for local interpretation. Outcomes follow Tables 2 (counties), C.3 (advisers) and 3 (mutual fund managers); advisers are weighted by their share of the county's ETF holdings in the quarter before the episode. Panel D uses managers whose offices are all within 50 miles of the disaster. Interpretation effects are basis points per 10-percentage-point increase. Panel A reports the effect of chain size on the interpretation share, in percentage points, per log point at the outlet level (newspaper and calendar-quarter fixed effects) and per standard deviation at the county level. First-stage $F$ is the Kleibergen–Paap rk Wald statistic. The Anderson–Rubin $p$-values and 90% sets remain valid when the instrument is weak. Standard errors and first-stage $F$ statistics are clustered by county (in Panel D, the manager's county) and, at the outlet level, by newspaper. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

FIGURE 6: NATIONAL ATTENTION, NATIONAL INTERPRETATION AND ESG FUND FLOWS




*Notes.* Responses of flows into ESG-labelled funds relative to other funds to a one-standard-deviation national disaster-attention surprise, at the 25th and 75th percentiles of national interpretation (17.9% and 33.9%), in basis points of fund assets. The surprise is the prediction error of an autoregression of the monthly disaster-article count, in inverse hyperbolic sine, fitted on the previous 36 months, standardized. Months −2 to 0 are monthly flows. Windows 0–1 and 0–2 cumulate them. Flows are return-adjusted net flows over previous-month assets. US domestic-equity mutual funds and ETFs, 2015–2024, weighted by log lagged assets, with investment-category-by-month fixed effects and lagged size, age, returns and expenses. Thick and thin bars are 90% and 95% confidence intervals from Newey–West standard errors with 12 lags.

prices the added demand with van der Beck's (2026) calibration of how other investors accommodate ESG flows. Appendix J derives each step.

The first input is the allocation response. Raising local climate interpretation from its 25th to its 75th percentile in my sample, with newspaper coverage at its median and the remaining controls at their sample averages, moves interpretation by about one and a quarter standard deviations. Among affected counties, each standard deviation raises the green-ETF share by 26 basis points over the event quarter and the following quarter, so the move adds 32 basis points of ESG allocation per dollar of capital (Appendix J). I assume that affected and comparison advisers respond to the common movement in relative fund prices in the same way on average, so that this movement cancels in the comparison and the 32 basis points measure a shift in desired allocation at initial prices.

The second input is the capital that responds. Let $K$ be the capital of an investor population, $a$ the added ESG allocation per dollar of capital (the 32 basis points above), and $\alpha$ the *activated* share of that capital, whose holders respond. Assume additional ESG purchases are financed by reducing other holdings. Since the market portfolio already holds a share $s$ of ESG funds, selling it to buy ESG funds offsets part of the purchase, and the additional ESG demand $T$ and the reallocation $X$ from the market portfolio to the ESG portfolio are

$$T = \alpha K a, \qquad X = \frac{T}{1-s}. \tag{11}$$

I assume that activated investors respond to stronger climate interpretation as the affected advisers in my sample do, per dollar of capital, at the same level of attention and the same national information environment, and I set $\alpha = 1$ within each population. I apply the increment to three populations in turn. The first is green-ETF investors. The second is mutual-fund and ETF investors together, since relative flows into ESG mutual funds also respond to interpretation (Section 3). The third, the main case, is institutional equity investors, since managers of funds without an ESG mandate also tilt toward greener stocks (Section 4). For mutual-fund and ETF investors the reallocation is 0.32% of capital. Institutions held $24.7 trillion of the covered stocks in December 2022 (Appendix J).

The third input is the price response. Buying $X$ dollars of the ESG portfolio and selling $X$ dollars of the market portfolio generates net purchases of the stocks that ESG funds overweight and net sales of the stocks they underweight. Let $\tau_G$ be the total weight of green stocks in the ESG portfolio in excess of their weight in the market portfolio, $\tau_N$ the matching shortfall of nongreen stocks, and $V_G$ and $V_N$ the market values of the two groups. The reallocation adds $X\tau_G$ dollars of demand for green stocks and removes the same amount from nongreen stocks. Scaled by each group's market value, this is the relative flow $f^{\text{rel}}$ into green stocks, and the green-minus-nongreen return $R_G - R_N$ that it generates is the relative flow times the relative-return multiplier $M_{\text{ESG}}$,

$$f^{\text{rel}} = \frac{X\tau_G}{V_G} - \frac{X\tau_N}{V_N}, \qquad R_G - R_N = M_{\text{ESG}}\, f^{\text{rel}}, \tag{12}$$

the counterpart for the two groups of stocks of the market multiplier in Gabaix and Koijen (2023). I take $M_{\text{ESG}}$ from van der Beck's (2026) calibration of how other investors accommodate ESG flows, which combines the portfolio weight differences with holders' willingness to trade as prices change. With the December 2022 inputs, $M_{\text{ESG}} \simeq 1.12$.[64] I assume that other investors accommodate the reallocation as in that calibration, with their price responsiveness and ownership weights unchanged as the set of activated investors grows.

To compare with annual returns, assume that over a year the disasters that occur activate the whole

[64]Calibrated using van der Beck's (2026) December 2022 inputs. In dollars, the same responses give $0.72 of green-minus-nongreen valuation per dollar reallocated from the market portfolio to the ESG portfolio.

institutional investor base, each investor making the additional adjustment once, at unchanged attention and national news.[65] Applying the reallocation rate to the capital of institutions gives $79 billion of additional ESG demand against the market portfolio over the year, counting each dollar of capital once. Appendix Table J.1 shows that this demand adds $57 billion of green-minus-nongreen valuation and 34 basis points of return difference. As a reference point, van der Beck (2026) attributes 186 of the 223 basis points of the average annual green-minus-nongreen return to institutional ESG demand. The calibrated appreciation is 18% of the former and 15% of the latter (Appendix Table J.2). The magnitudes are suggestive, since they extend a local response to investors who were not exposed and price it with a multiplier estimated on other flows.

# 7 Mechanisms

In the previous sections, I established that the interpretation of a disaster drives portfolios towards sustainable assets and that the responses reflect sustainability-related demand shocks, after ruling out mechanical changes from prices, liquidity needs and fund flows, as well as the incentives of advisers and managers toward their clients (Sections 3.1, 3.4 and 4.3).[66] Moreover, under mild assumptions on the form of return expectations,[67] I can also rule out return expectations as the key driver (Section 3.4). Household portfolios tend to be inert and therefore reflect beliefs only weakly (Giglio et al., 2021). Given the sizeable response I observe, I ask how this interpretation-induced shift in the demand for sustainable assets arises despite the inertia, and whether it comes from preferences (ethical considerations) or beliefs (climate risk and hedging demand), the two main motives for ESG investing besides return expectations (Giglio et al., 2025).

## 7.1 Experiences, Salience and Interpretation

A large body of work shows that personally lived outcomes leave persistent and often domain-specific marks on beliefs and financial behavior (Malmendier, 2021; Malmendier and Wachter, 2022). Extreme weather events can become disproportionately influential on beliefs and on actions of those experiencing them directly or indirectly, because they are more salient, easier to recall, more affectively charged, or more cognitively available at the moment of judgment (Deryugina, 2013; Djourelova et al., 2024; Ballew et al., 2022).

The pattern I observe in green portfolio reallocation and mutual fund manager rebalancing is consistent with salience and experiential updating more than rational belief updating. First, I observe a sizeable response to a single event, which, even if extreme, is weak evidence about long-run climate trends, and therefore less likely to move beliefs via rational updating. Moreover, the patterns in my estimates are consistent with documented salience effects of disasters (Alok et al., 2020; Dessaint and Matray, 2017; Gallagher, 2014): both household advisers and mutual fund managers respond more strongly closer to the disaster (Tables I.1, 3, and G.7), and the response is persistent, especially among household advisers, for whom it remains positive through six quarters (Table I.2).[68]

[65] Appendix J repeats the exercise for green ETFs alone and for ESG-labelled mutual funds and ETFs together.

[66] These incentives could produce the same portfolio response without a shift in demand: advisers may sell their own products or rebalance to justify their fees, and managers may tilt the portfolio to show their investors a response to the disaster.

[67] The assumption is that expectations about other investors' demand follow national coverage and reading and past green returns and not the local reading of the event.

[68] Alok et al. (2020) find that nearby mutual fund managers underweight disaster-zone stocks more than distant managers, with this aversion declining over time and distance. Gallagher (2014) finds that flood insurance take-up spikes after a flood and then declines to baseline, while take-up in nonflooded communities in the same television market rises at one third the rate.

The salience of the disaster alone would not explain the interpretation gradient across investors with similar exposure to the same disaster. Theories of experience effects and memory imply that the same event can be assigned to different domains depending on prior category use, narratives and contextual cues (Malmendier and Wachter, 2022; Bordalo et al., 2025), and in the psychology of climate change, people interpret a personal experience of extreme weather through their existing mental representations and predispositions, which shape its effect on their decisions (Brügger et al., 2021; Myers et al., 2013). My results are consistent with this idea: interpretation operates through whether investors process the disaster as climate-relevant and not as an isolated local event.

In Section 5, I show suggestive causal evidence that the interpretation supplied by local outlets moves portfolios, also holding constant the salience of the disaster (Tables 4 and H.1). Here, to further isolate interpretation from salience, I run the same estimation approach on purchases of flood insurance, another active and impactful financial decision revealing beliefs about disaster or climate risk. I compare these purchases in affected counties with those in unaffected counties of the same states, separately after floods and wildfires (Appendix Table I.6).[69] Floods make flood risk salient directly; wildfires allow me to examine whether climate attribution is associated with demand for protection against another hazard. By month three, the estimated increase relative to unaffected counties is 41 versus 15 policies per 100,000 residents at the 75th and 25th percentiles of climate attribution after wildfires, compared with 43 versus 40 after floods. As in Gallagher (2014), take-up rises strongly after floods and, as expected, regardless of how the flood is interpreted, while after wildfires it rises with the climate interpretation of the event, so that climate interpretation matters for households' demand for protection beyond disaster salience.[70]

## 7.2 Survey Evidence on Taste or Belief Shocks

Survey evidence supports a belief-based interpretation centered on perceived climate harm, with weaker evidence for beliefs about human causes and climate-policy attitudes. I use the Climate Change in the American Mind surveys for 2009–2019, which measure climate beliefs, perceived risks and policy attitudes. Figure 7A shows that, comparing media markets within the same year, advisers hold larger green ETF shares where more residents expect global warming to harm Americans already or within ten years. Associations with human-causation beliefs and regulatory support are positive but imprecisely estimated.

Perceived harm also responds to disaster experience. Comparing respondents interviewed in the same media market and survey year in the four weeks before and after a flood begins, moderates and conservatives become more likely after the flood to report harm occurring now or within ten years. Among conservatives, estimated changes in human-causation beliefs and regulatory support are much smaller and statistically insignificant. This contrast indicates that the perceived immediacy of climate consequences connects disaster experience to green investment, while changes in causal beliefs and policy attitudes receive less consistent support.

The evidence on flood insurance in Section 7.1 also speaks to this question. First-time voluntary flood-insurance purchases provide concrete evidence that extreme weather events affect households' finan-

[69] I estimate local projections of cumulative first-time voluntary NFIP issuance per 100,000 residents from the disaster month through month $h$. County, episode and state-by-event-year fixed effects absorb persistent county differences and shared shocks and trends, so the identifying variation comes from how insurance purchases vary with climate attribution in affected relative to unaffected counties, conditional on total disaster and climate-only coverage. Standard errors are clustered by county.

[70] Wildfires can raise subsequent flood risk in some locations, depending on their terrain. However, the outcome is the change in take-up, so county fixed effects absorb how take-up usually responds in each location, and the comparison is within the same wildfire. The objective increase in flood risk does not depend on how local coverage reads the fire.

cial decisions through the climate risk-perception channel, since insurance decisions are not governed by sustainability preferences.

The suggestive evidence presented in this section points to at least some part played by beliefs.[71] Future work will tie the shift more tightly to one or the other. Although the origin of the shift is of academic interest, it does not change the role of interpretation, nor its implications in the aggregate.

# 8 Conclusion

This paper studies how the interpretation of an event shapes investors' portfolio decisions, separately from the information it reveals and the attention it receives. Using extreme weather events, I show that households and mutual fund managers reallocate toward sustainable assets when disasters are interpreted locally as evidence of climate change. This response is difficult to explain through differences in disaster attention, investors' prior political views and climate beliefs, or the direct effects of the disaster on asset fundamentals. Attention to a disaster increases trading activity without systematically shifting portfolios toward green assets, while its climate interpretation shapes the direction of reallocation. Changes in newspaper ownership further show that locally supplied interpretations can move portfolios independently of readers' prior beliefs and preferences. Evidence from surveys and insurance purchases is consistent with changes in perceived climate harm as a mechanism linking interpretation to financial decisions.

These responses extend beyond local portfolio choices. Climate interpretation is associated with greater inflows into ESG mutual funds, and a counterfactual exercise suggests that, if institutional investors responded as the affected advisers do, the resulting shifts in demand could have economically meaningful price effects. An interquartile increase in climate interpretation would imply 34 basis points of ESG-stock appreciation relative to other stocks, equivalent to 15% of the observed annual return spread. More broadly, the findings suggest that differences in how the same event is interpreted can have substantial consequences for capital allocation and asset prices, beyond the information the event conveys or the attention it attracts.

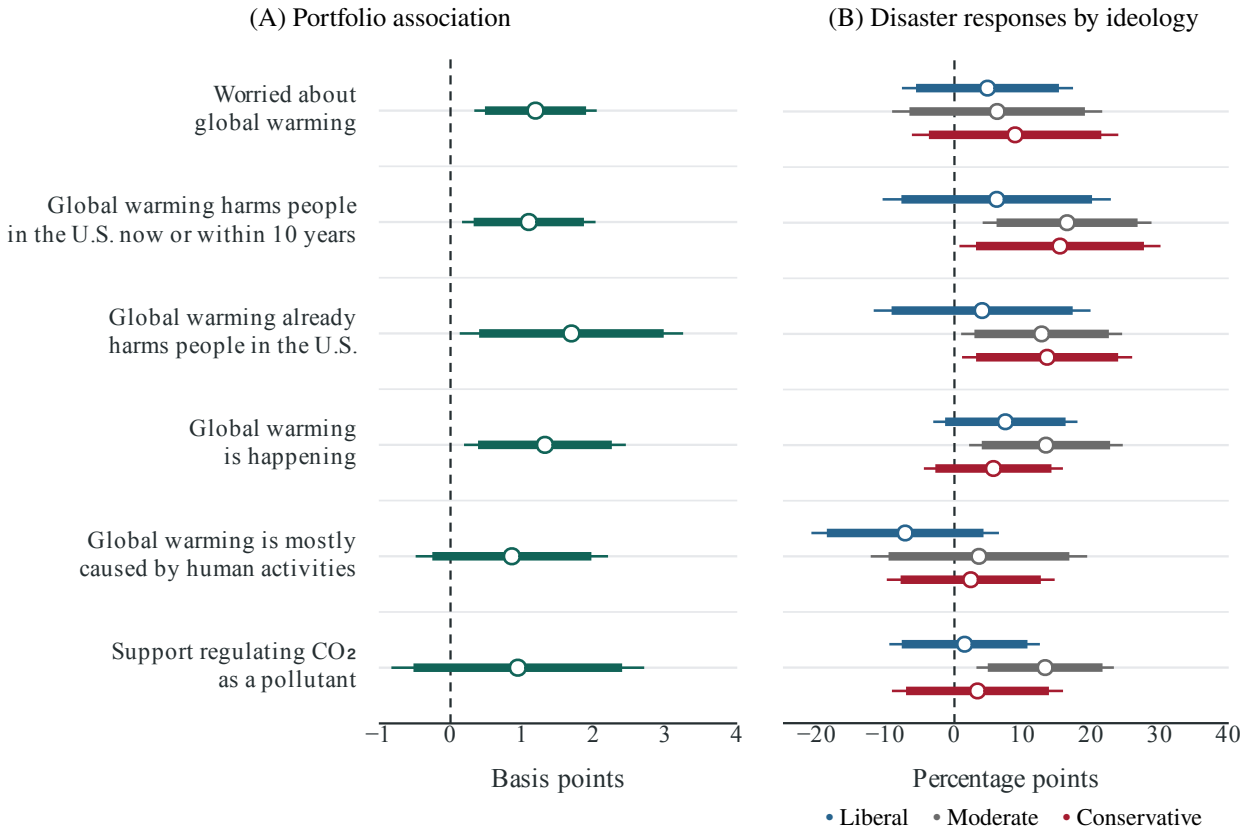
As climate change increases the frequency and severity of extreme weather events, understanding how their consequences propagate through financial markets becomes increasingly important. The findings suggest that the financial footprint of a disaster can extend well beyond its physical footprint. Previous research has examined how disasters affect firms exposed to physical risk and investors holding those firms (Alok et al., 2020; Dessaint and Matray, 2017; Kruttli et al., 2025). The results here highlight a different channel. When a disaster is interpreted locally as evidence of climate change, investors reallocate toward sustainable assets, even when those assets are not exposed to the event. Investors and regulators assessing the transmission of climate risk to asset prices therefore need to consider how disasters are interpreted locally, in addition to which firms are physically exposed.

The findings also speak to the ongoing debate about news deserts and the decline and consolidation of local journalism. Much of this debate concerns what communities lose when local newspapers disappear or reduce their coverage, including opportunities for local public discussion. Recent ownership changes have directly affected climate coverage, even at major national news organizations.[72] My findings highlight a

[71] Harm perceptions rise after floods among moderates and even conservatives, advisers hold more green ETFs where residents expect harm sooner, and insurance take-up after wildfires rises with the climate interpretation of the event.

[72] After Skydance acquired Paramount in 2025, the new owner cut staff across the company and installed a new editor-in-chief at CBS News, which dismissed its climate unit and reduced climate coverage by 61 percent over the following year (Media Matters, here). The Washington Post and NPR have since dissolved their climate desks amid cost cuts following Bezos's restructuring and the rescission of federal public-media funding (here, here).

FIGURE 7: CLIMATE BELIEFS, GREEN INVESTMENT, AND DISASTER RESPONSES




*Notes.* Panel A relates annual belief shares in each designated market area (DMA) to the green share of advisers' ETF holdings, with year fixed effects. Advisers are allocated to a DMA if they predominantly serve households and all their known U.S. offices lie within that DMA. Coefficients are basis points per 10-percentage-point higher belief share. Panel B reports post-disaster changes by ideology, in percentage points, using interviews in affected DMAs from 28 days before through 27 days after the start of the flood. Regressions include ideology indicators, DMA × survey-year, interview year-month and event fixed effects, and demographic and question-wording controls. Appendix Tables I.3 and I.4 report the estimates behind Panels A and B; Table I.5 reports them in standard deviations, with the response to the flood interacted with local interpretation.

*Worried*, "How worried are you about global warming?", equals one for somewhat or very worried. *Harm timing*, "When do you think global warming will start to harm people in the United States?", equals one for "In 10 years" or "They are being harmed now"; the second indicator counts only the latter. *Warming happening*, "Do you think that global warming is happening?" and "How sure are you that global warming is happening?", equals one for yes, with somewhat, very or extremely sure. *Human causation*, "Assuming global warming is happening, do you think it is. . . ", equals one for the answer "caused mostly by human activities". *Regulation*, "How much do you support or oppose the following policies?", equals one for somewhat or strong support of "Regulate carbon dioxide (the primary greenhouse gas) as a pollutant."

Other valid answers equal zero. Thick and thin lines show 90% and 95% confidence intervals, respectively; standard errors are clustered by DMA.

further consequence. When newspaper chains acquire local outlets, the share of disaster coverage linking extreme weather to climate change declines, and the portfolio responses of local investors weaken. This suggests that the erosion of local climate interpretation may have consequences beyond public discourse, reducing demand for sustainable investments and, potentially, the financial incentives for firms to improve their environmental performance.

## References


**Abowd, John M., Francis Kramarz, and David N. Margolis**, "High Wage Workers and High Wage Firms," *Econometrica*, 1999.

**Alekseev, Georgij, Stefano Giglio, Quinn Maingi, Julia Selgrad, and Johannes Stroebel**, "A Quantity-Based Approach to Constructing Climate Risk Hedge Portfolios," *Journal of Financial Economics*, forthcoming.

**Alok, Shashwat, Nitin Kumar, and Russ Wermers**, "Do Fund Managers Misestimate Climatic Disaster Risk?," *Review of Financial Studies*, 2020, *33* (3), 1146–1183.

**Ananyev, Maxim and Ekaterina Volkova**, "Media Ownership and Ideological Slant: Evidence from Australian Newspaper Mergers," *PLOS ONE*, 2024, *19* (12), e0315137.

**Anderson, Anders**, "Trading and Under-Diversification," *Review of Finance*, 2013, *17* (5), 1699–1741.

_ **and David T. Robinson**, "Populism, Polarization and Green Investment," Working Paper 32131, National Bureau of Economic Research 2026.

**Appel, Ian R., Todd A. Gormley, and Donald B. Keim**, "Passive Investors, Not Passive Owners," *Journal of Financial Economics*, 2016, *121* (1), 111–141.

**Ardia, David, Keven Bluteau, Kris Boudt, and Koen Inghelbrecht**, "Climate Change Concerns and the Performance of Green vs. Brown Stocks," *Management Science*, 2023.

**Ash, Elliott and Stephen Hansen**, "Text Algorithms in Economics," *Annual Review of Economics*, 2023, *15*, 659–688.

_ **, Anton Boltachka, Sergio Galletta, and Matteo Pinna**, "Media Bias and Climate Change Skepticism," Working Paper 05/2023, ETH Zurich, Center for Law & Economics 2023.

**Athey, Susan, Markus Mobius, and Jeno Pal**, "The Impact of Aggregators on Internet News Consumption," Working Paper 28746, National Bureau of Economic Research 2021.

**Baik, Bok, Jun-Koo Kang, and Jin-Mo Kim**, "Local Institutional Investors, Information Asymmetries, and Equity Returns," *Journal of Financial Economics*, 2010, *97* (1), 81–106.

**Bailey, Warren, Alok Kumar, and David Ng**, "Behavioral Biases of Mutual Fund Investors," *Journal of Financial Economics*, 2011, *102* (1), 1–27.

**Baker, Malcolm, Mark L. Egan, and Suproteem K. Sarkar**, "Demand for ESG," Working Paper 30708, National Bureau of Economic Research 2024.

**Bakkensen, Laura A. and Lint Barrage**, "Going Underwater? Flood Risk Belief Heterogeneity and Coastal Home Price Dynamics," *The Review of Financial Studies*, 2022, *35* (8), 3666–3709.

**Baldauf, Markus, Lorenzo Garlappi, and Constantine Yannelis**, "Does Climate Change Affect Real Estate Prices? Only If You Believe In It," *Review of Financial Studies*, 2020, *33* (3), 1256–1295.

**Ballew, Matthew T., Jennifer R. Marlon, Matthew H. Goldberg, Edward W. Maibach, Seth A. Rosenthal, Emily Aiken, and Anthony Leiserowitz**, "Changing Minds about Global Warming: Vicarious Experience Predicts Self-Reported Opinion Change in the USA," *Climatic Change*, 2022, *173*, 19.

**Barro, Robert J. and Tao Jin**, "On the Size Distribution of Macroeconomic Disasters," *Econometrica*, 2011, *79* (5), 1567–1589.

**Berg, Florian, Julian F. Kölbel, and Roberto Rigobon**, "Aggregate Confusion: The Divergence of ESG Ratings," *Review of Finance*, 2022, *26* (6), 1315–1344.

**Bernile, Gennaro, Alok Kumar, and Johan Sulaeman**, "Home away from Home: Geography of Information and Local Investors," *The Review of Financial Studies*, 2015, *28* (7), 2009–2049.

**Board of Governors of the Federal Reserve System**, "Changes in U.S. Family Finances from 2019 to 2022: Evidence from the Survey of Consumer Finances," Technical Report, Federal Reserve Board 2023.

**Bordalo, Pedro, Nicola Gennaioli, Giacomo Lanzani, and Andrei Shleifer**, "A Cognitive Theory of Reasoning and Choice," 2025.

**Brown, David C., Shaun William Davies, and Matthew C. Ringgenberg**, "ETF Arbitrage, Non-Fundamental Demand, and Return Predictability," *Review of Finance*, 2021, *25* (4), 937–972.

**Brügger, Adrian, Christina Demski, and Stuart Capstick**, "How Personal Experience Affects Perception of and Decisions Related to Climate Change: A Psychological View," *Weather, Climate, and Society*, 2021, *13* (3), 397–408.

**Calvet, Laurent E., John Y. Campbell, and Paolo Sodini**, "Fight or Flight? Portfolio Rebalancing by Individual Investors," *The Quarterly Journal of Economics*, 2009, *124* (1), 301–348.

**Calzada, Joan and Ricard Gil**, "What Do News Aggregators Do? Evidence from Google News in Spain and Germany," *Marketing Science*, 2020.

**Ceccarelli, Marco, Stefano Ramelli, and Alexander F. Wagner**, "Low Carbon Mutual Funds," *Review of Finance*, 2024, *28* (1), 45–74.

**Cengiz, Doruk, Arindrajit Dube, Attila Lindner, and Ben Zipperer**, "The Effect of Minimum Wages on Low-Wage Jobs," *The Quarterly Journal of Economics*, 2019, *134* (3), 1405–1454.

**Choi, Darwin, Zhenyu Gao, and Wenxi Jiang**, "Attention to Global Warming," *Review of Financial Studies*, 2020, *33* (3), 1112–1145.

**Cookson, J. Anthony, Joseph E. Engelberg, and William Mullins**, "Does Partisanship Shape Investor Beliefs? Evidence from the COVID-19 Pandemic," *Review of Asset Pricing Studies*, 2020, *10* (4), 863–893.

**Coval, Joshua and Erik Stafford**, "Asset Fire Sales (and Purchases) in Equity Markets," *Journal of Financial Economics*, 2007, *86* (2), 479–512.

**Deryugina, Tatyana**, "How Do People Update? The Effects of Local Weather Fluctuations on Beliefs about Global Warming," *Climatic Change*, 2013, *118* (2), 397–416.

**Dessaint, Olivier and Adrien Matray**, "Do Managers Overreact to Salient Risks? Evidence from Hurricane Strikes," *Journal of Financial Economics*, 2017, *126* (1), 97–121.

**Djourelova, Milena, Ruben Durante, Elliot Motte, and Eleonora Patacchini**, "Experience, Narratives, and Climate Change Beliefs," Discussion Paper DP18738, CEPR 2024.

**Druckman, James N. and Mary C. McGrath**, "The Evidence for Motivated Reasoning in Climate Change Preference Formation," *Nature Climate Change*, 2019, *9* (2), 111–119.

**Dube, Arindrajit, Daniele Girardi, Òscar Jordà, and Alan M. Taylor**, "A Local Projections Approach to Difference-in-Differences," *Journal of Applied Econometrics*, 2025.

**Engelberg, Joseph E. and Christopher A. Parsons**, "The Causal Impact of Media in Financial Markets," *Journal of Finance*, 2011, *66* (1), 67–97.

**Engle, Robert F., Stefano Giglio, Bryan Kelly, Heebum Lee, and Johannes Stroebel**, "Hedging Climate Change News," *The Review of Financial Studies*, 2020, *33* (3), 1184–1216.

**Ewens, Michael, Arpit Gupta, and Sabrina T. Howell**, "Local Journalism under Private Equity Ownership," Working Paper 29743, National Bureau of Economic Research February 2022.

**Ewing, Sean**, "Profit Over Public Good: The Impact of Investment Firm Ownership on Local News and Political Behavior," *Political Communication*, 2026, *43* (3), 501–538.

**Fedyk, Anastassia**, "Disagreement after News: Gradual Information Diffusion or Differences of Opinion?," *Review of Asset Pricing Studies*, 2021, *11* (3), 465–501.

**Fich, Eliezer M. and Guosong Xu**, "Do Salient Climatic Risks Affect Shareholder Voting?," *Review of Finance*, 2025, *29* (2), 567–602.

**Fiordelisi, Franco, Giuseppe Galloppo, and Viktoriia Paimanova**, "Climate Change Shocks and Socially Responsible Investments," *Business Ethics, the Environment & Responsibility*, 2023, *32* (1), 40–56.

**Gabaix, Xavier and Ralph S. J. Koijen**, "In Search of the Origins of Financial Fluctuations: The Inelastic Markets Hypothesis," 2023.

**Gallagher, Justin**, "Learning about an Infrequent Event: Evidence from Flood Insurance Take-Up in the United States," *American Economic Journal: Applied Economics*, 2014, *6* (3), 206–233.

**Gantchev, Nickolay, Mariassunta Giannetti, and Rachel Li**, "Sustainability or Performance? Ratings and Fund Managers' Incentives," *Journal of Financial Economics*, 2024, *155*, 103831.

**Gao, Pengjie, Chang Lee, and Dermot Murphy**, "Financing Dies in Darkness? The Impact of Newspaper Closures on Public Finance," *Journal of Financial Economics*, 2020, *135* (2), 445–467.

**Garz, Marcel and Jonna Rickardsson**, "Ownership and Media Slant: Evidence from Swedish Newspapers," *Kyklos*, 2023, *76* (1), 18–40.

**Giglio, Stefano, Matteo Maggiori, Johannes Stroebel, and Stephen Utkus**, "Five Facts about Beliefs and Portfolios," *American Economic Review*, 2021, *111* (5), 1481–1522.

_ , _ , _ , **Zhenhao Tan, Stephen Utkus, and Xiao Xu**, "Four Facts about ESG Beliefs and Investor Portfolios," *Journal of Financial Economics*, 2025, *164*, 103984.

**Hartzmark, Samuel M. and Abigail B. Sussman**, "Do Investors Value Sustainability? A Natural Experiment Examining Ranking and Fund Flows," *Journal of Finance*, 2019, *74* (6), 2789–2837.

**Hausman, Jerry**, "Mismeasured Variables in Econometric Analysis: Problems from the Right and Problems from the Left," *Journal of Economic Perspectives*, 2001, *15* (4), 57–67.

**Hazlett, Chad and Matto Mildenberger**, "Wildfire Exposure Increases Pro-Environment Voting within Democratic but Not Republican Areas," *American Political Science Review*, 2020, *114* (4), 1359–1365.

**Investment Company Institute**, "2025 Investment Company Fact Book," Technical Report, Investment Company Institute 2025.

**IPCC**, "Summary for Policymakers," in Core Writing Team, H. Lee, and J. Romero, eds., *Climate Change 2023: Synthesis Report. Contribution of Working Groups I, II and III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*, Geneva, Switzerland: IPCC, 2023, pp. 1–34.

**Iyengar, Shanto**, *Is Anyone Responsible? How Television Frames Political Issues*, Chicago: University of Chicago Press, 1991.

**Jr., James M. Snyder and David Strömberg**, "Press Coverage and Political Accountability," *Journal of Political Economy*, 2010, *118* (2), 355–408.

**Kacperczyk, Marcin, Clemens Sialm, and Lu Zheng**, "Unobserved Actions of Mutual Funds," *Review of Financial Studies*, 2008, *21* (6), 2379–2416.

**Kandel, Eugene and Neil D. Pearson**, "Differential Interpretation of Public Signals and Trade in Speculative Markets," *Journal of Political Economy*, 1995, *103* (4), 831–872.

**Kang, Byoung Uk and Jonathan Sangwook Nam**, "Do Local Newspapers Matter to Institutional Investors?," *Contemporary Accounting Research*, 2025, *42* (3), 1713–1743.

**Kempf, Elisabeth, Mancy Luo, Larissa Schäfer, and Margarita Tsoutsoura**, "Political Ideology and International Capital Allocation," *Journal of Financial Economics*, 2023, *148* (2), 150–173.

**King, Gary, Benjamin Schneer, and Ariel White**, "How the News Media Activate Public Expression and Influence National Agendas," *Science*, 2017, *358* (6364), 776–780.

**Kline, Patrick, Raffaele Saggio, and Mikkel Sølvsten**, "Leave-Out Estimation of Variance Components," *Econometrica*, 2020.

**Koijen, Ralph S. J. and Motohiro Yogo**, "A Demand System Approach to Asset Pricing," *Journal of Political Economy*, 2019, *127* (4), 1475–1515.

_ **, Robert J. Richmond, and Motohiro Yogo**, "Which Investors Matter for Equity Valuations and Expected Returns?," *Review of Economic Studies*, 2024, *91* (4), 2387–2424.

**Kostovetsky, Leonard, Lin Peng, Christopher Rauh, and Muhammed Yönaç**, "Measuring Local Climate Change Attention: Does it Affect Households and Firms?," 2026. Forthcoming, Journal of Financial and Quantitative Analysis.

**Kruttli, Mathias S., Brigitte Roth Tran, and Sumudu W. Watugala**, "Pricing Poseidon: Extreme Weather Uncertainty and Firm Return Dynamics," *Journal of Finance*, 2025, *80* (2), 783–832.

**Leiserowitz, Anthony, John Kotcher, Seth Rosenthal, Emily Goddard, Jennifer Carman, Marija Verner, Teresa Myers, and Edward Maibach**, "Climate Change in the American Mind: Beliefs & Attitudes, Spring 2026," Yale Program on Climate Change Communication 2026.

**Lettau, Martin and Ananth Madhavan**, "Exchange-Traded Funds 101 for Economists," *Journal of Economic Perspectives*, 2018, *32* (1), 135–154.

**LSEG**, "Environmental, Social and Governance Scores from LSEG," Methodology, LSEG Data & Analytics October 2024.

**Malamud, Semyon**, "A Dynamic Equilibrium Model of ETFs," 2016.

**Malmendier, Ulrike**, "Experience Effects in Finance: Foundations, Applications, and Future Directions," *Review of Finance*, 2021, *25* (5), 1339–1363.

_ **and Jessica A. Wachter**, "Memory of Past Experiences and Economic Decisions," in "Oxford Handbook of Human Memory," Oxford University Press, 2022.

**Marshall, Ben R., Hung T. Nguyen, Nhut H. Nguyen, Nuttawat Visaltanachoti, and Martin Young**, "Do Climate Risks Matter for Green Investment?," *Journal of International Financial Markets, Institutions and Money*, 2021, *75*, 101438.

**Martin, Gregory J. and Joshua McCrain**, "Local News and National Politics," *American Political Science Review*, 2019, *113* (2), 372–384.

**Meeuwis, Maarten, Jonathan A. Parker, Antoinette Schoar, and Duncan Simester**, "Belief Disagreement and Portfolio Choice," *Journal of Finance*, 2022, *77* (6), 3191–3247.

**Misev, Marina A. and Patrick Balles**, "Natural Disasters, Investor Attention, and Non-Fundamental Green Asset Demand," WWZ Working Paper 2024/07, University of Basel, Faculty of Business and Economics 2024.

**Morris, Stephen and Hyun Song Shin**, "Social Value of Public Information," *American Economic Review*, 2002, *92* (5), 1521–1534.

**Moussawi, Rabih, Ke Shen, and Raisa Velthuis**, "The Role of Taxes in the Rise of ETFs," *Review of Financial Studies*, 2025, *38* (10), 2988–3039.

**Myers, Andrew C. W.**, "Press Coverage and Accountability in State Legislatures," *American Political Science Review*, 2026, *120* (1), 207–225.

**Myers, Teresa A., Edward W. Maibach, Connie Roser-Renouf, Karen Akerlof, and Anthony A. Leiserowitz**, "The Relationship between Personal Experience and Belief in the Reality of Global Warming," *Nature Climate Change*, 2013, *3*, 343–347.

**Pan, Yihui, Elena S. Pikulina, Stephan Siegel, and Tracy Yue Wang**, "Political Divide and Partisan Portfolio Disagreement," 2025.

**Pástor, Ľuboš, Robert F. Stambaugh, and Lucian A. Taylor**, "Dissecting Green Returns," *Journal of Financial Economics*, 2022, *146* (2), 403–424.

**Pew Research Center**, "Americans' Changing Relationship With Local News," Pew Research Center, Journalism 2024.

_ , "Friends, Family and Neighbors Are Americans' Most Common Source of Local News," Pew Research Center, Short Reads 2024.

**Quéré, Marianne Aubin Le, Ting-Wei Chiang, and Mor Naaman**, "Understanding Local News Social Coverage and Engagement at Scale during the COVID-19 Pandemic," *Proceedings of the International AAAI Conference on Web and Social Media*, 2022, *16* (1), 560–572.

**Riedl, Arno and Paul Smeets**, "Why Do Investors Hold Socially Responsible Mutual Funds?," *The Journal of Finance*, 2017, *72* (6), 2505–2550.

**Sammon, Marco and John Shim**, "Do Active Funds Do Better in What They Trade?," 2023.

**Schmidt-Engelbertz, Paul and Kaushik Vasudevan**, "Speculating on Higher-Order Beliefs," *Review of Financial Studies*, 2025, *38* (8), 2434–2466.

**Schwartzstein, Joshua and Adi Sunderam**, "Using Models to Persuade," *American Economic Review*, 2021, *111* (1), 276–323.

**Trenberth, Kevin E., John T. Fasullo, and Theodore G. Shepherd**, "Attribution of Climate Extreme Events," *Nature Climate Change*, 2015, *5* (8), 725–730.

**Tubaldi, Roberto**, "Mutual Funds' Fire Sales and the Real Economy: Evidence from Hurricanes," 2021.

**U.S. Census Bureau**, "Income in the United States: 2025," Current Population Reports P60-289, U.S. Census Bureau September 2026.

**van der Beck, Philippe**, "Flow-Driven ESG Returns," *The Journal of Finance*, 2026.

_ **, Lorenzo Bretscher, and Julie Zhiyu Fu**, "A Bound on Price Impact and Disagreement," 2025.

**Wagner, Alexander F., Richard J. Zeckhauser, and Alexandre Ziegler**, "Company Stock Price Reactions to the 2016 Election Shock: Trump, Taxes, and Trade," *Journal of Financial Economics*, 2018, *130* (2), 428–451.

# A Data Construction

TABLE A.1: DISASTER SAMPLES: HAZARD COMPOSITION AND EPISODE CHARACTERISTICS

*Panel A. Broad disaster sample, 2009–2024*

| Hazard | Episodes | Share (%) | Mean damage ($ millions) | Mean fatalities | Mean injuries | Median affected counties |
|---|---|---|---|---|---|---|
| Flood | 240 | 21.8 | 606.2 | 3.31 | 2.82 | 5.5 |
| Heat | 233 | 21.2 | 0.0 | 14.03 | 33.09 | 1.0 |
| Tornado | 129 | 11.7 | 260.1 | 8.26 | 81.91 | 7.0 |
| Winter weather | 109 | 9.9 | 89.2 | 4.27 | 34.61 | 7.0 |
| Hail | 89 | 8.1 | 324.2 | 0.11 | 1.60 | 3.0 |
| Drought | 75 | 6.8 | 268.0 | 0.00 | 4.67 | 5.0 |
| Wildfire | 61 | 5.6 | 613.8 | 4.61 | 6.44 | 2.0 |
| Hurricane | 57 | 5.2 | 1,799.3 | 8.39 | 4.49 | 8.0 |
| Severe storm | 38 | 3.5 | 95.3 | 2.26 | 27.16 | 5.5 |
| Wind | 29 | 2.6 | 268.5 | 3.83 | 6.48 | 12.0 |
| Coastal | 24 | 2.2 | 1,539.3 | 4.04 | 0.54 | 5.5 |
| Other | 15 | 1.4 | 75.7 | 8.07 | 31.00 | 2.0 |
| All episodes | 1,099 | 100.0 | 388.7 | 6.17 | 23.26 | 4.0 |

*Panel B. Selected flood episodes, 2009–2024*

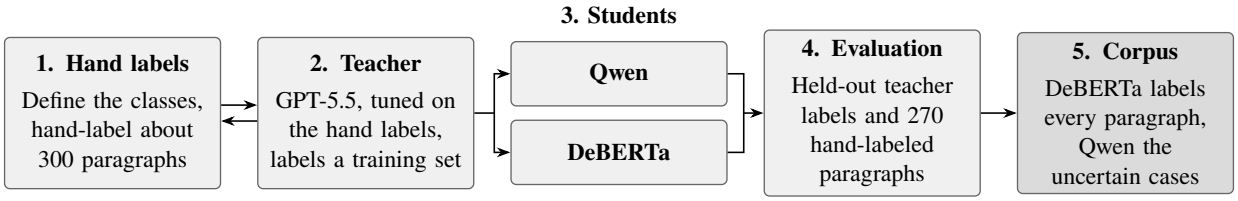
| Characteristic | Episodes | Mean | Std. dev. | P25 | Median | P75 | P90 |
|---|---|---|---|---|---|---|---|
| Damage ($ millions) | 1,490 | 101.59 | 1,546.87 | 1.57 | 3.36 | 10.74 | 46.75 |
| Fatalities | 1,490 | 0.85 | 3.41 | 0.00 | 0.00 | 0.00 | 2.00 |
| Injuries | 1,490 | 0.74 | 5.67 | 0.00 | 0.00 | 0.00 | 1.00 |
| Affected counties | 1,490 | 4.82 | 5.44 | 1.00 | 3.00 | 6.00 | 10.10 |
| Duration (days) | 1,490 | 4.49 | 6.52 | 1.00 | 2.00 | 4.00 | 11.00 |

*Panel C. Selected flood, hurricane, wildfire, and coastal episodes, 2009–2024*

| Characteristic | Episodes | Mean | Std. dev. | P25 | Median | P75 | P90 |
|---|---|---|---|---|---|---|---|
| Damage ($ millions) | 1,938 | 167.11 | 1,776.55 | 1.33 | 3.37 | 12.26 | 61.38 |
| Fatalities | 1,938 | 1.27 | 5.23 | 0.00 | 0.00 | 1.00 | 3.00 |
| Injuries | 1,938 | 1.38 | 6.68 | 0.00 | 0.00 | 0.00 | 3.00 |
| Affected counties | 1,938 | 4.50 | 5.56 | 1.00 | 3.00 | 6.00 | 10.00 |
| Duration (days) | 1,938 | 5.04 | 7.13 | 1.00 | 2.00 | 5.00 | 15.00 |

*Notes.* This table reports the hazard composition and episode characteristics of the disaster samples. Panel A describes the broad disaster sample. Panel B describes the flood episodes used in the main flood analysis. Panel C describes the flood, hurricane, wildfire, and coastal episodes used in the main analysis. Damage is expressed in 2023 dollars. Fatalities, injuries, and affected counties are episode-level measures; the affected-counties measure counts distinct U.S. counties linked to an episode. Other combines hazard categories represented by fewer than ten episodes.

FIGURE A.1: THE CLASSIFICATION PIPELINE




*Notes.* This figure shows the steps of the paragraph-level classification of disaster articles. I define the classes and hand-label about 300 paragraphs, which I use to tune the prompts of a large language model, GPT-5.5. This model labels a training set, on which I train two local models, DeBERTa and Qwen. I evaluate the local models on held-out labels of the large model and on 270 hand-labeled paragraphs (Appendix Tables A.2 and A.3). DeBERTa classifies the full corpus and passes the paragraphs it classifies with low confidence to Qwen.

TABLE A.2: CANDIDATE MODELS AGAINST THE REFERENCE LABELS

| Model | Disaster (1) | Specific event (2) | Climate link (3) | Central link (4) | Skeptical (5) | Human toll (6) | Flip (7) |
|---|---|---|---|---|---|---|---|
| Qwen3.5-9B-staged | .952 | .960 | .947 | .900 | .911 | .962 | .015 |
| DeBERTa-v3-large | .905 | .932 | .938 | .869 | .880 | .909 | .015 |
| gpt-5.4-mini | .877 | .917 | .906 | .838 | .818 | .954 | .053 |
| gpt-4.1-mini | .893 | .895 | .899 | .605 | .774 | .893 | .075 |
| gpt-4o-mini | .745 | .931 | .739 | .541 | .804 | .911 | .063 |
| Keyword rule | .809 | .468 | .856 | .802 | .597 | .780 | .215 |

*Notes.* This table reports the positive-class $F_1$ score of each candidate model against the reference labels on the 1,399 held-out paragraphs, the 639 random paragraphs and the 760 paragraphs of the hand-labeled articles. Climate link counts a passing or central link; central link counts only a central link. Flip is the share of paragraphs with an affirming or denying stance to which the model assigns the opposite stance. The keyword rule labels a paragraph as a disaster if it contains a natural-hazard term, as linked if it contains a climate term (climate change, global warming, greenhouse gases, carbon emissions, and similar), as skeptical if it contains a skeptic term (hoax, alarmist, so-called, exaggerated, natural cycles, and similar), as human toll if it contains a casualty or displacement term, and as a specific event if it names a storm or a date. It has no notion of a central link, so the central-link column repeats the link rule, and its stance is denying when a skeptic term appears and affirming otherwise.

TABLE A.3: CANDIDATE MODELS AGAINST HAND LABELS ON THE HELD-OUT PARAGRAPHS

| Model | Disaster (1) | Climate link (2) | Central link (3) | Skeptical (4) | Flip (5) |
|---|---|---|---|---|---|
| Qwen3.5-9B-staged | .888 | .876 | .727 | .918 | .009 |
| DeBERTa-v3-large | .876 | .868 | .682 | .871 | .027 |
| gpt-5.4-mini | .804 | .768 | .667 | .830 | .029 |
| gpt-4.1-mini | .869 | .846 | .636 | .800 | .059 |
| gpt-4o-mini | .667 | .638 | .526 | .844 | .077 |
| Keyword rule | .803 | .887 | .684 | .565 | .196 |

*Notes.* See notes to Table A.2. Scores are computed against my hand labels on the 270 held-out paragraphs, the hardest cases of the labeling set.

TABLE A.4: EXAMPLE PARAGRAPHS AND THEIR CLASSIFICATION

| **Climate link**<br>Hurricane Helene, Associated Press, October 9, 2024 | **Human toll and personal story**<br>Hurricane Katrina, *Wall Street Journal*, August 31, 2005 |
|---|---|
| "The scientists warned that **continued burning of fossil fuels will lead to more hurricanes like Helene**, with 'unimaginable' floods well inland, not just on coasts."<br><br>*Classification.* Specific disaster event; affirming climate link, central to the paragraph. The paragraph never mentions "climate change." | "After climbing out of one boat, Judy Martin took a few steps and then **collapsed onto the highway** [. . . ] Ms. Martin said that **her 95-year-old mother had drowned in their house** as the floodwaters rose around them."<br>*Classification.* Specific disaster event; human toll, told as one person's story; no climate link. |

*Notes.* This table shows two paragraphs from the coverage of disaster events and the labels the classifier assigns to them. Bold text marks the passages that determine the label.

TABLE A.5: AGREEMENT BETWEEN THE COUNTY NEWS CLASSES AND PUBLISHED NATIONAL CLIMATE-NEWS INDICES

| | EGKLS WSJ (monthly) | | MCCC (weekly) | | NYT (monthly) | |
|---|---|---|---|---|---|---|
| News class | Level (1) | Innov. (2) | Level (3) | Innov. (4) | Level (5) | Innov. (6) |
| *Panel A. Climate-topic classes* | | | | | | |
| Climate, no disaster, plus climate policy and green energy | .565 | .333 | −.011 | .313 | .159 | .126 |
| Climate coverage, no disaster | .564 | .411 | .227 | .291 | .305 | .202 |
| Climate policy and green energy | .550 | .282 | −.027 | .309 | .142 | .106 |
| Climate or peripheral (disaster-contaminated) | .318 | .253 | −.462 | .299 | −.580 | .064 |
| *Panel B. Disaster classes* | | | | | | |
| Disaster coverage with a climate attribution | .475 | .333 | .647 | .278 | .709 | .237 |
| All disaster coverage | .088 | .037 | −.516 | .019 | −.553 | −.059 |
| Disaster policy, preparedness, insurance | −.067 | .020 | −.592 | .046 | −.683 | −.132 |
| *Panel C. Pooled series, denominator and placebo* | | | | | | |
| All scored coverage except irrelevant | .207 | .099 | −.534 | .149 | −.583 | −.034 |
| All scored coverage | .140 | .070 | −.571 | .127 | −.628 | −.045 |
| Irrelevant coverage (placebo) | −.086 | −.054 | −.633 | −.002 | −.709 | −.098 |
| Periods (levels / innovations) | 210 / 209 | | 1,074 / 916 | | 288 / 287 | |

*Notes.* This table reports the agreement between the county news classes and three published national climate-news indices. Each class is aggregated to a county panel, standardized within county, and reduced to a national common factor by principal components across counties. "Level" is the correlation of that first principal component with the published index in levels; "Innov." is the correlation of their innovations. Innovations are AR(1) forecast errors from the prior 156 weeks for the weekly series; for MCCC and EGKLS they are the innovation series the authors themselves publish.

Column frequencies follow the published series. EGKLS is monthly and ends in June 2017; MCCC is daily, aggregated here to weeks; the NYT index is daily.

The placebo row is coverage the classifier marks as unrelated to climate.

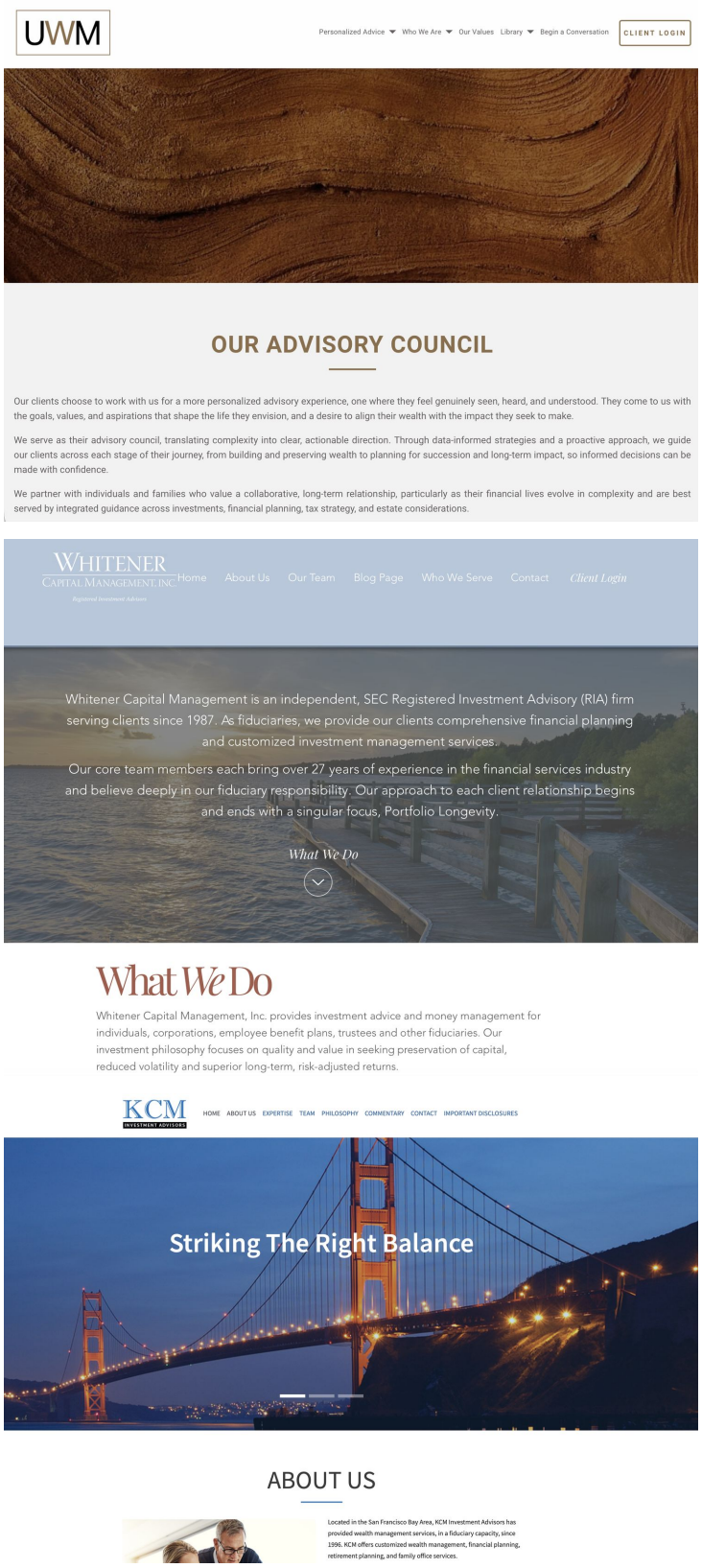
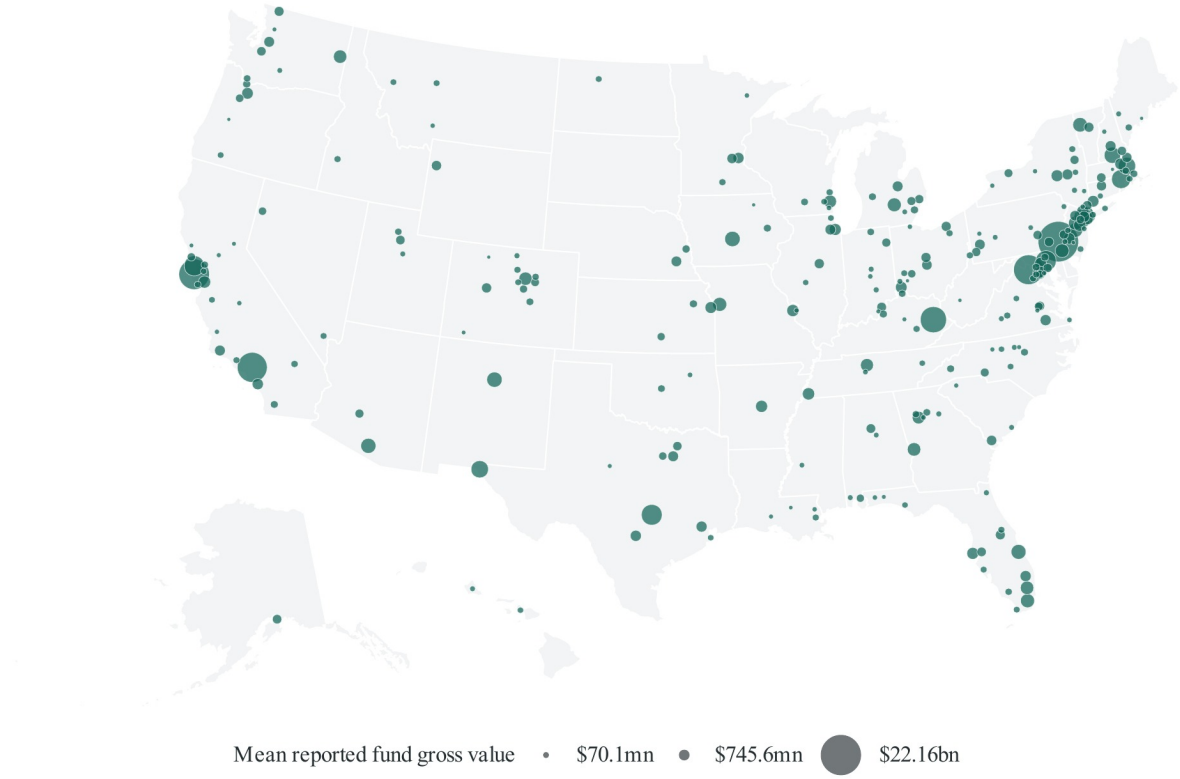



FIGURE A.2: THREE ILLUSTRATIVE LOCAL REGISTERED INVESTMENT ADVISERS

*Notes.* This figure shows the public web pages of three advisers in the household-serving sample, chosen across its size range. Urban Wealth Management is near the bottom of the range, Whitener Capital at the median, and KCM Investment Advisors near the top.

FIGURE A.3: MUTUAL-FUND MANAGEMENT GEOGRAPHY, 2009–2024




*Notes.* This figure maps the counties from which mutual funds are managed. Funds whose managers or managing firms span several counties are assigned proportional county weights.

# B Conceptual Framework and Empirical Strategy

I use a portfolio-choice model to describe how interpreting a disaster as evidence of climate change affects demand for ESG funds. The model links the local allocation response to investors' preferences and belief updating. It then provides the additional allocation for the exercise in Appendix J, holding attention and the wider information environment fixed.

## B.1 Portfolio choice and beliefs

Consider an investor choosing the share $q \in [0,1]$ of eligible wealth invested in a representative ESG portfolio. The remaining share, $1-q$, is invested in an other-fund portfolio. Let $P > 0$ be the price of the ESG portfolio relative to the other-fund portfolio, whose price is normalized to one. Initial eligible capital is normalized to one. If $q_0$ is the inherited ESG share at relative price $P^0$, wealth before rebalancing is $w(P) = q_0 P/P^0 + 1 - q_0$. The chosen shares purchase $qw(P)/P$ units of the ESG portfolio and $(1-q)w(P)$ units of the other portfolio, so the budget is satisfied without external deposits or withdrawals.

There are two future climate states, adverse $h$ and benign $l$. In state $k$, the ESG and other portfolios pay $X_{Ek} > 0$ and $X_{Ok} > 0$ per unit, respectively. The investor incurs background loss $L_k$ and receives a nonfinancial benefit $\psi g_k$ per ESG unit in consumption units: $\psi$ measures preference strength and $g_k$ the state-specific benefit. Let

$$r_k(P) = \frac{X_{Ek} + \psi g_k}{P} - X_{Ok}, \qquad k \in \{h, l\}, \tag{B.1}$$

be the payoff, including the holdings benefit, from shifting one dollar from the other-fund portfolio to the ESG portfolio. I assume $r_h(P) > 0 > r_l(P)$ over the relevant price range. The ESG tilt provides insurance in the adverse climate state at a cost in the benign state.

The other-fund portfolio costs one, so $X_{Ok}$ is its return per dollar in state $k$, the benchmark return. The ESG portfolio costs $P$, so $(X_{Ek} + \psi g_k)/P$ is its return per dollar, the economic payoff plus the nonfinancial benefit, and $r_k(P)$ in equation (B.1) is the difference between the two. For perceived probability $\pi$ of the adverse state, the investor has exponential utility of end-of-period consumption with risk aversion $\rho > 0$ and solves

$$\begin{aligned} \max_{0 \le q \le 1} \quad & -\pi \exp\left\{-\rho \left[ w(P)\left((1-q)X_{Oh} + q\,\frac{X_{Eh} + \psi g_h}{P}\right) - L_h \right]\right\} \\ & -(1-\pi)\exp\left\{-\rho \left[ w(P)\left((1-q)X_{Ol} + q\,\frac{X_{El} + \psi g_l}{P}\right) - L_l \right]\right\}, \end{aligned} \tag{B.2}$$

which, collecting the terms in $q$ and using $r_k(P)$, is

$$\begin{aligned} \max_{0 \le q \le 1} \quad & -\pi \exp\{-\rho\,[w(P)X_{Oh} - L_h + w(P)q r_h(P)]\} \\ & -(1-\pi)\exp\{-\rho\,[w(P)X_{Ol} - L_l + w(P)q r_l(P)]\}. \end{aligned} \tag{B.3}$$

The news changes the perceived probability of each climate state, while asset payoffs and losses within each state, as well as investors' preferences, remain unchanged. The following derivation considers interior portfolios.

For episode $e$, let $f_i(z_e \mid k)$ be investor $i$'s subjective likelihood of disaster news $z_e$ in state $k$. Define the likelihood ratio $\mathrm{LR}_{i,e} = f_i(z_e \mid h)/f_i(z_e \mid l)$ and its logarithm $\ell_{i,e} = \log \mathrm{LR}_{i,e}$.

The event indicator $\eta_e$ is zero before disaster news and one afterward. Let $\pi_{0i} \in (0,1)$ and $\pi_{1i}$ be the

prior and posterior probabilities of the adverse state, with odds $\omega_{0i} = \pi_{0i}/(1-\pi_{0i})$ and $\omega_{1i} = \pi_{1i}/(1-\pi_{1i})$. Bayes' rule gives

$$\omega_{1i} = \omega_{0i}\mathrm{LR}_{i,e}^{\eta_e}, \qquad \log\omega_{1i} = \log\omega_{0i} + \eta_e \ell_{i,e}. \tag{B.4}$$

Positive $\ell_{i,e}$ raises perceived adverse-climate risk. Interpretation may reflect investors' own models or supplied narratives. The benchmark shares preferences, prior odds $\omega_{0i} = \omega_0$, and initial normalized portfolios; investors do not infer additional information from prices.

## B.2 The predicted portfolio response

Dividing expected utility by $1-\pi$ leaves the optimal portfolio unchanged. Beliefs then enter through the odds of the adverse state relative to the benign state, $\omega = \pi/(1-\pi)$. Differentiating with respect to $q$ and dividing by $\rho w(P)$ gives the optimality condition

$$\begin{aligned} &\omega r_h(P) e^{-\rho[w(P)X_{Oh} - L_h + w(P)q r_h(P)]} \\ &\quad = [-r_l(P)] e^{-\rho[w(P)X_{Ol} - L_l + w(P)q r_l(P)]}. \end{aligned} \tag{B.5}$$

Denote the optimal ESG share by $q^*(P,\omega)$. Taking logarithms gives a linear equation in this share. Because news changes log odds by $\eta_e \ell_{i,e}$, the solution implies

$$q^*(P,\omega_{1i}) - q^*(P,\omega_0) = \eta_e \chi(P)\ell_{i,e}, \qquad \chi(P) = \frac{1}{\rho w(P)[r_h(P) - r_l(P)]} > 0. \tag{B.6}$$

Here $\chi(P)$ is the change in the ESG share per unit increase in log odds, holding the price fixed. The formula applies to finite belief updates on the interior support.

Let $q_0 = q^*(P^0,\omega_0)$ be the initial optimal share. At initial prices, the event changes desired allocation by

$$\begin{aligned} u_{i,e} &\equiv q^*(P^0,\omega_{1i}) - q_0 = \eta_e \chi(P^0)\ell_{i,e} = \eta_e \gamma_{i,e}, \\ \gamma_{i,e} &\equiv \chi(P^0)\ell_{i,e}. \end{aligned} \tag{B.7}$$

Thus $u_{i,e} = 0$ before news ($\eta_e = 0$) and $u_{i,e} = \gamma_{i,e}$ after news ($\eta_e = 1$). The event response $\gamma_{i,e}$ combines portfolio responsiveness and the belief update at initial prices.

At post-news relative price $P_e$, the observed share change is $\Delta q_{i,e} = q^*(P_e,\omega_{1i}) - q_0$. Adding and subtracting the post-news allocation at initial prices gives

$$\Delta q_{i,e} = u_{i,e} + \underbrace{q^*(P_e,\omega_{1i}) - q^*(P^0,\omega_{1i})}_{\text{response to the price change after news}}. \tag{B.8}$$

The ratio $P_e/P^0$ is the gross relative-price return, and $(P_e - P^0)/P^0$ is the corresponding net return. Thus $\Delta p_e = \log(P_e/P^0) = \log(1 + (P_e - P^0)/P^0) \approx (P_e - P^0)/P^0$. No relative-price change means gross return one, net return zero, and $\Delta p_e = 0$. The expansion around $\Delta p_e = 0$ therefore concerns small relative-price returns. Holding updated beliefs fixed gives

$$\begin{aligned} \Delta q_{i,e} &= u_{i,e} - \zeta_{i,e}^q \Delta p_e + o(|\Delta p_e|), \\ \zeta_{i,e}^q &= -\left.\frac{\partial q^*(P,\omega_{1i})}{\partial \log P}\right|_{P=P^0}. \end{aligned} \tag{B.9}$$

The coefficient $\zeta^q_{i,e}$ measures the response of the portfolio share to prices. Its independence from the news is an additional local restriction.

**Assumption: price responsiveness.** Disaster exposure and stronger climate interpretation shift desired allocations but do not change investors' responsiveness to prices. Conditional on the controls, average price responsiveness does not systematically differ with disaster exposure or interpretation.[73]

I impose this assumption on the local price-response term in (B.9). The common price movement then contributes an episode component to the conditional mean, while differences associated with disaster exposure and interpretation reflect $u_{i,e}$. For holdings valued at initial prices, I impose the corresponding conditional-average restriction on the price response of that measured share.[74]

## B.3 From the model to the estimating equation

Let $A_{i,e}$ denote attention to disaster news, $I_{i,e}$ its climate interpretation, and $Z_e$ the wider information environment. Write the belief update as $\ell_{i,e} = \ell_i(A_{i,e}, I_{i,e}, Z_e)$ and the allocation response as $u_{i,e}(A, I, Z) = \eta_e \chi(P^0)\ell_i(A, I, Z)$.

For county $c$, the impact outcome $\Delta q_{c,e} = q_{c,0} - q_{c,-1}$ is the ESG-share change from the pre-event quarter $-1$ to the event quarter 0. Let $I_{c,e}$ be the circulation-weighted average of outlets' shares of disaster articles linking the event to climate change. Write $I^*_{c,e} = (I_{c,e} - \bar{I})/\sigma_I$, where $\bar{I}$ and $\sigma_I$ are its sample mean and standard deviation. Let $A_{c,e}$ be local disaster-news coverage and $A^*_{c,e}$ its standardized $\log(1 + A_{c,e})$. Set $H_{c,e} = 1$ for affected counties and zero for clean comparison counties in the same episode. All these counties are observed around a disaster, so $\eta_e = 1$.

Let $\ell_{c,e}$ be the conditional average belief update and $\ell^0_{c,e}$ its value at mean interpretation. Holding attention and controls fixed, I assume a unit increase in $I^*$ changes the belief update by $\gamma^\ell_{1,e}$ among affected investors and $\gamma^\ell_{0,e}$ among comparison investors:

$$\ell_{c,e} = \ell^0_{c,e} + \gamma^\ell_{H_{c,e},e} I^*_{c,e}. \tag{B.10}$$

In the common-response benchmark, $u_{c,e} = \chi(P^0)\ell_{c,e}$. Write $\overline{\Delta q}_{c,e}$ for the predicted share change conditional on exposure, interpretation, attention, controls, and episode. Under the price-response assumption, comparing counties $c$ and $c'$ at the same controls within an episode gives

$$\overline{\Delta q}_{c,e} - \overline{\Delta q}_{c',e} \simeq u_{c,e} - u_{c',e} = \chi(P^0)\big(\ell_{c,e} - \ell_{c',e}\big). \tag{B.11}$$

The approximation inherits the price remainder in (B.9). Individual price responses may differ; the assumption concerns their conditional average in the comparison.

Let $b_e = -\bar{\zeta}^q_e \Delta p_e$ denote the common conditional price contribution, where $\bar{\zeta}^q_e$ is average price respon-

[73] A related benchmark in van der Beck et al. (2025), Assumption 1, allows heterogeneous price elasticities that are independent of the process generating demand shifts across investors.

[74] Let $q_i = q^*(P, \omega_{1i})$. The share of these holdings valued at $P^0$ is $q^{\text{FP}}_i = (P^0/P)q_i/[1 - q_i + (P^0/P)q_i]$. At $P = P^0$, it equals $q_i$, so its initial-price information shift is also $u_{i,e}$. Its price response is obtained by expanding this measured share; its coefficient generally differs from $\zeta^q_{i,e}$.

siveness. Substituting the interpretation rule into the allocation response gives

$$\begin{aligned}\overline{\Delta q}_{c,e} \simeq b_e + \chi(P^0)\ell^0_{c,e} + \chi(P^0)\gamma^\ell_{0,e} I^*_{c,e} \\ + \chi(P^0)(\gamma^\ell_{1,e} - \gamma^\ell_{0,e}) H_{c,e} I^*_{c,e}.\end{aligned} \tag{B.12}$$

Assume a linear baseline response. Let $\lambda_e$ denote episode effects, $\delta_H$ the exposure coefficient, $\beta_A$ the comparison-county attention slope, and $\delta_A$ its affected-county increment. Remaining controls and other fixed effects $\mathbf{W}_{c,e}$ have coefficients $\theta$:

$$\begin{aligned}b_e + \chi(P^0)\ell^0_{c,e} = \lambda_e + \delta_H H_{c,e} + \beta_A A^*_{c,e} \\ + \delta_A H_{c,e} A^*_{c,e} + \mathbf{W}'_{c,e}\theta.\end{aligned} \tag{B.13}$$

Episode effects absorb $b_e$; portfolio responsiveness $\chi(P^0)$ remains in the interpretation response.

With interpretation slope $\beta_I$ in comparison counties, affected-county increment $\delta_I$, and regression residual $\varepsilon_{c,e}$, the estimating equation is

$$\begin{aligned}\Delta q_{c,e} = \lambda_e + \delta_H H_{c,e} + \beta_A A^*_{c,e} + \beta_I I^*_{c,e} \\ + \delta_A H_{c,e} A^*_{c,e} + \delta_I H_{c,e} I^*_{c,e} + \mathbf{W}'_{c,e}\theta + \varepsilon_{c,e}.\end{aligned} \tag{B.14}$$

The residual includes individual deviations from the conditional price response and has zero conditional mean in the maintained first-order representation. Matching the interpretation terms gives

$$\beta_I = \chi(P^0)\gamma^\ell_{0,e}, \qquad \delta_I = \chi(P^0)(\gamma^\ell_{1,e} - \gamma^\ell_{0,e}). \tag{B.15}$$

This coefficient mapping assumes that the response products are common across episodes. Otherwise, the pooled coefficients summarize the heterogeneous products using the county–episode variation remaining after controls and fixed effects; they are not generally arithmetic averages.

The mapping therefore uses the portfolio model, the local price approximation and price-response assumption, the linear interpretation and baseline relations, and common response products across episodes. Under these assumptions, the interpretation effect among affected investors is

$$\beta_I + \delta_I = \underbrace{\chi(P^0)}_{\substack{\text{portfolio response} \\ \text{to a belief update}}} \underbrace{\gamma^\ell_{1,e}}_{\substack{\text{belief response} \\ \text{to interpretation}}} = \frac{\gamma^\ell_{1,e}}{\rho[r_h(P^0) - r_l(P^0)]}. \tag{B.16}$$

It measures the additional desired ESG share at initial prices from one standard deviation more climate interpretation, holding attention and controls fixed. Here $w(P^0) = 1$ by the initial wealth normalization.

For stronger interpretation $I^1$ relative to baseline $I^0$ in the same county and event, define $a^I_{i,e} = u_{i,e}(A, I^1, Z) - u_{i,e}(A, I^0, Z)$. Appendix J aggregates its estimated counterpart,

$$\widehat{a}^I = (\widehat{\beta}_I + \widehat{\delta}_I)\frac{I^1 - I^0}{\sigma_I}. \tag{B.17}$$

## B.4 County and adviser portfolio aggregation

For a common set of advisers observed at both endpoints, let $\omega_{ac,t}$ denote adviser $a$'s share of county ETF holdings and $g_{at}$ its green ETF share. The change in the county green share satisfies

$$\Delta g_c = \underbrace{\sum_{a \in c} \omega_{ac,0} \Delta g_a}_{\text{within-adviser changes at pre-disaster weights}} + \underbrace{\sum_{a \in c} (\omega_{ac,1} - \omega_{ac,0}) g_{a,1}}_{\text{changes in advisers' relative ETF holdings}} . \tag{B.18}$$

Both components can reflect household investment decisions. The county outcome captures their combined contribution, whereas the pre-disaster-weighted average of adviser changes captures the first component. The two changes coincide when the second component is zero; unchanged adviser weights are a sufficient condition. Holding ETF prices fixed does not itself hold advisers' relative ETF holdings fixed, since purchases and sales can change those holdings.

Regression equivalence additionally requires the same estimation support and outcome window, regressors common to advisers within each county-event observation, and compatible regression weights. If the county-event regression weight is $q_{ce}$, the corresponding adviser weight is $q_{ce}\omega_{ac,0}$, with adviser shares summing to one on the estimation sample. County-event regressors and fixed effects must be repeated identically across its advisers. Under these conditions, the weighted adviser regression has the same coefficient estimates as the county regression using $\sum_a \omega_{ac,0}\Delta g_a$ as its outcome. The coefficients also equal those from the actual county outcome when the second component is zero. More generally, the difference in a particular coefficient equals the coefficient from regressing that second component on the same regressors with the same county-event weights. Standard-error equivalence requires compatible clustering and finite-sample adjustments as well.

The reported regressions are complementary specifications. The county outcome compares the average green share in quarters 0 through 2 with its average in quarters $-2$ and $-1$, while the adviser outcome compares quarters $+1$ and $-1$. Adviser financial controls and, in one column, adviser fixed effects also distinguish the specifications. Their similar magnitudes are therefore an empirical comparison rather than an exact regression identity.

# C Appendix to Section 2

TABLE C.1: SCALE AND HOUSEHOLD RELEVANCE OF THE ADVISER ETF CHANNEL

| | All registered advisers | Household-serving advisers |
|---|---|---|
| *Panel A. ETF adoption among registered investment advisers, 2009–2024* | | |
| Registered investment advisers | 8,489 | 4,439 |
| Adviser-quarters holding any ETF (%) | 72.1 | 87.4 |
| Advisers ever holding any ETF (%) | 87.1 | 97.3 |
| Adviser-quarters holding a green ETF (%) | 16.4 | 19.8 |
| Advisers ever holding a green ETF (%) | 36.3 | 45.6 |
| | All single-county advisers | Household-serving single-county advisers |
| *Panel B. Portfolio allocation among single-county ETF holders, 2009–2024* | | |
| ETF value / observed 13F value (%) | 8.1 | 17.7 |
| ETF value / (ETF + directly held stocks) (%) | 8.4 | 18.3 |
| Median adviser-quarter ETF value / observed 13F value (%) | 19.1 | 28.1 |
| Adviser-quarters with ETF value above directly held stocks (%) | 34.8 | 39.6 |
| Green-ETF value / ETF value (%) | 0.9 | 1.1 |
| Green-ETF value / observed 13F value (%) | 0.07 | 0.19 |
| | Share of total market (%) | Share of observed Form 13F value (%) |
| *Panel C. Share of the U.S. ETF market by reporting sector, 2024Q4 ($10,294.1 billion)* | | |
| All registered advisers | 41.1 | 67.0 |
| Household-serving single-county advisers | 8.1 | 13.1 |
| Other registered advisers | 33.1 | 53.8 |
| Other Form 13F institutions | 20.3 | 33.0 |
| Households and other non-13F holders, residual | 38.6 | — |
| *Panel D. Share of the U.S. green-ETF market by reporting sector, 2024Q4 ($91.7 billion)* | | |
| All registered advisers | 45.2 | 71.6 |
| Household-serving single-county advisers | 12.8 | 20.3 |
| Other registered advisers | 32.4 | 51.3 |
| Other Form 13F institutions | 17.9 | 28.4 |
| Households and other non-13F holders, residual | 36.9 | — |

*Notes.* This table reports the scale of the adviser ETF channel and its relevance for households. Advisers serving predominantly household accounts have a majority of individual clients; single-county advisers are represented in one U.S. county. Panel A covers registered investment advisers, including relying advisers, during 2009–2024. Panel B covers adviser-quarters with positive ETF holdings; aggregate ratios pool market values, while the median is across adviser-quarters. In Panels C and D, the first column uses December 2024 CRSP ETF market capitalization and the second uses CRSP-matched ETF value reported on Form 13F. Residuals equal market value not observed on Form 13F. Green ETFs follow the paper's classification.

TABLE C.2: SCALE OF THE REGISTERED-ADVISER SAMPLE AT THE END OF 2024

| | All registered advisers | Household-serving, single-county |
|---|---|---|
| Regulatory assets under management ($ billions) | 73,545.3 | 5,319.3 |
| Share of all registered advisers' regulatory assets (%) | 100.0 | 7.2 |
| Client accounts (millions) | 49.75 | 12.25 |
| ETF holdings ($ billions) | 3,957.5 | 828.7 |
| Green-ETF holdings ($ billions) | 38.2 | 11.7 |

*Notes.* This table reports the aggregate scale of the two adviser groups of Table 1 in the fourth quarter of 2024. Regulatory assets under management cover all client types and are not household assets. Client accounts are not households, since one household may hold several accounts. ETF holdings are Form 13F ETF positions valued at the last December CRSP price of the year. Green ETFs follow the paper's classification.

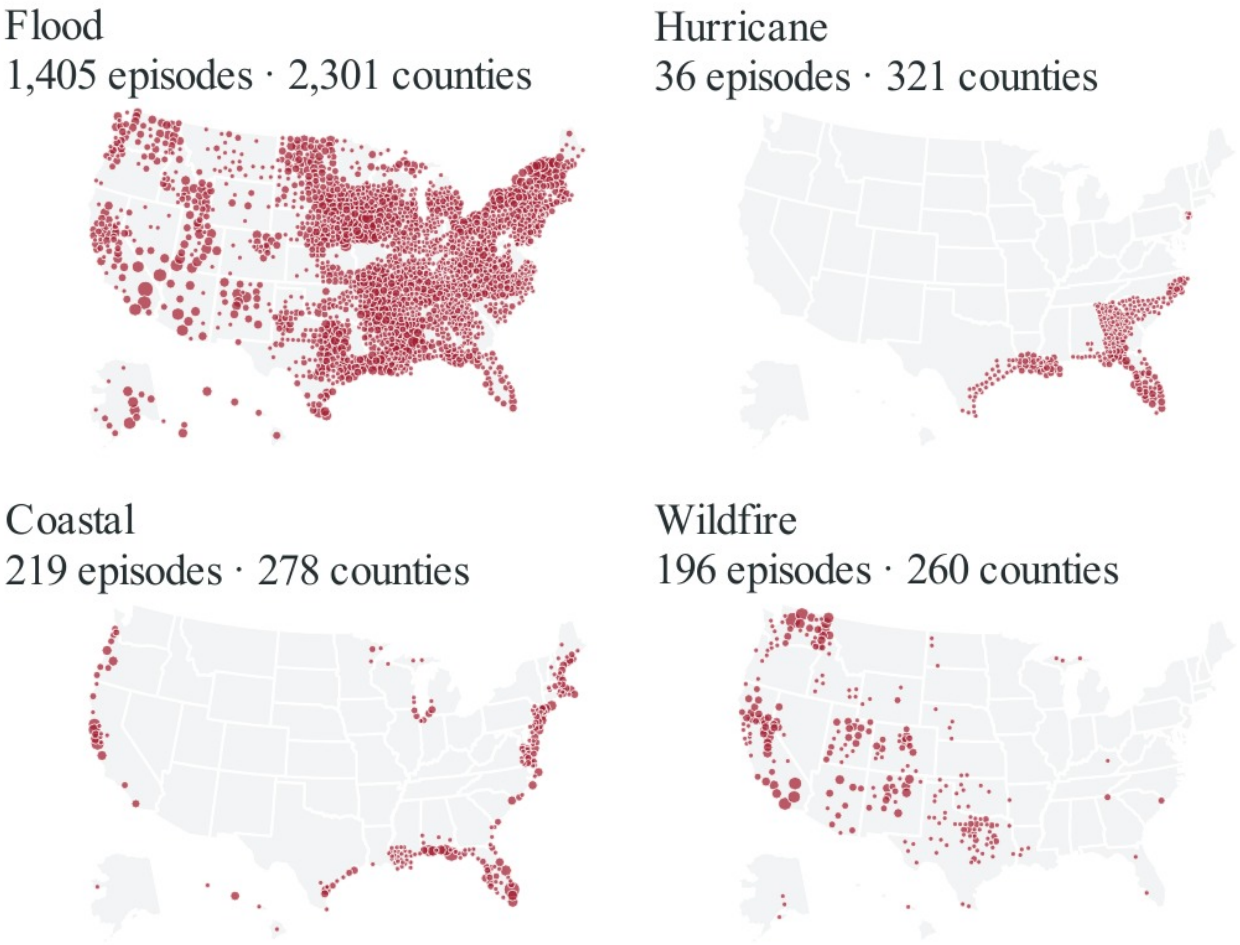



FIGURE C.1: GEOGRAPHIC DISTRIBUTION OF WITHIN-HAZARD TOP-SALIENCE EPISODES

*Notes.* This figure maps the U.S. counties affected by climate-hazard episodes beginning in 2009–2024. Point area is the county-level number of episodes. The maps keep episodes ranking in the top 15 percent of salience within their hazard type.

TABLE C.3: ADVISER GREEN-ETF PORTFOLIO RESPONSE

| | *Dependent variable:* | | | | | |
|---|---|---|---|---|---|---|
| | Green-ETF share response (bp) | | | Entry into green ETFs (pp) | | |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| *Panel A. Advisers weighted by their share of the county's ETF holdings* | | | | | | |
| Affected × Interpretation | 23.21*** | 27.19*** | 26.81*** | 5.20** | 5.06** | 6.07** |
| | (7.76) | (8.25) | (8.94) | (2.23) | (2.53) | (2.41) |
| Affected × Attention | 10.68** | 11.02* | 12.53* | −.42 | .84 | 1.15 |
| | (5.18) | (5.98) | (6.94) | (1.11) | (1.14) | (1.17) |
| Total Interpretation effect among affected advisers | 22.81*** | 26.78*** | 26.57*** | 5.27** | 5.13** | 6.20** |
| | (7.73) | (8.24) | (9.00) | (2.20) | (2.50) | (2.40) |
| Observations | 465,701 | 465,701 | 465,641 | 350,373 | 350,373 | 350,322 |
| $R^2$ | .090 | .090 | .132 | .196 | .196 | .375 |
| Affected adviser-episodes | 1,156 | 1,156 | 1,096 | 857 | 857 | 806 |
| *Panel B. Advisers weighted equally* | | | | | | |
| Affected × Interpretation | 14.55** | 14.12** | 15.06** | 2.52* | 3.32** | 4.00** |
| | (6.97) | (6.83) | (7.52) | (1.30) | (1.43) | (1.61) |
| Affected × Attention | 15.30*** | 16.71*** | 20.55*** | −.58 | .27 | .64 |
| | (3.76) | (4.02) | (4.83) | (.95) | (1.07) | (1.24) |
| Total Interpretation effect among affected advisers | 14.37** | 13.95** | 14.65* | 2.45* | 3.24** | 3.94** |
| | (6.90) | (6.77) | (7.46) | (1.31) | (1.44) | (1.61) |
| Observations | 465,701 | 465,701 | 465,641 | 350,373 | 350,373 | 350,322 |
| $R^2$ | .018 | .018 | .155 | .075 | .075 | .346 |
| Affected adviser-episodes | 1,156 | 1,156 | 1,096 | 857 | 857 | 806 |
| Episode and quarter FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| County FE | ✓ | ✓ | | ✓ | ✓ | |
| Adviser FE | | | ✓ | | | ✓ |
| Affected × hazard × year FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Financial and demographic controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| County damages, fatalities and injuries | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Political and belief controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Lagged climate attention | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Affected × controls | | ✓ | ✓ | | ✓ | ✓ |

*Notes.* See notes to Table 2. The unit is the adviser-episode. In columns 1–3 the outcome is the change in the adviser green-ETF share from the quarter before the episode to the quarter after it, at pre-event prices, in basis points. In columns 4–6 the sample is restricted to adviser-episodes with no green ETF in the quarter before the episode, and the outcome is 100 times an indicator for a positive green-ETF weight in the quarter after it, at pre-event prices (a linear probability model; a positive weight is a weight above $10^{-8}$). Panel A weights each adviser-episode by its share of the county's ETF holdings in the quarter before the episode; Panel B weights advisers equally. Financial controls are the logs of the adviser's ETF holdings and of its total Form 13F holdings in the quarter before the episode; Affected, Interpretation, Attention and the remaining controls are those of the county. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

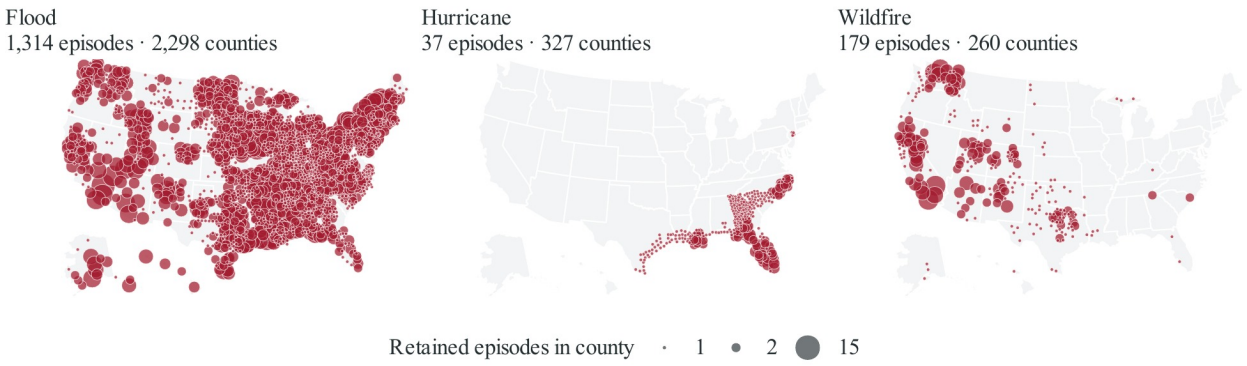



FIGURE C.2: GEOGRAPHIC DISTRIBUTION OF SELECTED FLOOD, HURRICANE, AND WILDFIRE EPISODES
*Notes.* This figure maps the counties affected by the flood, hurricane, and wildfire episodes in the non-overlapping analysis samples, 2009–2024. Point area is the number of episodes in each county. The three panels share one sample definition.

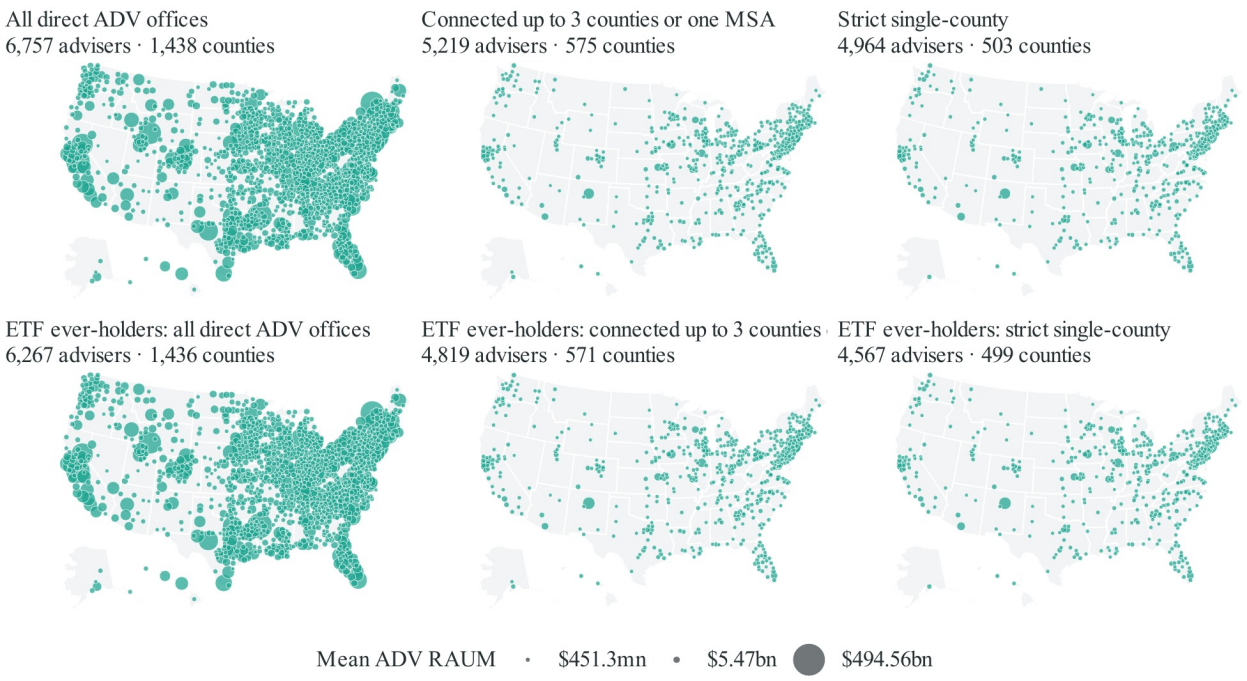
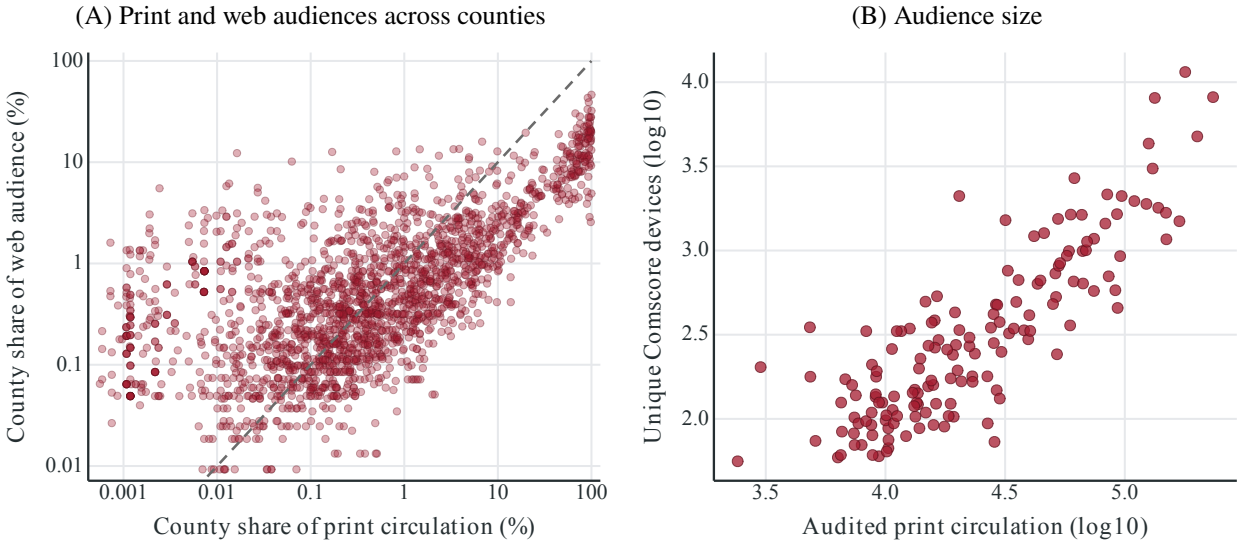



FIGURE C.3: REGISTERED INVESTMENT-ADVISER GEOGRAPHY AND ETF PARTICIPATION

*Notes.* This figure maps the direct Form ADV principal or branch offices observed in 2009–2024 of full registered investment advisers, excluding hedge-fund managers and exempt reporting advisers. From left to right, the top row shows all direct Form ADV offices, footprints contained in at most three connected counties or one metropolitan statistical area, and strict single-county footprints. The bottom row repeats those panels for advisers that held a positive ETF position at least once during 2009–2024. Point area is mean Form ADV regulatory assets under management within each county and panel.

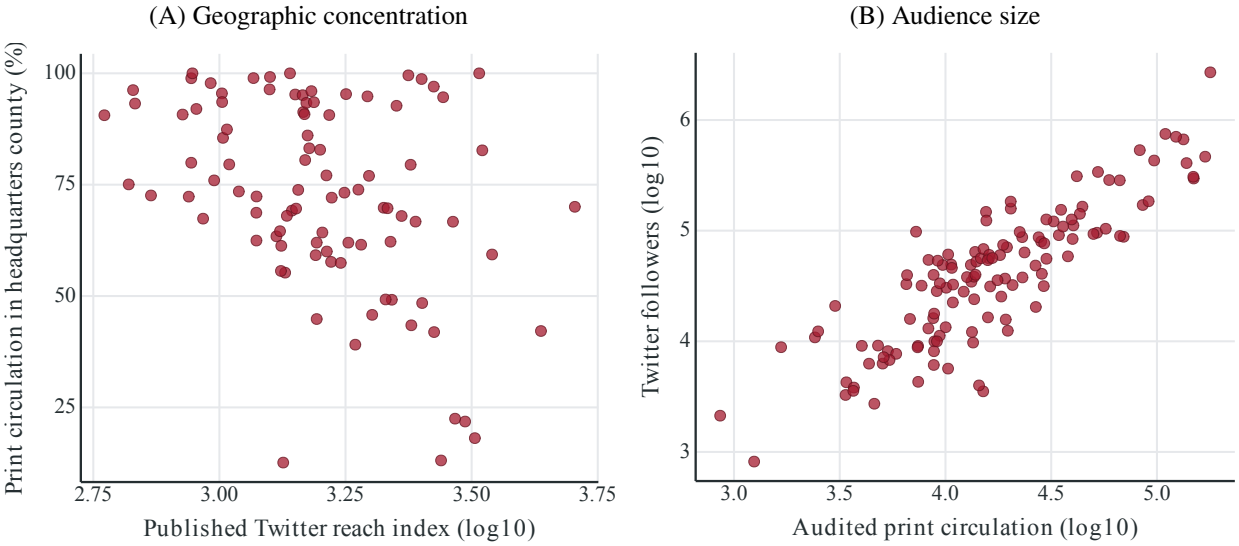
# D Appendix to Section 2.2

FIGURE D.1: PRINT CIRCULATION AND OBSERVED WEB AUDIENCES



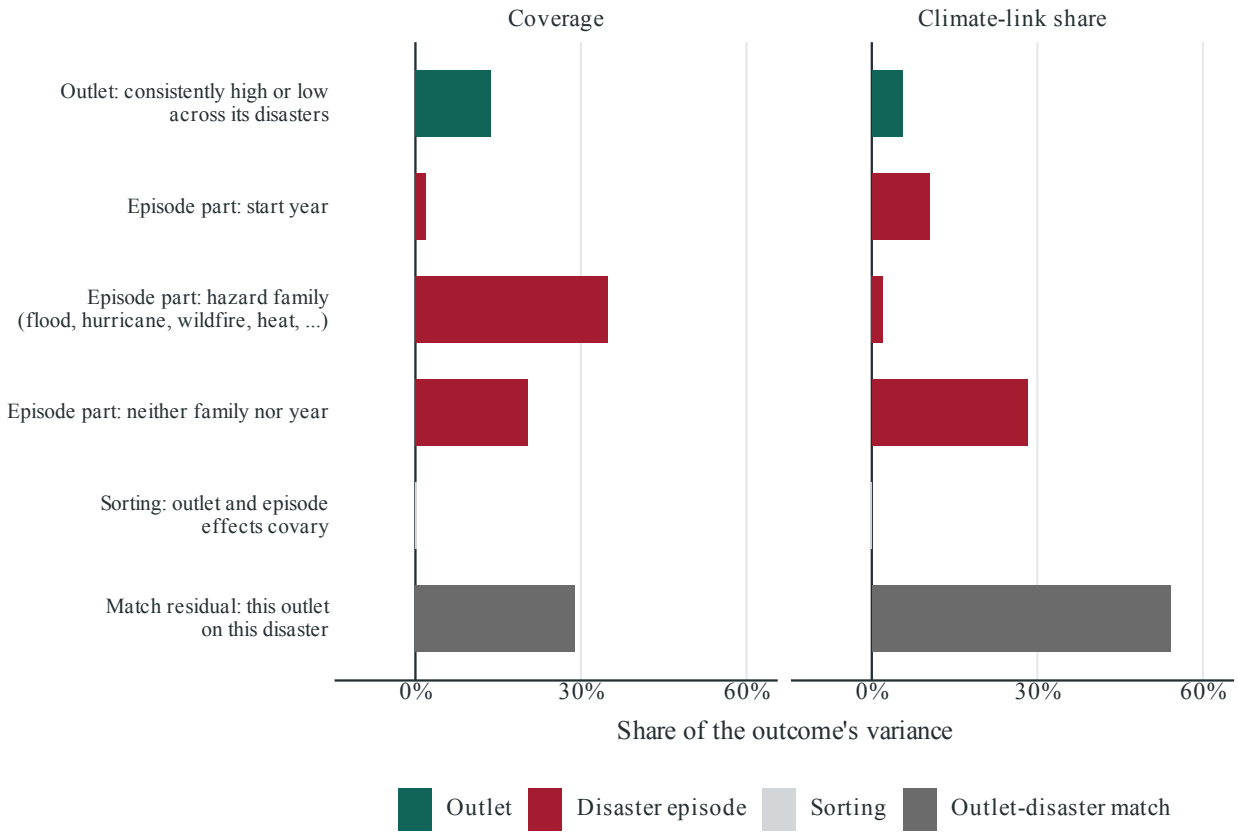

*Notes.* This figure plots observed Comscore web audiences against independently audited print circulation for 153 local newspapers. Panel A displays 2,114 newspaper–county pairs with positive observations in both sources on logarithmic axes labelled in percentages. Each share divides the county's audience by that newspaper's audience across all observed counties in the respective source; shares are computed before restricting to common counties. The dashed line denotes equal shares. The Pearson correlation of the untransformed shares is 0.822 and the Spearman correlation is 0.628. Panel B compares audited circulation with distinct outlet-wide Comscore devices (Spearman correlation 0.823). Comscore covers 2020; print averages available audits during 2015–2025. Newspapers require at least 50 unique web devices. Ambiguous shared print-audit mappings and one newspaper with invalid negative audited circulation are excluded. Web observations count distinct panel devices within each newspaper–county pair over the year, regardless of visit frequency; a device can appear in multiple counties. Common counties account for an average 96.1% of observed print circulation and 25.5% of observed county web-device counts per newspaper.

FIGURE D.2: PRINT CIRCULATION AND TWITTER AUDIENCES




*Notes.* This figure plots published Twitter audience measures against audited print circulation for matched local publications. Panel A compares the published population-reach index of Le Quéré et al. (2022) with the share of observed audited circulation in the newspaper's headquarters county (93 newspapers; Spearman correlation −0.360). Higher index values indicate a geographically broader Twitter audience. Panel B compares total Twitter followers with total observed audited print circulation (127 newspapers; Spearman correlation 0.839). Each newspaper receives equal weight. Twitter counts are recorded in 2020 or 2022; the print footprint averages available audits during 2015–2025. The county-share denominator is the newspaper's observed audited circulation across all available counties. Ambiguous publication matches, conflicting headquarters locations in Panel A, and one publication with invalid negative audited circulation are excluded.

FIGURE D.3: VARIANCE DECOMPOSITION OF COVERAGE AND CLIMATE INTERPRETATION



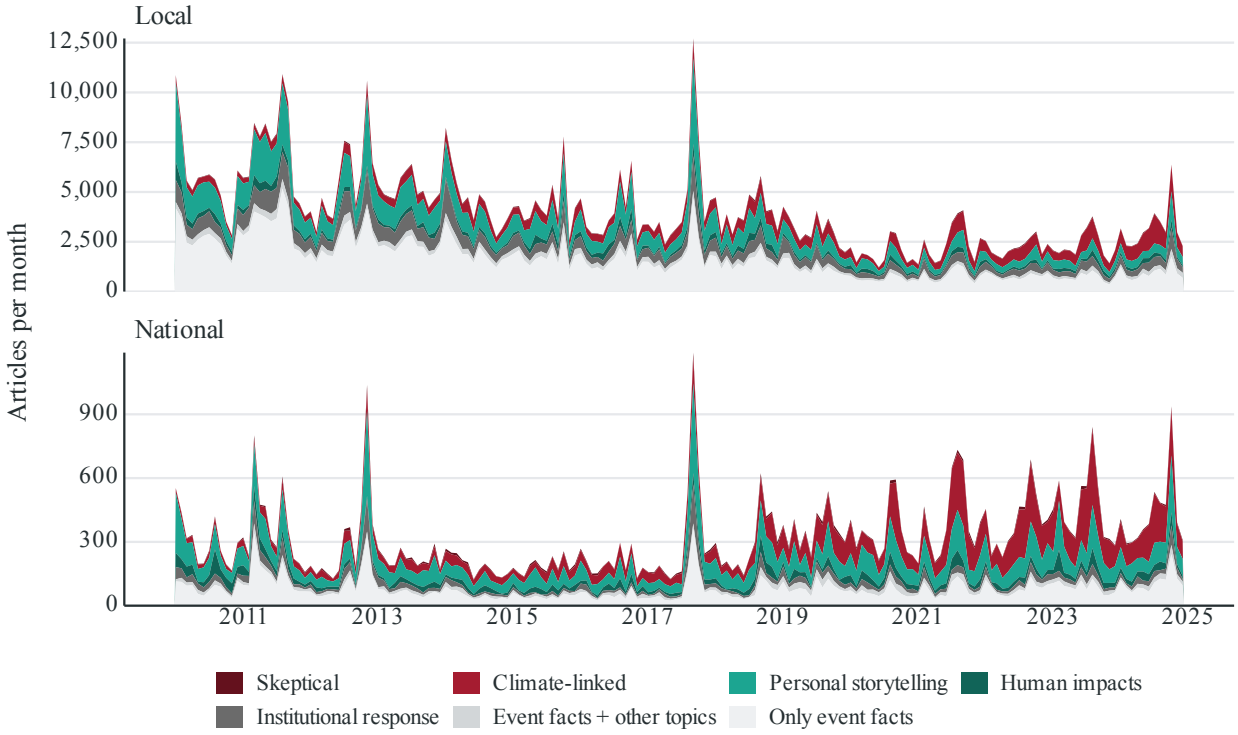

*Notes.* This figure decomposes the variance of each outcome across outlet–disaster pairs, in weeks 0 to 3 from the disaster start. Coverage is the log of one plus the number of articles on the disaster, including outlets with no article. Climate-link share is the share of these articles that link the disaster to climate change, among outlets with at least one article. Bars are shares of total variance.

TABLE D.1: COMPOSITION OF DISASTER COVERAGE: CHAIN-OWNED AND INDEPENDENT LOCAL OUTLETS

| | *Dependent variable: article belongs to the category (percent)* | | | | | | |
|---|---|---|---|---|---|---|---|
| | Skeptical | Climate-linked | Personal storytelling | Human impacts | Institutional response | Event facts + other topics | Only event facts |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
| *Panel A. No fixed effects* | | | | | | | |
| Local, chain-owned | −0.66*** | −4.22** | 0.31 | 1.01* | 0.24 | −0.85 | 4.16* |
| | (0.24) | (2.10) | (2.00) | (0.57) | (0.91) | (1.52) | (2.15) |
| Constant | 1.54 | 21.15 | 20.64 | 4.90 | 12.86 | 6.55 | 32.36 |
| | (0.23) | (2.07) | (1.98) | (0.54) | (0.86) | (1.51) | (1.97) |
| Observations | 383,872 | 383,872 | 383,872 | 383,872 | 383,872 | 383,872 | 383,872 |
| $R^2$ | .0003 | .0009 | .0000 | .0001 | .0000 | .0001 | .0005 |
| *Panel B. Calendar-quarter fixed effects* | | | | | | | |
| Local, chain-owned | −0.67*** | −3.83** | 0.30 | 1.02* | 0.26 | −0.90 | 3.81* |
| | (0.23) | (1.66) | (1.76) | (0.58) | (0.90) | (1.51) | (2.14) |
| Observations | 383,872 | 383,872 | 383,872 | 383,872 | 383,872 | 383,872 | 383,872 |
| $R^2$ | .0016 | .0333 | .0103 | .0046 | .0021 | .0019 | .0093 |
| Calendar-quarter FEs | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

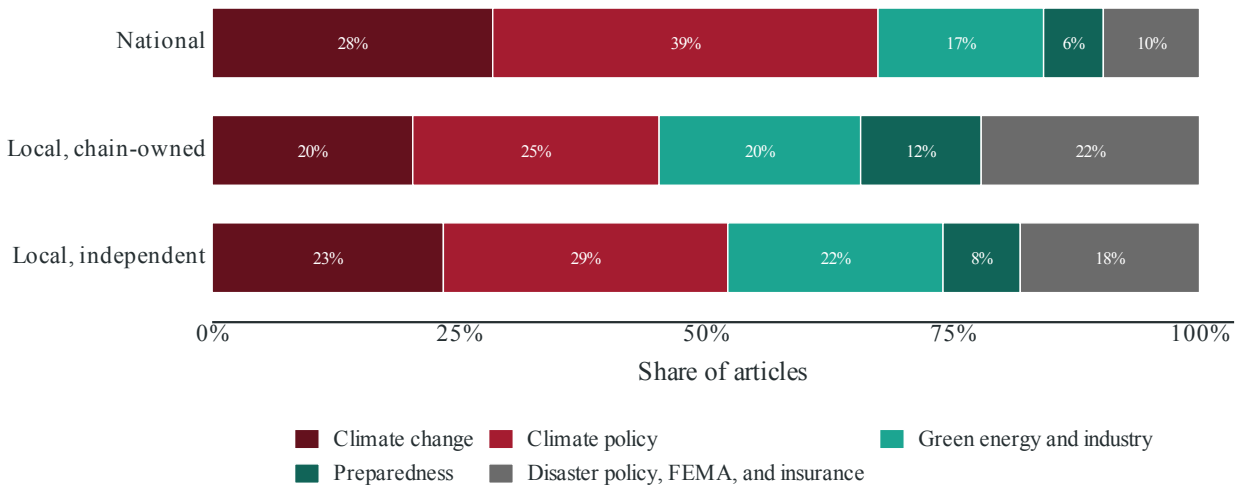
*Notes.* This table reports regressions of the category of each disaster article published by a local outlet from 2015 through 2024 on the outlet's ownership, one regression per column. The dependent variable equals 100 when the article belongs to the category named in the column header and zero otherwise; the categories are those of Figure 2 and are mutually exclusive. Local, chain-owned equals one when the outlet's owner in the quarter of publication holds two or more newspaper titles, and zero when it holds one, so the constant in Panel A is the category's share among independent outlets. Panel B adds calendar-quarter fixed effects. The sample has 370 outlets. Standard errors, in parentheses, are clustered by outlet. Stars are reported for the chain coefficient only. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

FIGURE D.4: MONTHLY COMPOSITION OF DISASTER COVERAGE




*Notes.* See notes to Figure 2. This figure plots monthly counts of disaster articles by category from 2010 through 2024, counting an article once for each outlet that publishes it, and the local and national panels use separate vertical scales.

FIGURE D.5: CLIMATE AND DISASTER-RELATED COVERAGE BEYOND INDIVIDUAL EVENTS BY OUTLET OWNERSHIP




*Notes.* This figure plots the composition of the climate- and disaster-related articles from 2015 through 2024 that are not centered on a disaster event, for national outlets, chain-owned local outlets and independent local outlets; such articles may still mention disasters. Shares are calculated among 413,255 articles, of which 47,911 are national, 324,050 chain-owned local and 41,294 independent local. Outlet classes are defined in the notes to Figure 2. Categories are mutually exclusive, and disaster policy, FEMA and insurance are combined as institutional management of disaster risk.

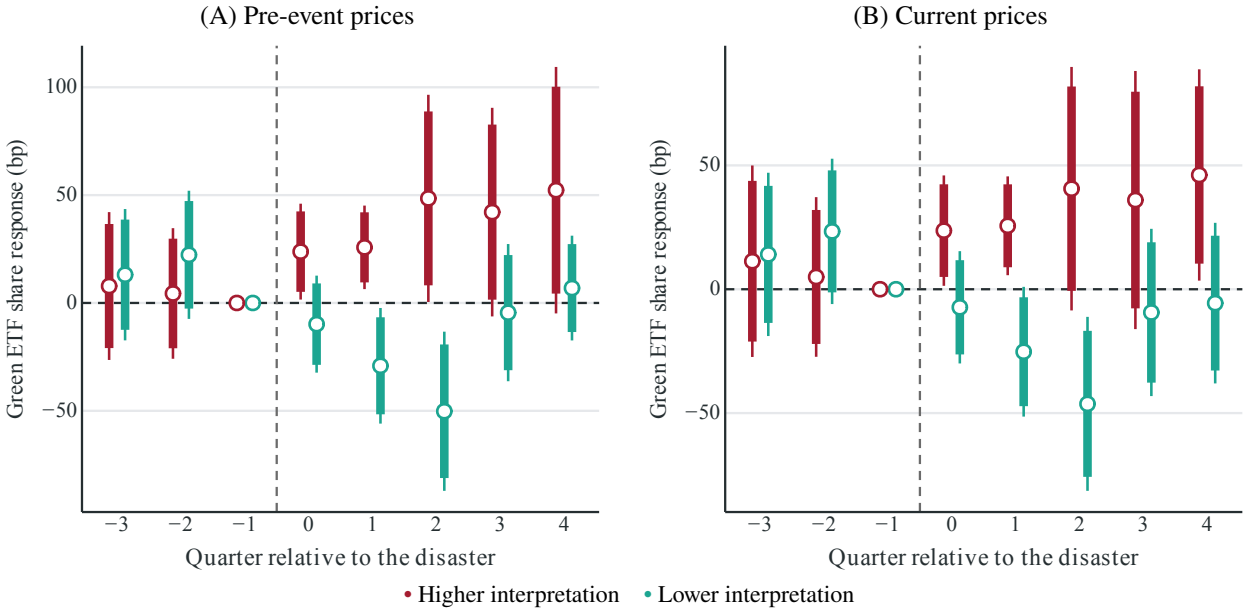
# E Appendix to Section 3

TABLE E.1: COUNTY GREEN-ETF PORTFOLIO RESPONSE

| | *Dependent variable:* | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Green-ETF share response (basis points) | | | | | | | | | |
| | Pre-event prices | | | | | Current prices | | | | |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) | (10) |
| Affected × Interpretation | 25.48** | 25.86** | 22.46* | 30.31** | 22.76* | 25.87** | 25.58** | 22.53* | 29.50** | 22.55* |
| | (10.13) | (12.33) | (12.18) | (12.58) | (12.44) | (10.46) | (12.50) | (12.25) | (12.62) | (12.28) |
| Affected × Attention | .86 | 4.91 | 3.64 | 1.69 | −9.36 | .46 | 4.55 | 3.40 | 1.79 | −9.16 |
| | (10.93) | (10.92) | (10.30) | (10.06) | (10.01) | (11.17) | (10.84) | (10.24) | (10.08) | (10.11) |
| Total Interpretation effect among affected counties | 26.70** | 27.08** | 23.47* | 31.32** | 22.73* | 27.02** | 26.73** | 23.43* | 30.43** | 22.54* |
| | (10.45) | (12.44) | (12.26) | (12.53) | (12.31) | (10.75) | (12.61) | (12.32) | (12.58) | (12.16) |
| Observations | 105,662 | 105,662 | 105,662 | 105,662 | 105,662 | 105,662 | 105,662 | 105,662 | 105,662 | 105,662 |
| $R^2$ | .028 | .029 | .031 | .032 | .102 | .030 | .030 | .033 | .033 | .106 |
| Affected county-episodes | 244 | 244 | 244 | 244 | 244 | 244 | 244 | 244 | 244 | 244 |
| Episode and quarter FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| State FE | ✓ | ✓ | ✓ | ✓ | | ✓ | ✓ | ✓ | ✓ | |
| County FE | | | | | ✓ | | | | | ✓ |
| Affected × hazard × year FE | | ✓ | ✓ | ✓ | ✓ | | ✓ | ✓ | ✓ | ✓ |
| Financial and demographic controls | | | ✓ | ✓ | ✓ | | | ✓ | ✓ | ✓ |
| County damages, fatalities and injuries | | | ✓ | ✓ | ✓ | | | ✓ | ✓ | ✓ |
| Political and belief controls | | | | ✓ | ✓ | | | | ✓ | ✓ |
| Lagged climate attention | | | | ✓ | ✓ | | | | ✓ | ✓ |
| Affected × controls | | | | ✓ | ✓ | | | | ✓ | ✓ |

*Notes.* This table reports estimates of the county green-ETF share response to disasters. The outcome is the average county green-ETF share over the quarter of the episode and the two following quarters minus its average over the two quarters before, in basis points, at pre-event prices, held at their level in the quarter before the episode (columns 1–5), and at current prices (columns 6–10). The sample covers high-salience flood, hurricane and coastal episodes, 2015–2024. Affected is one for counties in the episode footprint. Interpretation is the climate-interpreted share of local coverage of the episode and Attention the log volume of that coverage, both over the first four weeks of the episode and standardized; one standard deviation of Interpretation is 15.7 percentage points. Each regression also includes Interpretation and Attention and, in columns 1 and 6, Affected; their coefficients are not reported. The total Interpretation effect among affected counties is the sum of the Interpretation coefficient and its Affected interaction, with its standard error from their joint covariance. Financial controls are lagged log portfolio assets; demographic controls are income, college share, white share, population and population density; political and belief controls are the prior Democratic vote share and the county share worried about global warming; lagged climate attention is the share of local coverage on climate policy and green topics in the quarter before the episode. Standard errors, in parentheses, are clustered by county. A nonparametric bootstrap that resamples counties with replacement (999 draws) gives $p = .036$ for Affected × Interpretation in column 4 and $p = .032$ in column 9. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

FIGURE E.1: DISASTER RESPONSE OF COUNTY GREEN ETF SHARES BY LOCAL INTERPRETATION, BY QUARTER




*Notes.* This figure plots the conditional disaster responses of the county green ETF share for counties with local interpretation above and below the median, by quarter relative to the flood, evaluated at a common reference. Panel A uses pre-event prices and Panel B current prices. Changes are relative to quarter −1, normalized to zero. The regressions include calendar-quarter and disaster-episode fixed effects and Affected × year terms. Controls include lagged assets, prior politics, demographics, continuous interpretation and attention. Thin and thick bars are 95 and 90 percent confidence intervals, with two-way clustering by county and disaster episode. The joint test that the gap between the two groups is zero in quarters −4 to −2 gives $p = 0.463$ (A) and 0.397 (B).

TABLE E.2: LOCAL CLIMATE INTERPRETATION AND GREEN ETF DEMAND AFTER FLOODS, CALENDAR-QUARTER TIMING

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| *Panel A. Green ETF share change at pre-event prices (basis points)* | | | | | |
| Affected × Interpretation (1 SD) | 42.07** | 43.89** | 46.76** | 46.76** | 47.92** |
| | (18.14) | (18.76) | (20.51) | (20.50) | (20.67) |
| Interpretation (1 SD) | −6.71 | −6.87 | −7.47 | −7.46 | −7.20 |
| | (4.23) | (4.25) | (4.38) | (4.39) | (4.34) |
| Affected | 13.63 | 15.61 | 9.04 | 9.04 | 12.86 |
| | (27.20) | (26.65) | (27.05) | (27.06) | (28.18) |
| $R^2$ | .094 | .103 | .105 | .105 | .106 |
| Mean dependent variable (bp) | 1.06 | 1.06 | 1.06 | 1.06 | 1.06 |
| *Panel B. Green ETF share change at current prices (basis points)* | | | | | |
| Affected × Interpretation (1 SD) | 42.62** | 44.47** | 47.43** | 47.41** | 48.92** |
| | (18.19) | (18.78) | (20.58) | (20.56) | (20.84) |
| Interpretation (1 SD) | −6.74 | −6.89 | −7.51 | −7.45 | −7.19 |
| | (4.20) | (4.21) | (4.34) | (4.35) | (4.30) |
| Affected | 13.51 | 15.53 | 8.85 | 8.86 | 12.55 |
| | (27.84) | (27.29) | (27.60) | (27.54) | (28.55) |
| $R^2$ | .098 | .107 | .109 | .109 | .111 |
| Mean dependent variable (bp) | 1.21 | 1.21 | 1.21 | 1.21 | 1.21 |
| Share observations | 77,571 | 77,571 | 77,571 | 77,571 | 77,571 |
| Counties | 319 | 319 | 319 | 319 | 319 |
| Disaster episodes | 632 | 632 | 632 | 632 | 632 |
| County fixed effects | ✓ | ✓ | ✓ | ✓ | ✓ |
| Calendar-quarter fixed effects | ✓ | ✓ | ✓ | ✓ | ✓ |
| Disaster-episode fixed effects | ✓ | ✓ | ✓ | ✓ | ✓ |
| Affected × year effects | ✓ | ✓ | ✓ | ✓ | ✓ |
| Prior green-share change | | ✓ | ✓ | ✓ | ✓ |
| Lagged portfolio assets | | | ✓ | ✓ | ✓ |
| Predetermined political controls | | | | ✓ | ✓ |
| Predetermined demographic controls | | | | | ✓ |

*Notes.* See notes to Table E.3. Climate interpretation is measured over the event's calendar quarter. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

TABLE E.3: COUNTY GREEN-ETF PORTFOLIO RESPONSE, 2018–2024 SAMPLE

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| *Panel A. Pre-event prices* | | | | | |
| Affected × Interpretation | 36.01*** (13.20) | 38.71*** (13.06) | 38.73*** (13.05) | 38.80*** (12.99) | 43.05*** (13.26) |
| Affected | -8.88 (48.92) | -6.11 (47.97) | -13.74 (48.64) | -14.04 (48.10) | -16.39 (50.19) |
| Interpretation | -4.50* (2.44) | -4.62* (2.50) | -4.66* (2.47) | -4.59* (2.47) | -4.08* (2.44) |
| Attention | -12.72** (5.48) | -13.26** (5.58) | -12.66** (5.57) | -12.68** (5.58) | -12.78** (5.85) |
| Affected × Attention | 33.21* (17.64) | 34.13* (18.58) | 33.94* (19.16) | 34.16* (19.11) | 34.82* (20.09) |
| Observations | 59,330 | 59,330 | 59,330 | 59,330 | 59,330 |
| *Panel B. Current prices* | | | | | |
| Affected × Interpretation | 34.78*** (13.12) | 37.46*** (12.97) | 37.48*** (12.96) | 37.62*** (12.85) | 42.56*** (13.39) |
| Affected | -10.64 (51.07) | -7.89 (50.11) | -15.61 (50.71) | -16.16 (50.04) | -17.83 (51.66) |
| Interpretation | -4.81** (2.43) | -4.93** (2.48) | -4.97** (2.46) | -4.85* (2.46) | -4.34* (2.41) |
| Attention | -13.39** (5.55) | -13.92** (5.65) | -13.31** (5.64) | -13.35** (5.65) | -13.52** (5.91) |
| Affected × Attention | 34.30* (18.55) | 35.22* (19.46) | 35.02* (20.05) | 35.43* (20.01) | 35.76* (20.93) |
| Observations | 59,330 | 59,330 | 59,330 | 59,330 | 59,330 |
| County fixed effects | Yes | Yes | Yes | Yes | Yes |
| Quarter and episode fixed effects | Yes | Yes | Yes | Yes | Yes |
| Affected × year effects | Yes | Yes | Yes | Yes | Yes |
| Prior green-share change | | Yes | Yes | Yes | Yes |
| Lagged log portfolio assets | | | Yes | Yes | Yes |
| Political controls | | | | Yes | Yes |
| Demographic controls | | | | | Yes |

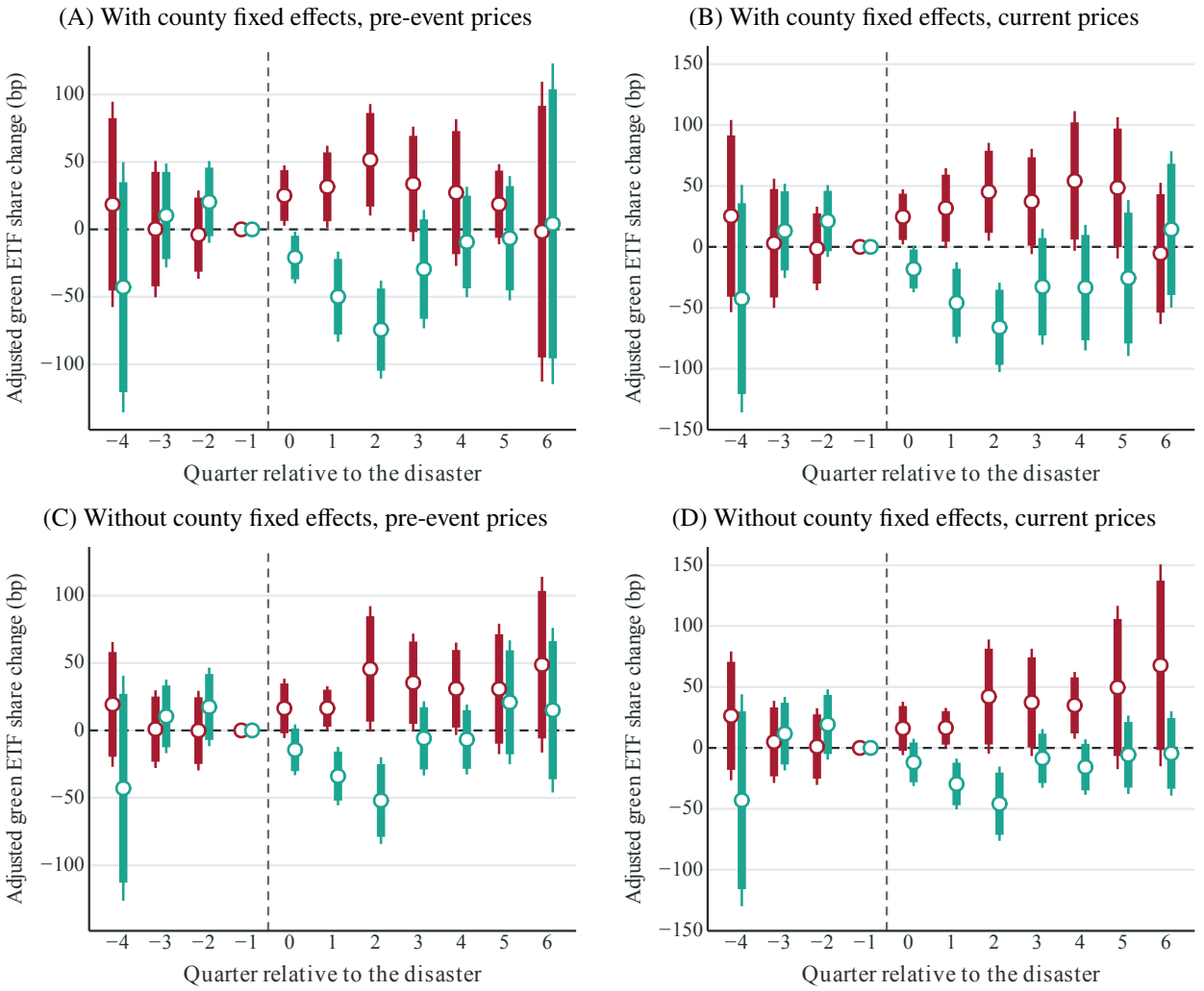
*Notes.* This table reports estimates of the county green-ETF share response on the 2018–2024 sample. The outcome is the change in green-ETF share from $t-1$ to $t+1$, in basis points, at pre-event prices (Panel A) and current prices (Panel B). Interpretation and log total disaster-news attention are measured over event weeks 0–3 and standardized. The sample covers high-salience flood, hurricane and coastal episodes, 2018Q1–2024Q4. Standard errors, in parentheses, are clustered by county and episode. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

TABLE E.4: ADVISER-LEVEL LOCAL CLIMATE INTERPRETATION AND GREEN ETF DEMAND AFTER FLOODS, CALENDAR-QUARTER TIMING

| | (1) | (2) | (3) | (4) |
|---|---|---|---|---|
| *Panel A. Green ETF share change at pre-event prices (basis points)* | | | | |
| Affected × Interpretation (1 SD) | 10.41* | 10.11* | 10.05* | 10.07* |
| | (5.39) | (5.68) | (5.69) | (5.67) |
| Interpretation (1 SD) | −1.05 | −1.19 | −1.17 | −1.32 |
| | (1.52) | (1.53) | (1.54) | (1.55) |
| Affected | 1.95 | 2.27 | 2.23 | .77 |
| | (4.24) | (4.71) | (4.69) | (4.88) |
| $R^2$ | .069 | .144 | .144 | .144 |
| Mean dependent variable (bp) | −.62 | −.62 | −.62 | −.62 |
| *Panel B. Green ETF share change at current prices (basis points)* | | | | |
| Affected × Interpretation (1 SD) | 10.48** | 10.19* | 10.13* | 10.15* |
| | (5.29) | (5.55) | (5.56) | (5.55) |
| Interpretation (1 SD) | −.84 | −.97 | −.95 | −1.11 |
| | (1.73) | (1.74) | (1.75) | (1.74) |
| Affected | 1.16 | 1.46 | 1.42 | −.15 |
| | (4.47) | (4.90) | (4.89) | (5.15) |
| $R^2$ | .067 | .133 | .133 | .133 |
| Mean dependent variable (bp) | −.58 | −.58 | −.58 | −.58 |
| Share observations | 621,761 | 621,761 | 621,761 | 621,761 |
| Advisers | 2,765 | 2,765 | 2,765 | 2,765 |
| Disaster episodes | 632 | 632 | 632 | 632 |
| Adviser fixed effects | ✓ | ✓ | ✓ | ✓ |
| Calendar-quarter fixed effects | ✓ | ✓ | ✓ | ✓ |
| Disaster-episode fixed effects | ✓ | ✓ | ✓ | ✓ |
| Affected × year effects | ✓ | ✓ | ✓ | ✓ |
| Prior green-share change | | ✓ | ✓ | ✓ |
| Lagged portfolio assets | | | ✓ | ✓ |
| Predetermined political controls | | | | ✓ |

*Notes.* This table reports the adviser-level green-ETF share response to the climate interpretation of local disaster coverage after flood, hurricane and coastal episodes. Interpretation is the circulation-weighted average across local outlets of the outlet-quarter share of disaster-related articles that connect the disaster to climate change. The Interpretation row reports the slope among unaffected observations linked to the episode; the Affected row reports the difference between affected and unaffected observations in the reference year, at mean interpretation and coverage; their interaction is the additional interpretation slope among affected observations. Share outcomes equal $Y_{t+1} - Y_{t-1}$. Panel A holds prices fixed at pre-event values; Panel B uses current prices. Column (1) includes year-specific effects for affected observations. Column (2) adds the previous green-share change; column (3) adds lagged log portfolio assets; column (4) adds predetermined political controls. Adviser-level demographics are not included. All columns include coverage, its interaction with Affected, adviser, calendar-quarter, and disaster-episode fixed effects, and standard errors clustered by adviser location. $R^2$ includes absorbed fixed effects. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

FIGURE E.2: FLOODS, HURRICANES AND COASTAL EPISODES: HIGH- AND LOW-INTERPRETATION AFFECTED COUNTIES




*Notes.* This figure plots the disaster responses of affected counties with high and low interpretation, split at the median of interpretation, which is held fixed across quarters, for each quarter from −4 to +6 relative to the episode, with quarter −1 normalized to zero. The sample covers flood, hurricane and coastal episodes and begins in 2015. Panels A and B include county, calendar-quarter and episode fixed effects, and panels C and D calendar-quarter and episode fixed effects only; panels A and C use pre-event prices and panels B and D current prices. All specifications include Affected × year terms, interpretation, attention, Affected × Attention and the remaining controls, and leave out the previous green-share change. Each group's interpretation is set at the group's weighted mean, and attention, the other controls and the fixed effects at a common reference for all affected counties in each quarter. The number of observations varies across quarters. Thick and thin bars are pointwise 90% and 95% confidence intervals, with standard errors clustered by county and episode. The joint test that the gap between the two series is zero in quarters −4, −3 and −2 has $p = 0.368$ (A), 0.327 (B), 0.468 (C) and 0.439 (D).

TABLE E.5: COUNTY GREEN-HOLDING MARGINS

| | (1) | (2) | (3) | (4) | (5) |
|---|---|---|---|---|---|
| *Share of advisers holding green, all advisers (percentage points)* | | | | | |
| Affected × Interpretation | 5.76*** (1.40) | 5.77*** (1.40) | 5.76*** (1.41) | 5.74*** (1.44) | 5.96*** (1.53) |
| Affected | 3.27 (3.25) | 3.27 (3.23) | 2.87 (3.18) | 2.99 (3.24) | 2.95 (3.02) |
| Interpretation | -0.51* (0.29) | -0.51* (0.29) | -0.51* (0.29) | -0.53* (0.29) | -0.56* (0.29) |
| Attention | -0.54 (0.58) | -0.54 (0.58) | -0.51 (0.59) | -0.50 (0.59) | -0.58 (0.59) |
| Affected × Attention | -0.51 (1.28) | -0.50 (1.29) | -0.51 (1.31) | -0.60 (1.31) | -0.81 (1.23) |
| Observations | 59,326 | 59,326 | 59,326 | 59,326 | 59,326 |
| *Share holding green, continuing in the same county (percentage points)* | | | | | |
| Affected × Interpretation | 5.94*** (1.75) | 5.94*** (1.74) | 5.94*** (1.74) | 5.91*** (1.75) | 5.89*** (1.86) |
| Affected | 2.75 (3.29) | 2.75 (3.26) | 2.50 (3.21) | 2.51 (3.29) | 2.22 (3.06) |
| Interpretation | -0.37 (0.28) | -0.37 (0.28) | -0.37 (0.28) | -0.40 (0.28) | -0.42 (0.28) |
| Attention | 0.16 (0.52) | 0.16 (0.52) | 0.18 (0.52) | 0.19 (0.52) | 0.10 (0.53) |
| Affected × Attention | 0.11 (1.14) | 0.11 (1.15) | 0.10 (1.17) | 0.03 (1.16) | -0.14 (1.09) |
| Observations | 56,693 | 56,693 | 56,693 | 56,693 | 56,693 |
| *County entry (percentage points)* | | | | | |
| Affected × Interpretation | 1.64 (2.15) | 1.55 (2.27) | 1.46 (2.28) | 1.42 (2.28) | 0.99 (1.95) |
| Affected | 4.22 (9.66) | 4.38 (9.67) | 4.83 (10.02) | 5.52 (10.37) | 3.66 (9.68) |
| Interpretation | 0.53 (0.48) | 0.54 (0.49) | 0.54 (0.49) | 0.51 (0.48) | 0.52 (0.49) |
| Attention | 0.54 (1.40) | 0.54 (1.40) | 0.52 (1.39) | 0.51 (1.38) | 0.60 (1.38) |
| Affected × Attention | -1.24 (2.76) | -1.40 (2.96) | -1.46 (2.95) | -1.71 (2.93) | -2.02 (2.80) |
| Observations | 34,389 | 34,389 | 34,389 | 34,389 | 34,389 |
| *County incumbent share (basis points)* | | | | | |
| Affected × Interpretation | 41.51** (16.00) | 42.35*** (15.80) | 32.62** (13.48) | 32.26** (13.66) | 44.82** (17.77) |
| Affected | -66.71 (59.78) | -57.57 (58.65) | -75.47** (37.51) | -86.67** (41.88) | -102.07* (56.50) |
| Interpretation | -10.26** (4.73) | -10.27** (4.97) | -11.06** (5.00) | -11.41** (5.03) | -10.52** (5.08) |
| Attention | -26.04** (10.28) | -27.23** (10.86) | -26.75** (11.39) | -26.32** (11.44) | -22.14* (12.43) |
| Affected × Attention | 55.87*** (17.57) | 53.63*** (18.19) | 45.54** (18.89) | 46.44** (19.12) | 43.79** (21.09) |
| Observations | 24,934 | 24,934 | 24,934 | 24,934 | 24,934 |
| County fixed effects | Yes | Yes | Yes | Yes | Yes |
| Quarter and episode fixed effects | Yes | Yes | Yes | Yes | Yes |
| Affected × year effects | Yes | Yes | Yes | Yes | Yes |
| Prior green-share change | | Yes | Yes | Yes | Yes |
| Lagged log portfolio assets | | | Yes | Yes | Yes |
| Political controls | | | | Yes | Yes |
| Demographic controls | | | | | Yes |

*Notes.* This table reports the response of the county green-holding margins. Outcomes are changes from $t-1$ to $t+1$ at pre-event prices. Interpretation and log total disaster-news attention are measured over event weeks 0–3. The sample covers high-salience flood, hurricane and coastal episodes, 2018Q1–2024Q4. Entry is conditional on no green holdings before the event; incumbent changes are conditional on positive green holdings before the event. Share-of-adviser outcomes use the advisers observed at both $t-1$ and $t+1$. Standard errors, in parentheses, are clustered by county and episode. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

TABLE E.6: EVENT TIMING AND THE PORTFOLIO RESPONSE

| | Green-ETF share response (basis points) | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | Without affected × hazard × year FE | | | | Table 2, column 4 | | | |
| | 0–30 days | | 31–91 days | | 0–30 days | | 31–91 days | |
| | Episode quarter (1) | Next quarter (2) | Episode quarter (3) | Next quarter (4) | Episode quarter (5) | Next quarter (6) | Episode quarter (7) | Next quarter (8) |
| *Panel A. Pre-event prices* | | | | | | | | |
| Affected × Interpretation | 13.87 | 27.23 | 28.18* | 38.92** | 8.74 | 22.31* | 25.85** | 48.17** |
| | (8.73) | (16.59) | (14.38) | (19.10) | (6.35) | (13.18) | (12.17) | (20.02) |
| Affected × Attention | 7.30 | 7.48 | −5.01 | −15.36 | 10.33 | 6.84 | −3.48 | −16.51 |
| | (7.65) | (12.49) | (10.02) | (10.27) | (6.53) | (11.16) | (12.41) | (10.33) |
| Total Interpretation effect among affected counties | 16.22* | 30.17* | 27.35* | 38.57** | 11.08* | 25.24** | 25.02** | 47.80** |
| | (8.49) | (15.61) | (14.42) | (19.21) | (5.96) | (11.99) | (12.21) | (19.73) |
| *Panel B. Current prices* | | | | | | | | |
| Affected × Interpretation | 13.15 | 25.55 | 27.78* | 38.85** | 8.23 | 21.48* | 25.36** | 47.81** |
| | (8.72) | (16.23) | (14.13) | (18.73) | (6.43) | (12.69) | (11.87) | (19.18) |
| Affected × Attention | 6.76 | 6.07 | −4.20 | −14.53 | 9.65 | 5.01 | −2.95 | −15.52 |
| | (7.45) | (11.66) | (9.68) | (10.19) | (6.44) | (10.40) | (12.23) | (10.06) |
| Total Interpretation effect among affected counties | 15.35* | 28.52* | 27.05* | 38.31** | 10.42* | 24.44** | 24.63** | 47.25** |
| | (8.44) | (15.27) | (14.17) | (18.91) | (6.01) | (11.53) | (11.92) | (18.94) |
| Observations | 37,714 | 34,765 | 67,853 | 62,759 | 37,712 | 34,764 | 67,853 | 62,758 |
| $R^2$ | .024 | .032 | .018 | .027 | .024 | .032 | .018 | .028 |
| Affected county-episodes | 100 | 95 | 144 | 129 | 98 | 94 | 144 | 128 |
| Affected × hazard × year FE | | | | | ✓ | ✓ | ✓ | ✓ |
| Episode and quarter FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| State FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Financial and demographic controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| County damages, fatalities and injuries | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Political and belief controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Lagged climate attention | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Affected × controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

*Notes.* This table reports estimates of the county green-ETF share response by the day on which the episode starts within its calendar quarter: in the last 30 days of the quarter (columns 1, 2, 5 and 6) or 31 to 91 days before its end (columns 3, 4, 7 and 8). The outcome is the change in the county green-ETF share from the quarter before the episode to the quarter of the episode (columns 1, 3, 5 and 7) or to the following quarter (columns 2, 4, 6 and 8), in basis points, at pre-event prices (Panel A) and current prices (Panel B). Columns 1–4 replace the affected × hazard × year fixed effects with the Affected indicator. The sample covers high-salience flood, hurricane and coastal episodes, 2015–2024. Interpretation, Attention and the controls are those of Table 2, column 4, and Interpretation is per standard deviation; Panel B uses the observations of Panel A. Standard errors, clustered by county, are in parentheses. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

# F Appendix to Section 3.4

TABLE F.1: WHO RESPONDS TO CLIMATE INTERPRETATION: HETEROGENEITY OF THE INTERPRETATION EFFECT

| | Interactions with: | | | At the mean | | |
|---|---|---|---|---|---|---|
| Characteristic | Affected | Interpretation | Affected × Interpretation | Affected × Interpretation | Affected | N |
| *Panel A. Beliefs and ambient coverage* | | | | | | |
| Share worried about global warming | -13.9 (23.4) | -1.8 (2.7) | 22.4* (13.3) | 36.7*** (10.1) | -14.2 (45.8) | 59,330 |
| Perceived scientific consensus | -0.7 (23.5) | -0.0 (2.8) | 9.9 (16.4) | 42.8*** (12.8) | -16.6 (46.9) | 59,330 |
| Climate-policy news share (same quarter) | 34.5** (17.1) | -5.3* (2.8) | 4.3 (14.5) | 32.6*** (12.1) | 0.5 (54.3) | 59,330 |
| Climate-policy news share (previous quarter) | 28.6* (16.9) | -1.1 (2.5) | 10.8 (15.0) | 43.3*** (15.6) | -4.5 (53.3) | 59,330 |
| News circulation per resident | 18.7 (14.3) | -3.1 (3.0) | -0.2 (18.6) | 43.1*** (13.5) | -21.5 (48.5) | 59,330 |
| *Panel B. Politics and socioeconomic characteristics* | | | | | | |
| County Democratic vote | -8.2 (27.1) | -3.6 (2.8) | 0.9 (11.0) | 45.2*** (11.5) | -13.9 (44.9) | 59,330 |
| Median income | -29.0 (25.0) | 0.6 (2.3) | 17.5 (14.1) | 47.2*** (15.0) | -28.7 (48.6) | 59,330 |
| College share | -28.2 (19.1) | 0.6 (2.2) | 4.6 (10.4) | 39.3*** (12.7) | -19.9 (51.3) | 59,330 |
| *Panel C. Exposure and experience* | | | | | | |
| Episode damages | -24.9* (13.5) | -0.1 (1.4) | 10.1 (11.9) | 41.1*** (12.7) | -12.7 (49.3) | 59,330 |
| County damages | -28.3 (17.6) | — | 13.1 (12.6) | 41.0*** (12.2) | -9.2 (50.7) | 59,330 |
| ETF turnover before 2018 (high vs low) | 100.4* (56.1) | -5.4 (6.8) | 31.8 (53.2) | 49.1*** (18.3) | 45.8 (53.6) | 43,558 |
| History of the same disaster type (previous 60 months) | 4.3 (16.3) | -7.1 (5.2) | 2.9 (11.0) | 42.2** (18.6) | -27.4 (47.8) | 59,330 |
| No history of the same disaster type (previous 60 months) | -7.5 (56.6) | -6.8 (8.1) | 124.8** (63.3) | 42.2** (18.6) | -27.4 (47.8) | 59,330 |
| History of other disaster types (previous 60 months) | -13.2 (13.0) | -4.4 (3.0) | -0.3 (10.1) | 44.0*** (13.8) | -7.8 (50.3) | 59,330 |
| History of climate-related disasters (previous 60 months) | -9.5 (12.4) | -5.3* (3.1) | -4.6 (10.4) | 48.5*** (15.6) | -10.0 (49.6) | 59,330 |
| Prior climate episodes | -24.9 (23.9) | -6.7 (5.1) | -2.0 (13.7) | 42.3** (16.5) | -1.1 (53.7) | 59,330 |
| *Panel D. Population and density* | | | | | | |
| Population | 2.3 (15.9) | 1.4 (3.6) | -16.5 (10.3) | 39.9*** (13.7) | -16.1 (46.3) | 59,330 |
| Population density | -10.1 (19.1) | 0.2 (4.0) | -10.9 (12.7) | 31.1** (15.6) | -9.1 (43.9) | 59,330 |

*Notes.* This table reports, for each county characteristic, a separate regression of the change in the county green-ETF portfolio share at pre-event prices, in basis points. Columns (1)–(3) report the characteristic's interactions with the affected indicator, climate interpretation, and their interaction. Columns (4) and (5) report the Affected × Interpretation coefficient and the Affected coefficient from the same regression, evaluated at the sample mean of the characteristic (the sample share for indicators). Interpretation and log total disaster-news attention are measured over event weeks 0–3. Models include the baseline controls and county, quarter, and episode fixed effects, and standard errors are clustered by county and episode. Continuous characteristics are centered and divided by their sample standard deviation, and indicators are not scaled; the indicator for no history of the same disaster type is not centered, so its column (4) slope is evaluated at counties with such history. Perceived scientific consensus uses the last county value before 2018. The climate-policy news share is the local-news share discussing climate policy or green issues, measured in the same quarter or one quarter earlier. News circulation is local-outlet circulation per county resident. ETF turnover before 2018 measures gross portfolio reallocation across all ETF positions, absolute position changes divided by lagged assets, averaged within county before 2018; counties are split into terciles, with the low tercile omitted and the middle tercile controlled. County damages are zero for every unaffected county, so their interaction with interpretation is not identified and is shown as a dash; the other cells of that row come from one regression that also controls for episode damages. Disaster-history counts enter in logarithms, with the history of the same disaster type centered among counties with such history. The sample covers flood, hurricane and coastal episodes, 2018Q1–2024Q4. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

TABLE F.2: EARTHQUAKE PLACEBO: LOCAL GREEN-ETF SHARE AND CLIMATE INTERPRETATION

| | *Dependent variable:* | |
|---|---|---|
| | Green-ETF share change at pre-event prices (bp) | |
| Interpretation | Event quarter | Mean through $q-1$ |
| | (1) | (2) |
| Affected × Interpretation (bp per SD) | 1.01 | −.86 |
| | (4.53) | (7.26) |
| Affected | 10.00 | 10.42 |
| | (6.27) | (7.12) |
| Interpretation | −1.56 | 5.01 |
| | (2.66) | (15.42) |
| Coverage | 3.64 | 3.59 |
| | (6.33) | (5.26) |
| Affected × Coverage | 3.11 | 3.51 |
| | (7.83) | (6.68) |
| $p$-value: Affected × Interpretation | .824 | .906 |
| Observations | 180,934 | 195,425 |
| Treated counties | 17 | 18 |
| Treated county–quarters | 32 | 34 |
| Treated episodes | 52 | 54 |
| County, quarter and episode FE | ✓ | ✓ |

*Notes.* This table reports the county green-ETF share response to earthquakes (USGS ComCat). The outcome is the fixed-price share in quarter $q+1$ minus the share in quarter $q-1$, $q$ the earthquake quarter, in basis points. Interpretation is the county's climate interpretation of other disasters, in the event quarter (column 1) or averaged through $q-1$ (column 2, at least four earlier observations), standardized within sample; Coverage is standardized event-quarter coverage. All columns include county, calendar-quarter and earthquake-episode fixed effects. Episodes with a large climate disaster in an affected county in $q-1$ to $q+1$ are excluded. Standard errors, in parentheses, are clustered by county. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

TABLE F.3: DISASTER NEWS AND TURNOVER OF ETF AND TOTAL REPORTED PORTFOLIOS

| | *Outcome:* log(1 + turnover in bp), quarter $t+1$ | |
|---|---|---|
| | (1) ETFs | (2) All reported securities |
| Affected × Interpretation | 0.050 | -0.003 |
| | (0.065) | (0.055) |
| Unaffected × Interpretation | 0.028 | 0.032** |
| | (0.018) | (0.014) |
| Affected × Attention | 0.121* | 0.107** |
| | (0.062) | (0.054) |
| Unaffected × Attention | 0.091*** | 0.081*** |
| | (0.030) | (0.022) |
| Interpretation: affected minus unaffected | 0.023 | -0.035 |
| | (0.067) | (0.055) |
| Attention: affected minus unaffected | 0.031 | 0.026 |
| | (0.062) | (0.053) |
| Observations | 101,785 | 101,785 |
| $R^2$ | 0.135 | 0.188 |
| Affected county-episodes | 236 | 236 |
| Episode and quarter FE | ✓ | ✓ |
| State FE | ✓ | ✓ |
| Affected × hazard × year FE | ✓ | ✓ |
| Financial and demographic controls | ✓ | ✓ |
| County damages, fatalities and injuries | ✓ | ✓ |
| Political and belief controls | ✓ | ✓ |
| Lagged climate attention | ✓ | ✓ |
| Affected × controls | ✓ | ✓ |

*Notes.* This table reports the response of the turnover of advisers' ETF and total reported portfolios, with the slopes of affected and unaffected counties estimated jointly in each column. Turnover adapts Calvet et al. (2009, online Appendix, Section 6.1) to quarterly adviser holdings. For adviser $i$ and set of securities $B$, $T_{iB,t+1} = \sum_{j\in B} \widetilde{P}_{j,t}|Q_{ij,t+1} - Q_{ij,t}|/A_{iB,t}$, where $A_{iB,t} = \sum_{j\in B} MV_{ij,t}$ is the starting reported value of the securities in $B$. County turnover is $T_{cB,t+1} = \sum_{i\in c} A_{iB,t}T_{iB,t+1}/\sum_{i\in c} A_{iB,t}$. Absolute changes are computed within each adviser before county aggregation, so opposing trades of different advisers both count. No factor of one-half is applied. The outcome is $\log(1 + 10^4 T_{cB,t+1})$, with the log applied after county aggregation, following the log(1 + turnover) approach in Anderson (2013). Column (1) covers ETFs and column (2) all reported security classes; both columns use the same advisers, with valid reports in adjacent quarters and unchanged county assignments, and the same county-episodes. Prices are security-quarter median unit values at $t$, with $t+1$ values used when unavailable; pairs with any unpriced position are excluded. The horizon is the end of event quarter $t$ to the end of $t+1$. The sample covers high-salience flood, hurricane and coastal episodes in 2015–2024. Attention and Interpretation are measured over the first four weeks and standardized. Controls and fixed effects follow Table 2, column (4). Standard errors of the difference rows use the full joint covariance. Standard errors are clustered by county. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

TABLE F.4: LOCAL CLIMATE INTERPRETATION AND ADVISER PORTFOLIOS NEAR DISASTERS

| | *Dependent variable:* | | | | |
|---|---|---|---|---|---|
| | Green-ETF share response (bp) Pre-event prices | | | | |
| | (1) | (2) | (3) | (4) | (5) |
| *Panel A.* $h = 0$ | | | | | |
| Interpretation effect among: | | | | | |
| Affected or nearby advisers, 0–50 miles | 7.97*** | 8.21** | 6.69*** | 6.64*** | 7.32** |
| | (2.75) | (3.18) | (2.44) | (2.53) | (3.09) |
| Observations | 6,481 | 6,481 | 6,481 | 6,481 | 6,481 |
| $R^2$ | .069 | .074 | .100 | .101 | .272 |
| Advisers | 1,030 | 1,030 | 1,030 | 1,030 | 1,030 |
| ZIP codes | 431 | 431 | 431 | 431 | 431 |
| Counties | 124 | 124 | 124 | 124 | 124 |
| Episodes | 303 | 303 | 303 | 303 | 303 |
| *Panel B.* $h = 1$ | | | | | |
| Interpretation effect among: | | | | | |
| Affected or nearby advisers, 0–50 miles | 11.75** | 11.60** | 11.59** | 10.70** | 13.12* |
| | (4.57) | (5.79) | (5.42) | (4.99) | (7.85) |
| Observations | 5,889 | 5,889 | 5,889 | 5,889 | 5,889 |
| $R^2$ | .181 | .185 | .215 | .218 | .454 |
| Advisers | 992 | 992 | 992 | 992 | 992 |
| ZIP codes | 416 | 416 | 416 | 416 | 416 |
| Counties | 123 | 123 | 123 | 123 | 123 |
| Episodes | 295 | 295 | 295 | 295 | 295 |
| Episode × state FE | ✓ | ✓ | ✓ | | |
| Episode × band FE | | | ✓ | | |
| Episode × state × band FE | | | | ✓ | ✓ |
| County FE | ✓ | ✓ | ✓ | ✓ | |
| Adviser FE | | | | | ✓ |
| Attention controls | ✓ | ✓ | ✓ | ✓ | ✓ |
| Financial and demographic controls | ✓ | ✓ | ✓ | ✓ | ✓ |
| County damages, fatalities and injuries | ✓ | ✓ | ✓ | ✓ | ✓ |
| Political controls | ✓ | ✓ | ✓ | ✓ | ✓ |
| Lagged climate attention | ✓ | ✓ | ✓ | ✓ | ✓ |
| Group × controls | | ✓ | ✓ | ✓ | ✓ |

*Notes.* This table reports within-episode regressions of green-ETF portfolio changes among advisers in affected counties and nearby advisers within 50 miles of the affected area. The outcome is the share change from the quarter before the episode to the disaster quarter (Panel A) or the following quarter (Panel B), in basis points, with holdings valued at pre-event prices. Interpretation is the climate-interpreted share of disaster coverage at the adviser's ZIP over the episode's first four weeks. Coefficients are reported per ten-percentage-point increase and are common across affected and nearby advisers. All columns control for log disaster-news volume and its interaction with Affected; lagged log ETF and total Form 13F holdings; county income, college share, white share, population, density and prior Democratic vote share; county damages, fatalities and injuries; and lagged climate-policy news share. Group × controls allows financial, demographic and political coefficients to differ across affected, nearby 0–25 and nearby 25–50 mile bands, which also define the band fixed effects. Observations are weighted by advisers' shares of the pre-event ETF holdings of the advisers in the same ZIP code and episode. Standard errors, in parentheses, are clustered by the physical event behind each episode and by the ZIP code of the adviser's office. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

TABLE F.5: LOCAL CLIMATE INTERPRETATION AND ADVISER PORTFOLIOS NEAR DISASTERS

| | *Dependent variable:* | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | Green-ETF share response (bp) | | | | | | | |
| | Pre-event prices | | | | Current prices | | | |
| | $h = 0$ | | $h = 1$ | | $h = 0$ | | $h = 1$ | |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
| *Panel A. Advisers weighted by their share of the ZIP code's ETF holdings* | | | | | | | | |
| Interpretation effect among: | | | | | | | | |
| Affected advisers | 11.00** | 9.91** | 11.25 | 10.63 | 9.04* | 8.35* | 11.68 | 11.21 |
| | (5.50) | (4.94) | (8.21) | (8.21) | (4.61) | (4.24) | (8.47) | (8.19) |
| Nearby advisers, 0–25 miles | 8.95*** | 7.03*** | 14.37** | 11.86* | 8.38*** | 6.33** | 16.05** | 13.38 |
| | (2.64) | (2.59) | (5.59) | (6.61) | (2.73) | (2.60) | (7.13) | (8.30) |
| Nearby advisers, 25–50 miles | 6.40** | 8.85** | 9.41* | 11.62** | 5.78** | 8.33** | 11.10* | 12.96** |
| | (2.80) | (3.45) | (4.97) | (5.49) | (2.81) | (3.45) | (5.86) | (6.00) |
| Observations | 6,481 | 6,481 | 5,889 | 5,889 | 6,481 | 6,481 | 5,889 | 5,889 |
| $R^2$ | .069 | .074 | .181 | .185 | .067 | .072 | .176 | .180 |
| Advisers | 1,030 | 1,030 | 992 | 992 | 1,030 | 1,030 | 992 | 992 |
| ZIP codes | 431 | 431 | 416 | 416 | 431 | 431 | 416 | 416 |
| Counties | 124 | 124 | 123 | 123 | 124 | 124 | 123 | 123 |
| Episodes | 303 | 303 | 295 | 295 | 303 | 303 | 295 | 295 |
| *Panel B. Advisers weighted equally* | | | | | | | | |
| Interpretation effect among: | | | | | | | | |
| Affected advisers | 9.15** | 9.79* | 9.78 | 11.88 | 8.42* | 8.89* | 11.07 | 13.43* |
| | (4.45) | (5.44) | (6.89) | (7.56) | (4.59) | (4.98) | (7.61) | (7.89) |
| Nearby advisers, 0–25 miles | 8.19** | 7.41 | 10.58** | 8.17 | 8.23* | 7.44 | 13.31* | 10.97 |
| | (3.62) | (4.84) | (4.52) | (5.80) | (4.24) | (5.15) | (7.07) | (7.82) |
| Nearby advisers, 25–50 miles | 7.02** | 8.68** | 10.47** | 12.66** | 6.81* | 8.40* | 12.68** | 14.55** |
| | (3.40) | (4.27) | (4.12) | (5.01) | (3.70) | (4.38) | (5.85) | (5.99) |
| Observations | 6,481 | 6,481 | 5,889 | 5,889 | 6,481 | 6,481 | 5,889 | 5,889 |
| $R^2$ | .045 | .049 | .084 | .089 | .048 | .052 | .087 | .091 |
| Advisers | 1,030 | 1,030 | 992 | 992 | 1,030 | 1,030 | 992 | 992 |
| ZIP codes | 431 | 431 | 416 | 416 | 431 | 431 | 416 | 416 |
| Counties | 124 | 124 | 123 | 123 | 124 | 124 | 123 | 123 |
| Episodes | 303 | 303 | 295 | 295 | 303 | 303 | 295 | 295 |
| Episode × state FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| County FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Attention controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Financial and demographic controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| County damages, fatalities and injuries | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Political controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Lagged climate attention | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Group × controls | | ✓ | | ✓ | | ✓ | | ✓ |

*Notes.* This table reports adviser green-ETF share responses in the disaster quarter ($h = 0$) and cumulatively through the following quarter ($h = 1$), both relative to the quarter before the episode. Pre-event prices are held at their levels in that previous quarter. Nearby advisers are outside the disaster footprint. Interpretation is the climate-interpreted share of disaster coverage at the adviser's ZIP; estimates are per ten percentage points. Attention is the log volume of disaster coverage at that ZIP. Each regression jointly estimates the three group effects and includes Attention and Affected × Attention. Financial controls are the logs of pre-event ETF holdings and total Form 13F holdings. Demographic controls are income, college share, white share, population and density; political controls use the prior county Democratic vote share. Lagged climate attention is the prior-quarter share of coverage on climate policy and green topics. Even-numbered columns interact financial, demographic and political controls, including missing-value indicators, with the three groups. Panel A weights advisers by their shares of the pre-event ETF holdings of the advisers in the same ZIP code and episode; Panel B weights adviser–episodes equally. Standard errors, in parentheses, are clustered by the physical event behind each episode and by the ZIP code of the adviser's office. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

TABLE F.6: LOCAL CLIMATE INTERPRETATION AND ADVISER PORTFOLIOS: TWO-QUARTER AVERAGE

| | *Dependent variable:* | | | |
|---|---|---|---|---|
| | Green-ETF share response (bp) | | | |
| | Pre-event prices | | Current prices | |
| | (1) | (2) | (3) | (4) |
| *Panel A. Advisers weighted by their share of the ZIP code's ETF holdings* | | | | |
| Interpretation effect among: | | | | |
| Affected advisers | 11.49* | 10.42* | 10.64* | 9.85* |
| | (5.91) | (5.75) | (6.10) | (5.79) |
| Nearby advisers, 0–25 miles | 11.93*** | 9.49** | 12.46** | 9.85* |
| | (4.19) | (4.39) | (4.93) | (5.24) |
| Nearby advisers, 25–50 miles | 8.08** | 10.41** | 8.54* | 10.75** |
| | (3.96) | (4.25) | (4.35) | (4.57) |
| Observations | 5,889 | 5,889 | 5,889 | 5,889 |
| $R^2$ | .124 | .128 | .124 | .128 |
| Advisers | 992 | 992 | 992 | 992 |
| ZIP codes | 416 | 416 | 416 | 416 |
| Counties | 123 | 123 | 123 | 123 |
| Episodes | 295 | 295 | 295 | 295 |
| *Panel B. Advisers weighted equally* | | | | |
| Interpretation effect among: | | | | |
| Affected advisers | 9.64* | 10.60* | 10.09* | 11.24* |
| | (5.54) | (6.05) | (6.08) | (6.62) |
| Nearby advisers, 0–25 miles | 9.93** | 8.33* | 11.50** | 9.95 |
| | (3.85) | (4.94) | (5.69) | (6.63) |
| Nearby advisers, 25–50 miles | 9.70*** | 11.59** | 10.81** | 12.49** |
| | (3.63) | (4.50) | (4.77) | (5.32) |
| Observations | 5,889 | 5,889 | 5,889 | 5,889 |
| $R^2$ | .064 | .069 | .067 | .072 |
| Advisers | 992 | 992 | 992 | 992 |
| ZIP codes | 416 | 416 | 416 | 416 |
| Counties | 123 | 123 | 123 | 123 |
| Episodes | 295 | 295 | 295 | 295 |
| Episode × state FE | ✓ | ✓ | ✓ | ✓ |
| County FE | ✓ | ✓ | ✓ | ✓ |
| Attention controls | ✓ | ✓ | ✓ | ✓ |
| Financial and demographic controls | ✓ | ✓ | ✓ | ✓ |
| County damages, fatalities and injuries | ✓ | ✓ | ✓ | ✓ |
| Political controls | ✓ | ✓ | ✓ | ✓ |
| Lagged climate attention | ✓ | ✓ | ✓ | ✓ |
| Group × controls | | ✓ | | ✓ |

*Notes.* See notes to Table F.5. This table reports the average adviser green-ETF share in the disaster quarter and the following quarter, relative to the quarter before the episode. Both post-episode quarters must be observed. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

TABLE F.7: WHOSE VIEWS: COUNTY VIEWS AGAINST THE ADVISERS' OWN LEAN

| | Dependent variable: ETF green-share response, $h = 1$ (bp) | | | | | |
|---|---|---|---|---|---|---|
| | ETF-holdings weights | | | | Equal weights | |
| | Adviser lean (1) | County vote (2) | Both (3) | Both, giving (4) | Both (5) | Both, giving (6) |
| Interpretation response, at affected means | 19.40*** | 21.55*** | 21.65*** | 21.34*** | 5.44 | 5.25 |
| | (7.43) | (7.96) | (8.03) | (7.98) | (6.54) | (5.97) |
| Disaster response, at affected means | 5.91 | 13.24** | 13.77** | 8.12* | 0.43 | −2.58 |
| | (4.74) | (5.57) | (5.92) | (4.71) | (5.67) | (5.00) |
| Interpretation response × County Rep. vote | | 6.90* | 10.64* | | 10.52** | |
| | | (3.96) | (5.48) | | (5.15) | |
| Interpretation response × Adviser Rep. lean | −8.37 | | −13.84 | −5.08 | −7.82 | −6.34 |
| | (8.33) | | (10.31) | (9.56) | (8.62) | (8.09) |
| Disaster response × County Rep. vote | | 7.54 | 6.56 | | 3.41 | |
| | | (5.88) | (5.97) | | (3.85) | |
| Disaster response × Adviser Rep. lean | 1.52 | | 0.79 | −3.01 | −9.83* | −10.71** |
| | (5.17) | | (5.33) | (5.12) | (5.20) | (4.88) |
| Interpretation: adviser minus county | | | −24.48* | −2.49 | −18.34 | −12.58 |
| | | | (14.83) | (12.73) | (12.92) | (10.66) |
| Interpretation response × County Rep. giving | | | | −2.60 | | 6.25* |
| | | | | (5.01) | | (3.29) |
| Disaster response × County Rep. giving | | | | 11.49* | | 5.45** |
| | | | | (6.40) | | (2.69) |
| Observations | | | 465,705 | | | |
| Affected adviser-episodes (with an adviser lean) | | | 1158 (723) | | | |
| Advisers | | | 2130 | | | |

*Interpretation response by county political group (slope within group, group-specific disaster response removed)*

| | ETF-holdings weights | | | Equal weights | | |
|---|---|---|---|---|---|---|
| | Rep. share < .33 | .33–.50 | ≥ .50 | Rep. share < .33 | .33–.50 | ≥ .50 |
| Interpretation response | 9.21 | 41.29** | 21.47** | −4.42 | 17.24 | 5.35 |
| | (6.51) | (16.71) | (8.36) | (5.33) | (14.08) | (5.17) |

*Notes.* This table reports the adviser response to interpretation and to the disaster by the political lean of the county and of the adviser. Observations are adviser-episodes for flood, hurricane and coastal episodes, 2015Q1–2024Q4. All columns include episode and calendar-quarter fixed effects, affected-by-year terms, attention and its interactions, the previous green-share change, lagged portfolio size and an indicator for advisers with no measured lean, and standard errors are clustered by county. Interpretation and attention are per standard deviation. The adviser's Republican lean comes from the political donations of the individuals on the adviser's roster where a match exists, and from those of the firm's investment-decision staff otherwise; zero is a politically neutral firm. County Republican vote is the two-party presidential vote share, averaged over the sample elections, per standard deviation across the sample counties; the county giving tilt is defined the same way. The response rows are evaluated at the means of all county and adviser characteristics among affected advisers. ETF-holdings weights weight each adviser by its share of the county's ETF holdings, and equal weights count each adviser-episode once. The interpretation entries are differences in slopes between affected and comparison advisers; the affected-county responses discussed in Appendix F add the comparison-group slopes, with standard errors from the full covariance matrix. The lower panel re-estimates the model with disaster responses and interpretation slopes specific to each county political group (cutoffs on the county Republican vote share); equality of the three slopes is not rejected ($p = .145$ with ETF-holdings weights, $p = .117$ with equal weights). $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table F.7 reports the estimates for the disaster response at reference levels of interpretation and attention and for the response to interpretation among affected counties. The disaster response does not move significantly with either county politics or the adviser's own lean. The interpretation effect is estimated to be stronger in less Democratic counties. A one-standard-deviation higher Republican vote share moves a county from the average of 47 percent to 62 percent. Such counties respond to interpretation by 8.9 basis points per standard deviation more. This is about half the base response.

I divide the counties into three political groups, with a Republican vote share below one third, between one third and one half, and at least one half, and net out each group's own direct disaster response. Within groups, the response to interpretation is estimated to be largest in counties with a Republican vote share between one third and one half, at 40.5 basis points per standard deviation of interpretation, and is statistically significant. It is also present among stronger Republican-majority counties, at about half that size, while the estimate in the strongly Democratic group is positive but not statistically distinguishable from zero. The point estimates suggest that the combination of a disaster and a climate narrative is most persuasive where the population is politically mixed, although the differences across groups are not jointly statistically significant. This is consistent with the survey evidence of Section 7.2.

Table F.8: Interpretation, predetermined activity, and flows during the event, county level, 2015Q1–2024Q4

| Outcome | Affected × Interpretation | Affected × Coverage | Affected | Mean | N |
|---|---|---|---|---|---|
| *Panel A. Predetermined quantities* | | | | | |
| Predetermined ETF turnover, per quarter | -44.7 (229.1) | 170 (694) | -2,288 (1,393) | 3,699.5 | 79,683 |
| Predetermined total-book turnover, per quarter | 42.0 (91.1) | -281 (179) | 734 (341) | 2,171.5 | 79,712 |
| Predetermined ETF net flow, per quarter | -61.4 (211.6) | 612 (634) | -3,100 (1,419) | 1,594.4 | 79,683 |
| Predetermined total-book net flow, per quarter | -56.1 (44.0) | 163 (130) | 53.9 (134.4) | 288.2 | 79,712 |
| *Panel B. Event-window quantities* | | | | | |
| ETF net flow, $t..t+1$ | 277.7 (375.1) | 285 (302) | -1,896 (558) | 1,098.8 | 79,737 |
| Total-book net flow, $t..t+1$ | 32.0 (158.9) | 172 (210) | -1,550 (346) | 364.9 | 79,674 |
| ETF asset growth, $t..t+1$ (percent) | 4.2 (3.1) | 5.0 (2.4) | -23.1 (7.0) | 11.8 | 79,737 |
| Total-book asset growth, $t..t+1$ (percent) | 1.4 (1.7) | 2.3 (1.8) | -14.4 (3.5) | 7.5 | 79,674 |
| Green-ETF net purchases, $t..t+1$ | 14.6* (7.4) | 13.8 (7.8) | 2.6 (3.9) | 4.5 | 79,737 |
| Total-book turnover, $t..t+1$, per quarter | 166.8 (123.5) | -13.1 (139.7) | 1,653 (670) | 1,949.4 | 79,674 |

*Notes.* This table reports, in each row, the paper's county-level specification with a different outcome, the county mean over its advisers, weighted by lagged ETF assets, of the adviser-level quantity named, winsorized at the 1st and 99th percentiles. Predetermined quantities are per-quarter means over quarters $t-5$ to $t-2$; event-window quantities are cumulated over quarters $t$ to $t+1$, with turnover as a per-quarter mean. Turnover and net flow are in basis points of lagged assets, over the ETF positions or over the whole Form 13F report as named; asset growth is in percent; green-ETF net purchases are signed green-ETF position changes at lagged prices over lagged ETF assets. All rows include the baseline controls and county, quarter, and episode fixed effects, and standard errors are clustered by county and episode. The Affected coefficient is the reference-year term of the year-specific effects for affected counties. Stars are reported for Affected × Interpretation only. The sample covers flood, hurricane and coastal episodes, 2015Q1–2024Q4.

$^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table F.9: County holdings by asset class, in logs

| | Affected | Affected × Interpretation |
|---|---|---|
| log(1 + green ETF assets) | 0.125 | 0.769** |
| | (0.181) | (0.384) |
| log(1 + non-green ETF assets) | 0.043 | 0.043** |
| | (0.033) | (0.021) |
| log ETF assets | 0.039 | 0.047** |
| | (0.032) | (0.020) |
| log total observed holdings | -0.008 | 0.010 |
| | (0.021) | (0.019) |
| log stock holdings | -0.037* | 0.003 |
| | (0.021) | (0.027) |

*Notes.* This table reports county coefficients at $h = 1$ for the logs of ETF and total holdings, on the sample of all eligible disasters. The first column reports the Affected coefficient from the specification without interpretation, and the second the Affected × Interpretation coefficient from the specification with it. The green and non-green ETF rows use $\log(1+x)$, since these holdings can be zero; the other rows use logs. Total observed holdings are the total holdings in the FactSet Form 13F data. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table F.10: Direct trades in constituent stocks and implied purchases through green ETFs

| | All constituents | | Constituents held at $t-1$ | |
|---|---|---|---|---|
| | Dollars | Basis points of book | Dollars | Basis points of book |
| | (1) | (2) | (3) | (4) |
| *Panel A. Trades from $t-1$ to $t$* | | | | |
| Implied indirect purchase | 4.450*** | 1.724*** | 4.536** | 2.389** |
| | (1.408) | (0.310) | (1.961) | (1.165) |
| Implied indirect purchase, with lagged position control | 6.405*** | 1.712*** | 7.235*** | 2.368** |
| | (2.443) | (0.310) | (1.775) | (1.166) |
| Observations | 6,037,448 | 6,037,448 | 937,024 | 937,024 |
| Adviser-quarters | 11,516 | 11,516 | 9,601 | 9,601 |
| Advisers | 1,610 | 1,610 | 1,449 | 1,449 |
| *Panel B. Trades from $t-1$ to $t+1$* | | | | |
| Implied indirect purchase | 5.032*** | 1.845*** | 4.633 | 1.751*** |
| | (1.812) | (0.265) | (3.003) | (0.673) |
| Implied indirect purchase, with lagged position control | 8.260*** | 1.822*** | 8.906*** | 1.706** |
| | (2.253) | (0.269) | (1.473) | (0.679) |
| Observations | 7,011,653 | 7,011,653 | 1,080,221 | 1,080,221 |
| Adviser-quarters | 12,875 | 12,875 | 10,823 | 10,823 |
| Advisers | 1,601 | 1,601 | 1,441 | 1,441 |
| Adviser × quarter FE | X | X | X | X |
| Stock × quarter FE | X | X | X | X |

*Notes.* This table reports regressions, at the adviser-stock-quarter level, of advisers' direct trades in the constituent stocks of the green ETFs they buy on the purchases of those stocks implied by the ETF purchases. The sample covers household-serving single-county advisers of the paper sample, 2018Q1–2024Q4, in quarters with a positive net purchase of green ETFs; advisers that sponsor a green ETF are excluded. Direct holdings are U.S. common stocks under the CRSP security classification (ordinary equity common shares of U.S.-incorporated corporate issuers). The dependent variable is the adviser's direct trade in constituent stock $j$, the change in shares held valued at the price of quarter $t-1$; the second row adds the adviser's lagged position value in stock $j$ as a control. The regressor is the implied indirect purchase of stock $j$ through the green ETFs the adviser bought, the dollar purchase of each ETF times the stock's weight in that ETF's basket one quarter earlier. Columns (1) and (3) are in dollars; columns (2) and (4) divide both sides by the adviser's total holdings at $t-1$ and are in basis points. A constituent not held directly enters with a zero trade in columns (1)–(2); columns (3)–(4) keep constituents the adviser already held at $t-1$. Standard errors, two-way clustered by adviser and stock, are in parentheses. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

TABLE F.11: PRIOR TURNOVER AND THE INTERPRETATION RESPONSE, 2015–2024

| | *Outcome:* fixed-price change in green-ETF share (bp) | | | |
|---|---|---|---|---|
| | ETF turnover | | Whole-portfolio turnover | |
| Prior window | 4 quarters | 8 quarters | 4 quarters | 8 quarters |
| | (1) | (2) | (3) | (4) |
| Prior turnover | -0.22 | -0.47 | 1.12 | -0.59 |
| | (2.00) | (3.12) | (1.96) | (3.21) |
| Affected × Prior turnover | 25.59 | 23.18 | 9.70* | 8.28* |
| | (17.85) | (17.50) | (5.32) | (4.41) |
| Interpretation × Prior turnover | -3.11** | -3.33** | -2.39* | -2.00* |
| | (1.42) | (1.33) | (1.27) | (1.10) |
| Affected × Interpretation × Prior turnover | -8.40 | -9.58 | 6.55 | 4.44 |
| | (6.95) | (9.30) | (5.40) | (4.27) |
| Affected × Interpretation, at mean turnover | 4.13 | 3.56 | 3.90 | 4.18 |
| | (7.58) | (7.57) | (7.22) | (7.36) |
| Affected, at mean turnover | 2.17 | 2.28 | -4.12 | -3.63 |
| | (6.62) | (6.40) | (5.42) | (5.38) |
| Observations | 464,020 | 464,629 | 464,528 | 464,950 |
| Advisers | 2,061 | 2,065 | 2,063 | 2,064 |
| County clusters | 291 | 292 | 292 | 292 |
| Episodes | 974 | 974 | 974 | 974 |
| Adviser, quarter and episode FE | ✓ | ✓ | ✓ | ✓ |
| Baseline controls | ✓ | ✓ | ✓ | ✓ |
| Affected × year effects | ✓ | ✓ | ✓ | ✓ |

*Notes.* This table reports adviser-episode regressions of the change in the adviser's green-ETF portfolio share at pre-event prices from quarter $t-1$ to quarter $t+1$, in basis points, on prior turnover and its interactions, one regression per column, for flood, hurricane and coastal episodes in 2015Q1–2024Q4. Prior turnover is mean quarterly gross turnover in basis points of lagged assets, computed over ETF holdings or the whole Form 13F portfolio, over quarters $t-5$ to $t-2$ (four quarters) or $t-9$ to $t-2$ (eight quarters), using the quarters observed in each window. It enters as $\log(1+\text{prior turnover})$, centered and divided by its sample standard deviation. The table reports prior turnover, both lower-order interactions and the triple interaction; the last two coefficient rows are evaluated at mean prior turnover. The Affected row is the reference-year effect. Interpretation is standardized. The regression also includes interpretation, disaster coverage, coverage interacted with the affected indicator, year-specific effects for affected advisers, the previous green-share change, lagged log assets and the county's Democratic vote share, and adviser, quarter and episode fixed effects. Standard errors, clustered by the county of the adviser's location, are in parentheses. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

TABLE F.12: NATIONAL NEWS AND LOCAL INTERPRETATION AMONG AFFECTED COUNTIES AND ADVISERS

| | *Dependent variable:* Green-ETF share response (bp) | | |
|---|---|---|---|
| | County | Advisers, −1 to +1 | |
| | Pooled | ETF-holdings weights | Equal weights |
| | (1) | (2) | (3) |
| Baseline local interpretation | 31.32** | 26.78*** | 13.95** |
| | (12.53) | (8.24) | (6.77) |
| With national coverage | 27.28** | 22.44*** | 11.53** |
| | (12.26) | (6.27) | (4.91) |
| × national coverage | 26.85* | 22.88** | 15.35 |
| | (15.66) | (10.41) | (12.06) |
| With national interpretation | 22.41* | 23.64*** | 13.00* |
| | (12.82) | (8.38) | (6.61) |
| × national interpretation | 18.89 | −1.47 | 1.11 |
| | (12.48) | (5.61) | (4.67) |
| Observations | 105,662 | 465,701 | 465,701 |

*Notes.* This table reports the response to local interpretation among affected counties and advisers and its change with national news. Local-interpretation rows report the response to a one-standard-deviation increase, evaluated at mean national news; the interaction rows give its change per standard deviation of national news. National coverage and national interpretation enter separate regressions, both measured over weeks 0–3 across nine national outlets, as log one plus national disaster coverage and as the share of that coverage that links the disaster centrally to climate change. Both are interacted with affected status, local interpretation and local coverage, including their interactions with affected status. The county outcome is the mean share in quarters 0–2 minus its mean in quarters −2 and −1; adviser outcomes compare quarter +1 with quarter −1, at pre-event ETF prices. Within each column the sample is the same in every row, and each column includes the baseline controls and episode, quarter, affected-by-hazard-by-year and geographic fixed effects (state for counties, county for advisers). Column 2 weights advisers by their pre-event share of the county's ETF holdings, and column 3 weights them equally. Standard errors, clustered by county and computed from the full covariance matrix, are in parentheses. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

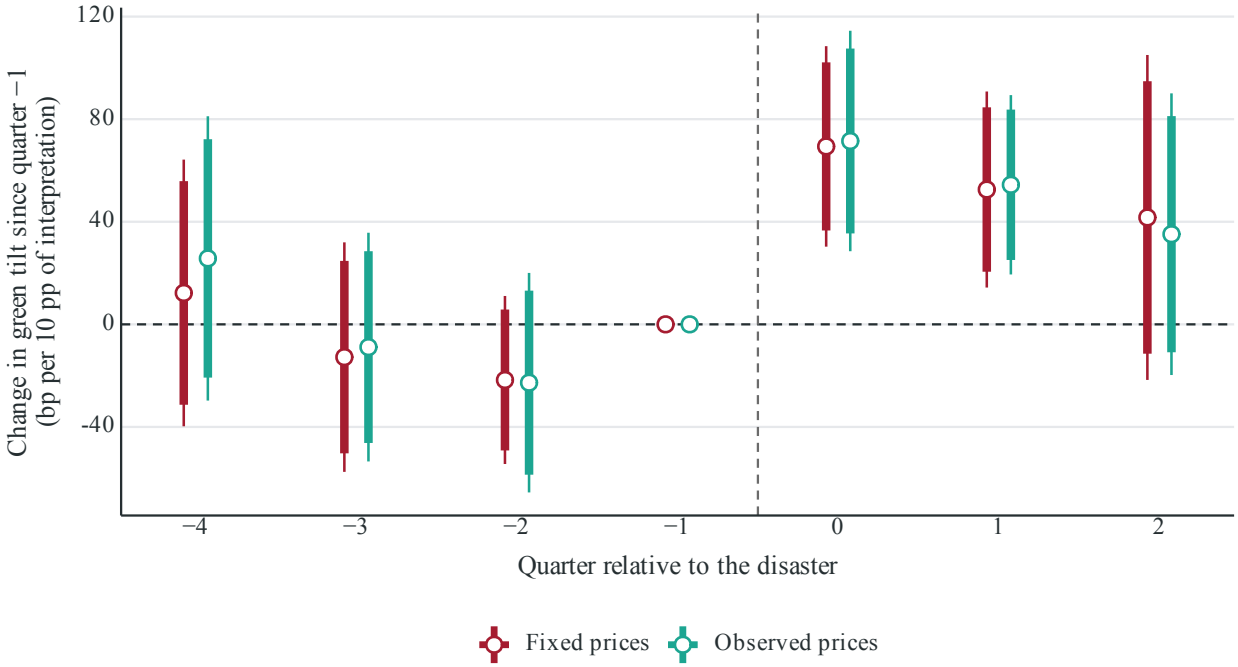
# G Appendix to Section 4

TABLE G.1: MUTUAL FUND PORTFOLIO RESPONSE

| | *Dependent variable:* response in basis points | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | Pre-event prices | | | | Current prices | | | |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
| *Panel A. Fund level: change in the fund's green tilt* | | | | | | | | |
| Interpretation | 69.34*** | 69.38*** | 66.25*** | 67.58*** | 71.03*** | 71.49*** | 68.74*** | 65.75*** |
| | (20.11) | (19.87) | (18.99) | (19.86) | (20.85) | (21.85) | (21.08) | (21.56) |
| Attention | −2.05 | −.85 | −1.09 | −1.95 | −4.52 | −3.79 | −4.01 | −4.43 |
| | (2.84) | (3.12) | (3.09) | (3.36) | (2.98) | (3.36) | (3.28) | (3.58) |
| Observations (fund-episodes) | 37,047 | 37,047 | 37,047 | 36,964 | 37,047 | 37,047 | 37,047 | 36,964 |
| Nearby fund-episodes | 1,712 | 1,712 | 1,712 | 1,710 | 1,712 | 1,712 | 1,712 | 1,710 |
| $R^2$ | .031 | .032 | .033 | .093 | .044 | .045 | .045 | .102 |
| Exposure × episode FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| County FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Fund style FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Fund FE | | | | ✓ | | | | ✓ |
| Fund controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Damage and coverage controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Exposure × county controls | | ✓ | ✓ | ✓ | | ✓ | ✓ | ✓ |
| Exposure × political controls | | | ✓ | ✓ | | | ✓ | ✓ |
| *Panel B. Stock level: change in the fund's weight in a stock, top minus bottom tercile* | | | | | | | | |
| Interpretation | | 3.57*** | 3.56*** | | | 3.66*** | 3.71*** | |
| | | (1.28) | (1.28) | | | (1.33) | (1.29) | |
| Attention | | .04 | .03 | | | −.09 | −.13 | |
| | | (.27) | (.28) | | | (.29) | (.30) | |
| Observations (fund-episode-stocks) | | 5,426,327 | 5,426,327 | | | 5,426,327 | 5,426,327 | |
| Fund-episodes | | 53,028 | 53,028 | | | 53,028 | 53,028 | |
| $R^2$ | | .577 | .577 | | | .586 | .586 | |
| Fund × episode FE | | ✓ | ✓ | | | ✓ | ✓ | |
| Fund style × stock × episode FE | | ✓ | ✓ | | | ✓ | ✓ | |
| Exposure × episode FE | | ✓ | ✓ | | | ✓ | ✓ | |
| Damage and coverage controls | | ✓ | ✓ | | | ✓ | ✓ | |
| Exposure × political controls | | | ✓ | | | | ✓ | |

*Notes.* This table reports estimates of equation (7). In Panel A the dependent variable is the change in the fund's green tilt over the episode quarter, the change in its weight in stocks in the top tercile of the Environmental Pillar score minus the change in its weight in bottom-tercile stocks; in Panel B it is the change in the fund's weight in each stock, and the rows report the response of top-tercile relative to bottom-tercile stocks. Rows report the effects among managers within 50 miles of the episode. Interpretation is per 10 percentage points and Attention per 10 log points. Fund controls are lagged log total net assets, log age and turnover; county controls are the county's interpretation of its other episodes, log fatalities, log population, log population density, log income, college share, unemployment rate and urban share; political controls are the county Democratic two-party vote share and the managers' political score. County and political controls enter separately for each band. Standard errors, in parentheses, are clustered by episode. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

FIGURE G.1: INTERPRETATION EFFECT AMONG NEARBY MANAGERS, BY QUARTER




*Notes.* This figure plots the effect of local climate interpretation on the green tilt of nearby managers by quarter relative to the disaster, per 10 percentage points, at pre-event and current prices. The outcome is the change in the tilt since the end of quarter −1, in the specification of Table G.1, Columns 2 and 6. Thin and thick bars are 95 and 90 percent confidence intervals, with standard errors clustered at the disaster level.

TABLE G.2: LOCAL HEAT AND THE ENVIRONMENTAL-PILLAR TILT OF MUTUAL FUNDS

| | *Dependent variable:* | |
|---|---|---|
| | Environmental-Pillar tilt response (basis points) | |
| | Pre-event prices | Current prices |
| | (1) | (2) |
| Heat exposure | 65.39* | 71.40** |
| | (37.74) | (34.46) |
| Observations (fund-quarters) | 53,802 | 53,802 |
| Funds | 2,335 | 2,335 |
| Location-weighted county effects | ✓ | ✓ |
| Fund style × quarter FE | ✓ | ✓ |
| Fund controls | ✓ | ✓ |
| Heat severity controls | ✓ | ✓ |

*Notes.* This table reports regressions of the quarterly change in the fund's Environmental-Pillar tilt on heat exposure, the share of the fund's offices in counties with a heat wave causing injuries or fatalities in the quarter (SHELDUS), on fund-quarters from 2010 to 2023. The tilt is the weight of top- minus bottom-tercile stocks, in basis points, at pre-event prices (column 1) and current prices (column 2). Fund controls are the past-year return and flow, expense ratio, turnover, log assets and log age; severity controls are the fatalities, injuries and damages of the heat events. Standard errors, clustered by fund and quarter, are in parentheses. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

TABLE G.3: MUTUAL FUND PORTFOLIO RESPONSE: ALL FUND-EPISODES AND FUND-EPISODES WITH POLITICAL MEASURES

| | *Dependent variable:* response in basis points | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Pre-event prices | | | | | Current prices | | | | |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) | (10) |
| *Panel A. Fund level: change in the fund's Environmental-Pillar tilt* | | | | | | | | | | |
| Interpretation | 50.25*** | 48.15*** | 69.34*** | 69.38*** | 52.03** | 42.47*** | 39.76** | 71.03*** | 71.49*** | 53.39** |
| | (12.80) | (12.51) | (20.11) | (19.87) | (20.69) | (15.46) | (15.37) | (20.85) | (21.85) | (22.64) |
| Attention | −1.06 | −1.06 | −2.05 | −.85 | −.15 | −2.27 | −2.32 | −4.52 | −3.79 | −1.91 |
| | (2.39) | (2.54) | (2.84) | (3.12) | (2.27) | (2.34) | (2.44) | (2.98) | (3.36) | (2.32) |
| Observations (fund-episodes) | 51,226 | 51,226 | 37,047 | 37,047 | 37,051 | 51,226 | 51,226 | 37,047 | 37,047 | 37,051 |
| Nearby fund-episodes | 2,280 | 2,280 | 1,712 | 1,712 | 1,712 | 2,280 | 2,280 | 1,712 | 1,712 | 1,712 |
| $R^2$ | .025 | .025 | .031 | .032 | .025 | .039 | .039 | .044 | .045 | .037 |
| Exposure × episode FE | ✓ | ✓ | ✓ | ✓ | | ✓ | ✓ | ✓ | ✓ | |
| Episode FE | | | | | ✓ | | | | | ✓ |
| Exposure × hazard × quarter FE | | | | | ✓ | | | | | ✓ |
| County FE | ✓ | ✓ | ✓ | ✓ | | ✓ | ✓ | ✓ | ✓ | |
| Fund style FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Fund controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Damage and coverage controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Exposure × county controls | | ✓ | | ✓ | ✓ | | ✓ | | ✓ | ✓ |
| Sample with political measures | | | ✓ | ✓ | ✓ | | | ✓ | ✓ | ✓ |
| *Panel B. Stock level: change in the fund's weight in a stock, High-E minus Low-E* | | | | | | | | | | |
| Interpretation | | 4.44*** | | 3.57*** | | | 4.51*** | | 3.66*** | |
| | | (1.28) | | (1.28) | | | (1.21) | | (1.33) | |
| Attention | | −.09 | | .04 | | | −.16 | | −.09 | |
| | | (.18) | | (.27) | | | (.19) | | (.29) | |
| Observations (fund-episode-stocks) | | 7,125,159 | | 5,426,327 | | | 7,125,159 | | 5,426,327 | |
| Fund-episodes | | 72,461 | | 53,028 | | | 72,461 | | 53,028 | |
| $R^2$ | | .539 | | .577 | | | .550 | | .586 | |
| Fund × episode FE | | ✓ | | ✓ | | | ✓ | | ✓ | |
| Fund style × stock × episode FE | | ✓ | | ✓ | | | ✓ | | ✓ | |
| Exposure × episode FE | | ✓ | | ✓ | | | ✓ | | ✓ | |
| Damage and coverage controls | | ✓ | | ✓ | | | ✓ | | ✓ | |
| Sample with political measures | | | | ✓ | | | | | ✓ | |

*Notes.* See notes to Table 3. Columns 1, 2, 6 and 7 use all fund-episodes, including managers and management companies without matched political donations; the other columns use the fund-episodes for which the political controls are observed, as in Table 3. Columns 5 and 10 replace the exposure-by-episode and county fixed effects with episode fixed effects and fixed effects for each exposure band, hazard and calendar quarter.

TABLE G.4: MUTUAL FUND PORTFOLIO RESPONSE: MANAGER LOCATION AND CONTINUOUS SCORE

| | *Dependent variable:* response in basis points | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | Pre-event prices | | | | | Current prices | | | | |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) | (9) | (10) |
| *Panel A. Funds with all offices in one county* | | | | | | | | | | |
| Interpretation | 57.75*** | 53.93*** | 81.00*** | 79.28*** | 74.74*** | 55.02*** | 50.86*** | 76.93*** | 75.17*** | 70.43*** |
| | (15.51) | (15.89) | (22.60) | (22.77) | (21.80) | (15.81) | (16.93) | (22.04) | (23.18) | (22.37) |
| Attention | −1.61 | −1.59 | −2.58 | −1.52 | −2.41 | −2.99 | −3.05 | −4.27 | −3.69 | −4.64 |
| | (2.79) | (2.82) | (3.26) | (3.49) | (3.43) | (2.83) | (2.83) | (3.40) | (3.73) | (3.56) |
| Observations (fund-episodes) | 37,590 | 37,590 | 25,559 | 25,559 | 25,559 | 37,590 | 37,590 | 25,559 | 25,559 | 25,559 |
| Nearby fund-episodes | 2,061 | 2,061 | 1,404 | 1,404 | 1,404 | 2,061 | 2,061 | 1,404 | 1,404 | 1,404 |
| *Panel B. Fund located at all its offices* | | | | | | | | | | |
| Interpretation | 53.10*** | 49.74*** | 72.30*** | 71.79*** | 68.23*** | 47.75*** | 43.91*** | 72.26*** | 71.63*** | 68.45*** |
| | (13.81) | (14.01) | (20.06) | (19.75) | (19.01) | (14.25) | (14.63) | (21.16) | (21.84) | (21.23) |
| Attention | −2.77 | −2.70 | −4.42 | −3.23 | −3.58 | −3.93 | −3.83 | −6.30* | −5.46 | −5.77 |
| | (2.90) | (3.03) | (3.13) | (3.46) | (3.46) | (2.98) | (3.01) | (3.34) | (3.64) | (3.58) |
| Observations (fund-episodes) | 51,252 | 51,252 | 37,080 | 37,080 | 37,080 | 51,252 | 51,252 | 37,080 | 37,080 | 37,080 |
| Nearby fund-episodes | 2,531 | 2,531 | 1,884 | 1,884 | 1,884 | 2,531 | 2,531 | 1,884 | 1,884 | 1,884 |
| *Panel C. Fund located at its office closest to the disaster* | | | | | | | | | | |
| Interpretation | 43.29*** | 38.62*** | 62.29*** | 63.50*** | 57.90*** | 45.16*** | 41.10*** | 65.08*** | 67.20*** | 61.61*** |
| | (14.02) | (13.95) | (17.88) | (17.39) | (16.69) | (14.01) | (13.83) | (18.70) | (19.15) | (18.48) |
| Attention | −2.42 | −2.03 | −2.54 | −1.21 | −1.86 | −3.99 | −3.44 | −4.79* | −3.44 | −4.10 |
| | (2.43) | (2.58) | (2.70) | (3.10) | (3.04) | (2.48) | (2.56) | (2.82) | (3.13) | (3.03) |
| Observations (fund-episodes) | 51,257 | 51,257 | 37,084 | 37,084 | 37,084 | 51,257 | 51,257 | 37,084 | 37,084 | 37,084 |
| Nearby fund-episodes | 2,389 | 2,389 | 1,784 | 1,784 | 1,784 | 2,389 | 2,389 | 1,784 | 1,784 | 1,784 |
| *Panel D. Continuous Environmental-Pillar score* | | | | | | | | | | |
| Interpretation | 14.64*** | 13.47*** | 18.54** | 19.37** | 17.66** | 13.55** | 12.03** | 20.48** | 21.30** | 19.54** |
| | (4.65) | (4.73) | (7.73) | (8.16) | (8.19) | (5.91) | (5.82) | (8.92) | (9.33) | (9.44) |
| Attention | −.18 | −.24 | −.28 | .09 | −.12 | −.35 | −.39 | −.92 | −.60 | −.79 |
| | (.86) | (1.04) | (1.12) | (1.40) | (1.49) | (.91) | (1.07) | (1.17) | (1.47) | (1.54) |
| Observations (fund-episodes) | 51,226 | 51,226 | 37,047 | 37,047 | 37,047 | 51,226 | 51,226 | 37,047 | 37,047 | 37,047 |
| Nearby fund-episodes | 2,280 | 2,280 | 1,712 | 1,712 | 1,712 | 2,280 | 2,280 | 1,712 | 1,712 | 1,712 |
| Exposure × episode FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| County FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Fund style FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Fund controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Damage and coverage controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Exposure × county controls | | ✓ | | ✓ | ✓ | | ✓ | | ✓ | ✓ |
| Exposure × political controls | | | | | ✓ | | | | | ✓ |
| Sample with political measures | | | ✓ | ✓ | ✓ | | | ✓ | ✓ | ✓ |

*Notes.* See notes to Table 3. Each panel changes one feature of the specification in Table 3. Panel A keeps the funds with all their offices in one county. Panels B and C place the fund at all its offices, with equal weight, or only at its office closest to the disaster. In Panel D the dependent variable sums, over the fund's scored stocks, the change in each stock's weight times the difference between its Environmental-Pillar score, from 0 to 1, and the fund's pre-event average score. Columns 1, 2, 6 and 7 use all fund-episodes, and the other columns the fund-episodes for which the political controls are observed.

TABLE G.5: MUTUAL FUND PORTFOLIO RESPONSE: TOP- AND BOTTOM-TERCILE WEIGHTS

| | *Dependent variable:* change in the fund's weight, basis points | | | |
|---|---|---|---|---|
| | Pre-event prices | | Current prices | |
| | Top tercile (1) | Bottom tercile (2) | Top tercile (3) | Bottom tercile (4) |
| Interpretation | 45.72*** | −23.66*** | 42.47*** | −29.02*** |
| | (13.37) | (9.13) | (14.12) | (9.97) |
| Attention | −.63 | .23 | −2.06 | 1.73 |
| | (1.97) | (1.79) | (2.02) | (1.94) |
| Observations (fund-episodes) | 37,047 | 37,047 | 37,047 | 37,047 |
| Nearby fund-episodes | 1,712 | 1,712 | 1,712 | 1,712 |
| $R^2$ | .029 | .034 | .038 | .045 |
| Exposure × episode FE | ✓ | ✓ | ✓ | ✓ |
| County FE | ✓ | ✓ | ✓ | ✓ |
| Fund style FE | ✓ | ✓ | ✓ | ✓ |
| Fund controls | ✓ | ✓ | ✓ | ✓ |
| Damage and coverage controls | ✓ | ✓ | ✓ | ✓ |
| Exposure × county controls | ✓ | ✓ | ✓ | ✓ |

*Notes.* See notes to Table 3. This table reports the regression of Table G.1, Columns 2 and 6, with the two parts of its dependent variable as separate outcomes, the change in the fund's weight in stocks in the top tercile of the Environmental-Pillar score and the change in its weight in stocks in the bottom tercile, over the episode quarter, in basis points, at pre-event and at current prices. The two parts are scaled so that their difference is the winsorized tilt of Table G.1; Column 1 minus Column 2 and Column 3 minus Column 4 reproduce its Columns 2 and 6. Rows report the effects among managers within 50 miles of the episode. Interpretation is per 10 percentage points and Attention per 10 log points. Standard errors, in parentheses, are clustered by episode. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

TABLE G.6: MUTUAL FUND PORTFOLIO RESPONSE BEFORE AND AFTER THE EPISODE

| | *Dependent variable:* | | | | | |
|---|---|---|---|---|---|---|
| | Environmental-Pillar tilt response (basis points) | | | | | |
| | Pre-event prices | | | Current prices | | |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Interpretation × Pre (−4, −3) | 3.04 | 13.86 | 11.83 | 15.38 | 22.21 | 19.82 |
| | (16.93) | (19.28) | (19.43) | (15.00) | (18.93) | (18.71) |
| Interpretation × Post (0 to 2) | 70.28*** | 71.03*** | 67.50*** | 61.26*** | 69.56*** | 68.29*** |
| | (20.04) | (20.56) | (19.72) | (20.25) | (21.88) | (21.35) |
| Observations (fund-episode quarters) | 251,856 | 251,856 | 251,856 | 251,856 | 251,856 | 251,856 |
| Fund-episodes | 37,066 | 37,066 | 37,066 | 37,066 | 37,066 | 37,066 |
| $R^2$ | .375 | .375 | .376 | .384 | .384 | .384 |
| Fund-episode FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Exposure × episode FE, by bin | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| County and fund style FE, by bin | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Fund controls, by bin | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Exposure × county controls, by bin | | ✓ | ✓ | | ✓ | ✓ |
| Exposure × political controls, by bin | | | ✓ | | | ✓ |

*Notes.* See notes to Table 3. This table reports the regression of Table 3, Panel A, on quarters −4 to 2 around each episode. The dependent variable is the change in the tilt since the end of quarter −1; every regressor is interacted with Pre (quarters −4 and −3) and Post (quarters 0 to 2), with quarters −2 and −1 as the base.

TABLE G.7: MUTUAL FUND PORTFOLIO RESPONSE BY DISTANCE BAND

| | *Dependent variable:* | | | | | |
|---|---|---|---|---|---|---|
| | Environmental-Pillar tilt response (basis points) | | | | | |
| | Pre-event prices | | | Current prices | | |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| *Panel A. Nearby managers (offices within 50 miles of the affected counties)* | | | | | | |
| Interpretation | 69.38*** | 66.25*** | 67.58*** | 71.49*** | 68.74*** | 65.75*** |
| | (19.87) | (18.99) | (19.86) | (21.85) | (21.08) | (21.56) |
| Fund-episodes in the band | 1,712 | 1,712 | 1,710 | 1,712 | 1,712 | 1,710 |
| *Panel B. Affected managers (offices in the affected counties)* | | | | | | |
| Interpretation | 75.15** | 73.98** | 81.18*** | 70.18** | 70.47** | 77.54** |
| | (30.54) | (31.05) | (31.23) | (30.06) | (30.45) | (31.32) |
| Fund-episodes in the band | 1,973 | 1,973 | 1,961 | 1,973 | 1,973 | 1,961 |
| *Panel C. Managers more than 100 miles away* | | | | | | |
| Interpretation | 3.70** | 3.70** | 3.34** | 3.43** | 3.45** | 3.16* |
| | (1.52) | (1.52) | (1.65) | (1.71) | (1.70) | (1.83) |
| Observations | 37,047 | 37,047 | 36,964 | 37,047 | 37,047 | 36,964 |
| $R^2$ | .032 | .033 | .093 | .045 | .045 | .102 |
| Exposure × episode FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| County FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Fund style FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Fund FE | | | ✓ | | | ✓ |
| Fund controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Damage and coverage controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Exposure × county controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Exposure × political controls | | ✓ | ✓ | | ✓ | ✓ |

*Notes.* See notes to Table 3. This table reports, from the regressions of Table G.1, columns 2 to 4 and 6 to 8, the effects among managers within 50 miles of the disaster zone (Panel A, as in Table 3), with an office in the disaster zone (Panel B), and more than 100 miles from it (Panel C).

TABLE G.8: MUTUAL FUND ACTIVE TRADING

| | *Dependent variable:* | | | |
|---|---|---|---|---|
| | Response (basis points of fund assets) | | | |
| | (1) | (2) | (3) | (4) |
| Interpretation | 30.37*** | 30.14*** | 28.81*** | 29.80*** |
| | (8.36) | (7.43) | (6.50) | (6.90) |
| Attention | .21 | 1.32 | 1.14 | .68 |
| | (1.36) | (1.52) | (1.48) | (1.63) |
| Nearby fund-episodes | 1,806 | 1,806 | 1,806 | 1,804 |
| Observations | 38,177 | 38,177 | 38,177 | 38,091 |
| $R^2$ | .025 | .026 | .026 | .087 |
| Exposure × episode FE | ✓ | ✓ | ✓ | ✓ |
| County FE | ✓ | ✓ | ✓ | ✓ |
| Fund style FE | ✓ | ✓ | ✓ | ✓ |
| Fund FE | | | | ✓ |
| Fund controls | ✓ | ✓ | ✓ | ✓ |
| Damage and coverage controls | ✓ | ✓ | ✓ | ✓ |
| Exposure × county controls | | ✓ | ✓ | ✓ |
| Exposure × political controls | | | ✓ | ✓ |

*Notes.* See notes to Table 3. The dependent variable is net active purchases of top- minus bottom-tercile stocks in the episode quarter, in basis points of lagged assets; as in Sammon and Shim (2023), the change in a continuing position is net of its proportional scaling with the fund's flow, and the whole change in a position opened or closed in the quarter counts as active.

TABLE G.9: DECOMPOSITION OF THE PORTFOLIO RESPONSE

| | *Dependent variable:* | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | Environmental-Pillar tilt response component (basis points) | | | | | | | |
| | Within distance band | | Across distance bands | | Within sector | | Across sectors | |
| | Pre-event | Current | Pre-event | Current | Pre-event | Current | Pre-event | Current |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
| *Panel A. Without political controls* | | | | | | | | |
| Interpretation | 69.90*** | 71.60*** | −.52 | −.11 | 34.48* | 31.73 | 34.90** | 39.76** |
| | (19.49) | (22.06) | (4.74) | (4.53) | (17.66) | (20.02) | (14.98) | (15.86) |
| Attention | −1.62 | −3.77 | .77 | −.02 | −2.30 | −3.61 | 1.45 | −.18 |
| | (3.21) | (3.34) | (.83) | (.82) | (2.78) | (3.00) | (2.12) | (2.25) |
| Nearby fund-episodes | 1,712 | 1,712 | 1,712 | 1,712 | 1,712 | 1,712 | 1,712 | 1,712 |
| Observations | 37,047 | 37,047 | 37,047 | 37,047 | 37,047 | 37,047 | 37,047 | 37,047 |
| $R^2$ | .033 | .043 | .027 | .029 | .030 | .038 | .026 | .030 |
| *Panel B. With political controls* | | | | | | | | |
| Interpretation | 66.33*** | 68.76*** | −.07 | −.02 | 31.65** | 28.54 | 34.60** | 40.19** |
| | (18.44) | (21.20) | (4.65) | (4.74) | (15.97) | (18.30) | (14.85) | (15.81) |
| Attention | −1.88 | −3.99 | .78 | −.02 | −2.64 | −4.01 | 1.55 | .00 |
| | (3.13) | (3.24) | (.80) | (.81) | (2.72) | (2.89) | (2.15) | (2.27) |
| Nearby fund-episodes | 1,712 | 1,712 | 1,712 | 1,712 | 1,712 | 1,712 | 1,712 | 1,712 |
| Observations | 37,047 | 37,047 | 37,047 | 37,047 | 37,047 | 37,047 | 37,047 | 37,047 |
| $R^2$ | .033 | .044 | .027 | .029 | .031 | .039 | .027 | .030 |
| Exposure × episode FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| County FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Fund style FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Fund controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Damage and coverage controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Exposure × county controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

*Notes.* See notes to Table 3. This table splits the dependent variable of Table G.1, columns 2 and 6 (Panel A) and 3 and 7 (Panel B), into the changes within and across firm-distance bands and GICS sectors.

TABLE G.10: MUTUAL FUND RESPONSE ACROSS STOCKS: FIRM LOCATION

| | *Dependent variable:* | | | |
|---|---|---|---|---|
| | Stock weight response, High-E minus Low-E (basis points) | | | |
| | Pre-event prices | | Current prices | |
| | (1) | (2) | (3) | (4) |
| *Panel A. Baseline* | | | | |
| Interpretation | 3.57*** | 3.56*** | 3.66*** | 3.71*** |
| | (1.28) | (1.28) | (1.33) | (1.29) |
| Observations | 5,426,327 | 5,426,327 | 5,426,327 | 5,426,327 |
| Fund-episodes | 53,028 | 53,028 | 53,028 | 53,028 |
| Fund × episode FE | ✓ | ✓ | ✓ | ✓ |
| Fund style × stock × episode FE | ✓ | ✓ | ✓ | ✓ |
| Exposure × political controls | | ✓ | | ✓ |
| *Panel B. Firm in the disaster zone* | | | | |
| Interpretation, firms outside the zone | 3.57*** | 3.54*** | 3.66*** | 3.69*** |
| | (1.11) | (1.09) | (1.12) | (1.07) |
| Interpretation, firms in the zone | 18.61** | 18.52** | 17.47** | 17.34** |
| | (7.48) | (7.50) | (7.50) | (7.50) |
| Observations | 5,272,599 | 5,272,599 | 5,272,599 | 5,272,599 |
| Fund-episodes | 53,028 | 53,028 | 53,028 | 53,028 |
| Fund × episode FE | ✓ | ✓ | ✓ | ✓ |
| Fund style × stock × episode FE | ✓ | ✓ | ✓ | ✓ |
| Exposure × political controls | | ✓ | | ✓ |
| *Panel C. Firm within 100 miles of the disaster zone* | | | | |
| Interpretation, firms more than 100 miles away | 3.50*** | 3.46*** | 3.55*** | 3.57*** |
| | (1.16) | (1.13) | (1.15) | (1.08) |
| Interpretation, firms within 100 miles | 6.44** | 6.48** | 6.48** | 6.62** |
| | (2.60) | (2.64) | (3.25) | (3.32) |
| Observations | 5,272,599 | 5,272,599 | 5,272,599 | 5,272,599 |
| Fund-episodes | 53,028 | 53,028 | 53,028 | 53,028 |
| Fund × episode FE | ✓ | ✓ | ✓ | ✓ |
| Fund style × stock × episode FE | ✓ | ✓ | ✓ | ✓ |
| Exposure × political controls | | ✓ | | ✓ |

*Notes.* See notes to Table 3, Panel B. In Panels B and C every term is interacted with an indicator for the location of the firm's headquarters: in the disaster zone (Panel B) or within 100 miles of it (Panel C).

TABLE G.11: MANAGERS OR HEADQUARTERS: WHO RESPONDS

| | *Dependent variable:* | |
|---|---|---|
| | Environmental-Pillar tilt response, pre-event prices (basis points) | |
| | (1) | (2) |
| *Panel A. Nearby managers, headquarters within 100 miles* | | |
| Interpretation | 66.21*** | 65.87*** |
| | (21.00) | (20.72) |
| Attention | −2.64 | −1.52 |
| | (3.06) | (3.24) |
| Fund-episodes (episodes) | 1,245 (50) | 1,245 (50) |
| *Panel B. Nearby managers, headquarters more than 100 miles away* | | |
| Interpretation | 71.19*** | 69.07*** |
| | (20.73) | (21.62) |
| Attention | −.07 | 1.54 |
| | (3.60) | (4.02) |
| Fund-episodes (episodes) | 418 (46) | 418 (46) |
| *Panel C. Headquarters within 50 miles, managers more than 100 miles away* | | |
| Interpretation | 18.74 | 21.04 |
| | (32.42) | (33.06) |
| Attention | −11.87* | −11.97* |
| | (6.19) | (6.26) |
| Fund-episodes (episodes) | 482 (75) | 482 (75) |
| *Panel D. Panel B minus Panel C* | | |
| Interpretation | 52.45* | 48.04 |
| | (30.64) | (33.80) |
| Observations | 36,165 | 36,165 |
| Exposure × episode FE | ✓ | ✓ |
| County FE | ✓ | ✓ |
| Fund style FE | ✓ | ✓ |
| Fund controls | ✓ | ✓ |
| Damage and coverage controls | ✓ | ✓ |
| Exposure × county controls | | ✓ |

*Notes.* See notes to Table 3. Each column is one regression with the specification of Table 3, columns 1 and 2. Panels A to C report the effects of interpretation, per 10 percentage points, and of attention, per 10 log points, within each group of funds. Panels A and B report the total effect among nearby managers, as in Table 3, for funds whose headquarters county is within 100 miles of the affected counties, an affected county included, or more than 100 miles away, and therefore unaffected. Panel C reports the effect of the headquarters county's interpretation and attention for funds with no office within 100 miles and the headquarters county within 50 miles, an affected county included. The rows report slopes; nearby managers share the level of their distance band in each episode, with an additional level common to all episodes when the headquarters is more than 100 miles away, and the funds of Panel C have their own level in each episode. The sample is the fund-episodes for which the political controls are observed and the headquarters location is known.

TABLE G.12: MANAGERS OR HEADQUARTERS: WHO RESPONDS, ALL FUND-EPISODES

| | *Dependent variable:* | | | |
|---|---|---|---|---|
| | Environmental-Pillar tilt response, pre-event prices (basis points) | | | |
| | All funds | | Funds below median size | |
| | (1) | (2) | (3) | (4) |
| *Panel A. Nearby managers, headquarters within 100 miles* | | | | |
| Interpretation | 47.36*** | 45.22*** | 48.55*** | 53.73*** |
| | (13.03) | (12.56) | (12.37) | (13.52) |
| Attention | −1.52 | −1.52 | .09 | .85 |
| | (2.39) | (2.50) | (2.98) | (3.15) |
| Fund-episodes (episodes) | 1,707 (63) | 1,707 (63) | 765 (62) | 765 (62) |
| *Panel B. Nearby managers, headquarters more than 100 miles away* | | | | |
| Interpretation | 55.64*** | 52.08*** | 66.85*** | 75.34*** |
| | (16.18) | (16.87) | (22.52) | (25.43) |
| Attention | .75 | .98 | −1.85 | −1.05 |
| | (3.63) | (3.86) | (5.12) | (5.33) |
| Fund-episodes (episodes) | 510 (51) | 510 (51) | 252 (49) | 252 (49) |
| *Panel C. Headquarters within 50 miles, managers more than 100 miles away* | | | | |
| Interpretation | −22.06 | −20.41 | −38.77 | −37.00 |
| | (23.68) | (23.89) | (29.41) | (30.36) |
| Attention | −6.29 | −6.25 | −5.97 | −5.97 |
| | (5.80) | (5.85) | (5.45) | (5.50) |
| Fund-episodes (episodes) | 653 (83) | 653 (83) | 345 (75) | 345 (75) |
| *Panel D. Panel B minus Panel C* | | | | |
| Interpretation | 77.69*** | 72.49*** | 105.62*** | 112.35*** |
| | (26.44) | (27.40) | (35.14) | (36.44) |
| Observations | 50,046 | 50,046 | 25,008 | 25,008 |
| Exposure × episode FE | ✓ | ✓ | ✓ | ✓ |
| County FE | ✓ | ✓ | ✓ | ✓ |
| Fund style FE | ✓ | ✓ | ✓ | ✓ |
| Fund controls | ✓ | ✓ | ✓ | ✓ |
| Damage and coverage controls | ✓ | ✓ | ✓ | ✓ |
| Exposure × county controls | | ✓ | | ✓ |

*Notes.* See notes to Table 3. Each column is one regression with the specification of Table 3, columns 1 and 2. Panels A to C report the effects of interpretation, per 10 percentage points, and of attention, per 10 log points, within each group of funds. Panels A and B report the total effect among nearby managers, as in Table 3, for funds whose headquarters county is within 100 miles of the affected counties, an affected county included, or more than 100 miles away, and therefore unaffected. Panel C reports the effect of the headquarters county's interpretation and attention for funds with no office within 100 miles and the headquarters county within 50 miles, an affected county included. The rows report slopes; nearby managers share the level of their distance band in each episode, with an additional level common to all episodes when the headquarters is more than 100 miles away, and the funds of Panel C have their own level in each episode. The sample is all fund-episodes whose headquarters location is known, including those without the political controls; columns 3 and 4 keep the funds below the median size.

TABLE G.13: TROPICAL STORM ISAIAS: TILT OF NEARBY FUNDS

| Offices of the fund | Funds | Change in the tilt (basis points) | |
|---|---|---|---|
| | | Median | Mean |
| Somerset and Morris, New Jersey | 7 | 0 | +210 |
| Essex, New Jersey, and Westchester, New York | 14 | −40 | −162 |
| Essex, New Jersey | 11 | −99 | −220 |
| More than 100 miles away | 102 | −11 | −40 |

*Notes.* This table reports the change in the Environmental-Pillar tilt of mutual funds over the quarter of Tropical Storm Isaias, which hit New York City and Long Island in August 2020, in basis points at pre-event prices, on the fund-episodes of Table 3, column 2. Funds with offices within 50 miles of the footprint are grouped by the counties of their offices. In Somerset and Morris counties, 16 percent of local articles on the storm interpreted it as climate change; in Essex and Westchester counties, the shares were 8 and 2 percent. The last row reports the funds whose offices are all more than 100 miles away.

# H Appendix to Section 5

TABLE H.1: OUTLET-LEVEL FIRST STAGE OF NEWSPAPER-CHAIN SIZE

| | *Dependent variable:* | |
|---|---|---|
| | Disaster coverage | Interpretation share |
| | (1) | (2) |
| Log chain size | −0.061 | −0.023*** |
| | (0.041) | (0.005) |
| Observations | 6,711 | 6,711 |
| $R^2$ | .650 | .362 |
| First-stage $F$ | 2.20 | 20.85 |
| Newspaper and quarter FE | ✓ | ✓ |
| Disaster coverage control | | ✓ |

*Notes.* This table reports newspaper–quarter regressions of disaster coverage and of the interpretation share on log chain size, for chained newspapers with positive disaster coverage, 2015–2024 (263 newspapers, 40 quarters). Disaster coverage is $\log(1 + A)$, $A$ the number of disaster articles in the quarter; the interpretation share is the climate-interpreted share of those articles; log chain size is $\log(1 + S)$, $S$ the number of other papers of the same owner, under the owner in place before the quarter. Column 2 controls for disaster coverage. Standard errors are clustered by newspaper. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

Table H.2 replaces the chain-size instrument with the 2019 Gannett–GateHouse merger, which formed the largest owner of US dailies (Table H.3). The specification is that of Table 2, with the county's pre-merger Gannett circulation share interacted with an indicator for quarters after 2019 and with the affected and nearby indicators; nearby counties enter as in the instrumental-variable sample. After the merger the response of affected or nearby counties is 6 to 7 basis points lower per 10 percentage points of Gannett share, unchanged with the political triple interaction; the difference is in affected and nearby counties, with none further away.

TABLE H.2: NEWSPAPER OWNERSHIP, INTERPRETATION AND THE COUNTY GREEN-ETF RESPONSE

| | Floods, hurricanes and coastal episodes | | | | Flood episodes | | | |
|---|---|---|---|---|---|---|---|---|
| *Dependent variable:* | Interpretation share (pp) | Green-ETF share response (basis points) | | | Interpretation share (pp) | Green-ETF share response (basis points) | | |
| | (1) | (2) | (3) | (4) | (5) | (6) | (7) | (8) |
| Gannett share × Post | −2.15** | | | | −2.30** | | | |
| | (.94) | | | | (.96) | | | |
| Affected or Nearby × | | −6.05** | −5.79** | | | −6.97** | −6.83** | |
| Gannett share × Post | | (2.75) | (2.67) | | | (2.80) | (2.77) | |
| Affected × | | | | −6.44 | | | | −7.91 |
| Gannett share × Post | | | | (4.82) | | | | (5.17) |
| Nearby (0–50 miles) × | | | | −5.91* | | | | −6.75** |
| Gannett share × Post | | | | (3.27) | | | | (3.26) |
| Further (50–100 miles) × | | | | .11 | | | | −.79 |
| Gannett share × Post | | | | (3.23) | | | | (3.06) |
| Observations | 239 | 77,571 | 77,571 | 77,571 | 222 | 64,938 | 64,938 | 64,938 |
| $R^2$ | .770 | .093 | .093 | .093 | .708 | .100 | .100 | .100 |
| Affected or nearby county-episodes | 239 | 599 | 599 | 599 | 222 | 484 | 484 | 484 |
| Quarter FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Episode FE | | ✓ | ✓ | ✓ | | ✓ | ✓ | ✓ |
| County FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Hazard × distance band FE | ✓ | | | | ✓ | | | |
| Affected or Nearby × Republican share × Post | | | ✓ | | | | ✓ | |

*Notes.* This table reports estimates of the county interpretation share and green-ETF share response to disasters by the newspaper ownership of the county. The interpretation share is the share of the county's local coverage of the event that links it to climate change, in percentage points. The first stage (columns 1 and 5) uses the exposed county-episodes with local coverage of the event. The green-ETF share response, sample and fixed effects of the other columns are those of Table 2, at pre-event prices, and columns 5–8 keep flood episodes only. Gannett share is the share of the county's audited newspaper circulation held by Gannett before its 2019 merger with GateHouse, per 10 percentage points; Post is one from the first quarter of 2020. Affected is one for counties in the episode footprint, Nearby for counties within 50 miles of it and Further for counties 50 to 100 miles from it; columns 2, 3, 6 and 7 pool affected and nearby counties. Each regression also includes the lower-order interactions of its terms; their coefficients are not reported. Standard errors, in parentheses, are clustered by county. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

TABLE H.3: LARGEST OWNERS OF US DAILY NEWSPAPERS

| Parent owner | Daily newspapers |
|---|---|
| Gannett | 250 |
| Lee Enterprises | 90 |
| Alden Global Capital / Digital First Media | 56 |
| Chatham Asset Management / McClatchy | 30 |
| Hearst | 23 |
| Advance Local | 22 |
| Tribune | 10 |

*Notes.* This table lists the number of major US daily newspapers held by each of seven parent owners. The News Media Alliance counts 672 major US dailies, of which 382 are held by seven parents. Source: Harvard Future of Media Project, *Index of Seven Big Owners of Dailies*, static index time-stamped 11 May 2021.

TABLE H.4: CHAIN SIZE AND PREDETERMINED CHARACTERISTICS

| | *Dependent variable:* chain size |
|---|---|
| *Politics and beliefs* | |
| Democratic two-party share, latest presidential election | 0.057 |
| | (0.257) |
| Share worried about global warming | -0.214 |
| | (0.159) |
| *Investors* | |
| Green-ETF share change, two pre-event quarters | 0.001 |
| | (0.005) |
| Portfolio assets (log) | -0.033 |
| | (0.028) |
| *County* | |
| Population growth, four years | 0.024 |
| | (0.052) |
| Income growth, four years | 0.066 |
| | (0.056) |
| *Disaster* | |
| Fatalities in the county (log) | 0.013 |
| | (0.013) |
| Damage in the county (log) | -0.022 |
| | (0.030) |
| Fatalities outside the county (log) | -0.002 |
| | (0.013) |
| Damage outside the county (log) | 0.003 |
| | (0.017) |
| Counties in the footprint (log) | 0.007 |
| | (0.014) |
| Distance to the disaster (miles) | 0.022 |
| | (0.017) |
| Local disaster coverage (log) | -0.035 |
| | (0.025) |
| Audience overlap with the footprint | 0.005 |
| | (0.017) |
| Joint test, $p$-value | 0.66 |
| County-events | 694 |
| Counties | 184 |

*Notes.* This table reports, for each characteristic, the coefficient from a regression of chain size on that characteristic, both standardized, on the county-event sample of Table 4, with the fixed effects of that table (county, calendar quarter, hazard by distance band). The joint test is from the regression of chain size on all characteristics. Standard errors, in parentheses, are clustered by county. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

TABLE H.5: INSTRUMENTED INTERPRETATION, DROPPING PURCHASES OF INDEPENDENT NEWSPAPERS

| | Portfolio response (basis points) | | | |
|---|---|---|---|---|
| | Pre-event prices | | Current prices | |
| | (1) | (2) | (3) | (4) |
| *Panel A. County ETF holdings* | | | | |
| Interpretation (IV) | 26.98* | 26.71* | 29.19* | 28.89* |
| | (15.62) | (15.48) | (15.80) | (15.65) |
| Observations | 642 | 642 | 642 | 642 |
| First-stage $F$ | 9.94 | 9.89 | 9.94 | 9.89 |
| Anderson–Rubin $p$ | .030 | .030 | .022 | .022 |
| Anderson–Rubin 90% set | [6.1, 71.0] | [6.0, 70.4] | [7.9, 73.3] | [7.7, 72.6] |
| *Panel B. Adviser ETF holdings* | | | | |
| Interpretation (IV) | 20.65* | 20.58* | 23.76** | 23.69** |
| | (11.05) | (11.03) | (11.51) | (11.50) |
| Observations | 3,974 | 3,974 | 3,974 | 3,974 |
| County-events | 608 | 608 | 608 | 608 |
| First-stage $F$ | 10.05 | 10.03 | 10.05 | 10.03 |
| Anderson–Rubin $p$ | .022 | .022 | .017 | .017 |
| Anderson–Rubin 90% set | [5.6, 51.2] | [5.6, 51.0] | [7.6, 54.5] | [7.5, 54.4] |
| *Panel C. Mutual fund managers* | | | | |
| Interpretation (IV) | 109.24 | 100.34 | 119.20 | 104.81 |
| | (79.46) | (84.99) | (88.92) | (99.26) |
| Observations | 1,716 | 1,716 | 1,716 | 1,716 |
| County-events | 113 | 113 | 113 | 113 |
| First-stage $F$ | 6.11 | 5.85 | 6.11 | 5.85 |
| Anderson–Rubin $p$ | .138 | .198 | .100 | .214 |
| Anderson–Rubin 90% set | [−13.7, 373.9] | [−31.9, 392.4] | [0.2, 461.4] | [−33.4, 485.9] |
| Coverage, county fatalities and damages | ✓ | ✓ | ✓ | ✓ |
| Democratic vote share | ✓ | ✓ | ✓ | ✓ |
| Audience reach | | ✓ | | ✓ |
| Fund size, age and turnover (Panel C) | ✓ | ✓ | ✓ | ✓ |
| County, quarter and hazard × distance-band FE | ✓ | ✓ | ✓ | ✓ |

*Notes.* This table reports two-stage least squares estimates with the outcomes, samples, controls and fixed effects of Panels B to D of Table 4, dropping the county-events at which a newspaper serving the county passed, since the county's previous event, from an owner with no other newspaper in the sample to an owner with at least one. For managers, the newspapers are those serving the counties of the manager's offices, and the previous event is the previous disaster of the manager's county. Interpretation effects are basis points per 10-percentage-point increase. First-stage $F$ is the Kleibergen–Paap rk Wald statistic. Anderson–Rubin inference is robust to weak instruments. Standard errors and first-stage $F$ statistics are clustered by county (for managers, the manager's county). $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

TABLE H.6: INSTRUMENTED INTERPRETATION, CHAIN SIZE AT THE OWNER BEFORE THE PURCHASE

| | Portfolio response (basis points) | | | |
|---|---|---|---|---|
| | Pre-event prices | | Current prices | |
| | (1) | (2) | (3) | (4) |
| *Panel A. County ETF holdings* | | | | |
| Interpretation (IV) | 25.08* | 24.89* | 27.52* | 27.29* |
| | (14.79) | (14.72) | (15.18) | (15.11) |
| Observations | 694 | 694 | 694 | 694 |
| First-stage $F$ | 11.34 | 11.33 | 11.34 | 11.33 |
| Anderson–Rubin $p$ | .038 | .039 | .029 | .030 |
| Anderson–Rubin 90% set | [4.8, 64.0] | [4.7, 63.6] | [6.5, 67.0] | [6.3, 66.5] |
| *Panel B. Adviser ETF holdings* | | | | |
| Interpretation (IV) | 21.36** | 21.27* | 25.12** | 25.02** |
| | (10.80) | (10.80) | (11.49) | (11.49) |
| Observations | 4,342 | 4,342 | 4,342 | 4,342 |
| County-events | 660 | 660 | 660 | 660 |
| First-stage $F$ | 11.79 | 11.79 | 11.79 | 11.79 |
| Anderson–Rubin $p$ | .016 | .017 | .012 | .012 |
| Anderson–Rubin 90% set | [6.5, 49.3] | [6.4, 49.2] | [8.8, 53.9] | [8.7, 53.8] |
| *Panel C. Mutual fund managers* | | | | |
| Interpretation (IV) | 94.03 | 108.38 | 69.64 | 102.65 |
| | (78.18) | (80.93) | (90.86) | (98.58) |
| Observations | 2,052 | 2,052 | 2,052 | 2,052 |
| County-events | 136 | 136 | 136 | 136 |
| First-stage $F$ | 9.34 | 9.25 | 9.34 | 9.25 |
| Anderson–Rubin $p$ | .275 | .212 | .438 | .278 |
| Anderson–Rubin 90% set | [−66.9, 245.2] | [−45.4, 278.5] | [−89.6, 275.9] | [−58.3, 343.9] |
| Coverage, county fatalities and damages | ✓ | ✓ | ✓ | ✓ |
| Democratic vote share | ✓ | ✓ | ✓ | ✓ |
| Audience reach | | ✓ | | ✓ |
| Fund size, age and turnover (Panel C) | ✓ | ✓ | ✓ | ✓ |
| County, quarter and hazard × distance-band FE | ✓ | ✓ | ✓ | ✓ |

*Notes.* This table reports two-stage least squares estimates with the outcomes, samples, controls and fixed effects of Panels B to D of Table 4. Chain size is computed holding each newspaper bought from an owner with no other newspaper in the sample at that owner, from the county's first event after the purchase on. For managers, the newspapers are those serving the counties of the manager's offices, and the previous event is the previous disaster of the manager's county. Interpretation effects are basis points per 10-percentage-point increase. First-stage $F$ is the Kleibergen–Paap rk Wald statistic. Anderson–Rubin inference is robust to weak instruments. Standard errors and first-stage $F$ statistics are clustered by county (for managers, the manager's county). $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

TABLE H.7: INSTRUMENTED INTERPRETATION WITH AUDIENCE REACH AND SOCIAL CONNECTEDNESS

| | Portfolio response (basis points) | | | | | |
|---|---|---|---|---|---|---|
| | Pre-event prices | | | Current prices | | |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| *Panel A. County ETF holdings* | | | | | | |
| Interpretation (IV) | 22.47 | 21.95 | 21.18 | 24.60* | 24.20* | 23.36* |
| | (14.21) | (13.97) | (13.65) | (14.59) | (14.39) | (14.09) |
| Observations | 694 | 694 | 694 | 694 | 694 | 694 |
| First-stage $F$ | 9.33 | 9.55 | 9.69 | 9.33 | 9.55 | 9.69 |
| Anderson–Rubin $p$ | .067 | .070 | .077 | .054 | .056 | .063 |
| Anderson–Rubin 90% set | [2.3, 61.2] | [2.0, 59.4] | [1.5, 57.2] | [3.6, 63.9] | [3.4, 62.4] | [2.8, 60.1] |
| *Panel B. Adviser ETF holdings* | | | | | | |
| Interpretation (IV) | 19.18* | 19.27* | 19.02* | 22.38** | 22.51** | 22.24** |
| | (10.53) | (10.33) | (10.30) | (11.31) | (11.13) | (11.10) |
| Observations | 4,342 | 4,342 | 4,342 | 4,342 | 4,342 | 4,342 |
| County-events | 660 | 660 | 660 | 660 | 660 | 660 |
| First-stage $F$ | 9.44 | 9.67 | 9.80 | 9.44 | 9.67 | 9.80 |
| Anderson–Rubin $p$ | .032 | .026 | .029 | .027 | .022 | .024 |
| Anderson–Rubin 90% set | [4.5, 48.2] | [5.0, 47.7] | [4.6, 47.0] | [6.0, 52.5] | [6.6, 52.2] | [6.3, 51.5] |
| *Panel C. Mutual fund managers* | | | | | | |
| Interpretation (IV) | 209.87** | 170.49** | 202.72** | 164.27* | 97.54 | 153.58 |
| | (97.94) | (84.94) | (100.68) | (97.44) | (82.27) | (97.13) |
| Observations | 2,052 | 2,052 | 2,052 | 2,052 | 2,052 | 2,052 |
| County-events | 136 | 136 | 136 | 136 | 136 | 136 |
| First-stage $F$ | 11.13 | 10.78 | 9.69 | 11.13 | 10.78 | 9.69 |
| Anderson–Rubin $p$ | .043 | .062 | .037 | .073 | .223 | .074 |
| Anderson–Rubin 90% set | [47.1, 429.3] | [25.1, 357.3] | [48.6, 460.5] | [14.5, 400.3] | [−37.7, 286.0] | [12.3, 415.3] |
| Coverage, county fatalities and damages | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Democratic vote share | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Audience reach | ✓ | | ✓ | ✓ | | ✓ |
| SCI-reach | | ✓ | ✓ | | ✓ | ✓ |
| Fund controls (Panel C) | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| County, quarter, hazard×band FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

*Notes.* This table reports two-stage least squares estimates with the outcomes, samples and fixed effects of Panels B to D of Table 4; columns 1 and 4 are its columns 2 and 4. SCI-reach is the share of the county's Facebook social connectedness that falls in the counties affected by the disaster, the county itself excluded; for managers it is weighted by the manager's offices within 50 miles, as audience reach. Interpretation effects are basis points per 10-percentage-point increase. First-stage $F$ is the Kleibergen–Paap rk Wald statistic. Anderson–Rubin inference is robust to weak instruments. Standard errors and first-stage $F$ statistics are clustered by county (for managers, the manager's county). $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

TABLE H.8: INTERPRETATION AND COVERAGE JOINTLY INSTRUMENTED

| | Portfolio response (basis points) | | | |
|---|---|---|---|---|
| | Pre-event prices | | Current prices | |
| | (1) | (2) | (3) | (4) |
| *Panel A. First stages* | | | | |
| Interpretation on chain size | −8.81*** | −8.51*** | −8.81*** | −8.51*** |
| | (2.89) | (2.83) | (2.89) | (2.83) |
| Interpretation on audience reach | | −.05 | | −.05 |
| | | (1.05) | | (1.05) |
| Coverage on chain size | | −.21* | | −.21* |
| | | (.12) | | (.12) |
| Coverage on audience reach | | .13*** | | .13*** |
| | | (.04) | | (.04) |
| *Panel B. County ETF holdings* | | | | |
| Interpretation (IV) | 22.75 | 28.47 | 24.96* | 32.02* |
| | (14.30) | (17.57) | (14.69) | (18.60) |
| Coverage (IV) | | −17.46 | | −23.82 |
| | | (39.15) | | (42.12) |
| Observations | 694 | 694 | 694 | 694 |
| First-stage $F$ | 9.32 | 2.50 | 9.32 | 2.50 |
| Anderson–Rubin $p$ | .064 | .121 | .051 | .097 |
| Anderson–Rubin 90% set | [2.5, 61.9] | [−5, 570] | [3.9, 64.7] | [0, 675] |
| *Panel C. Adviser ETF holdings* | | | | |
| Interpretation (IV) | 19.34* | 22.89* | 22.54** | 26.41** |
| | (10.51) | (11.73) | (11.30) | (13.09) |
| Coverage (IV) | | −12.06 | | −14.61 |
| | | (37.90) | | (39.20) |
| Observations | 4,342 | 4,342 | 4,342 | 4,342 |
| County-events | 660 | 660 | 660 | 660 |
| First-stage $F$ | 9.43 | | 9.43 | |
| Anderson–Rubin $p$ | .029 | | .025 | |
| Anderson–Rubin 90% set | [4.7, 48.4] | | [6.3, 52.8] | |
| Coverage as a control | ✓ | | ✓ | |
| Coverage instrumented by audience reach | | ✓ | | ✓ |
| County fatalities and damages, Democratic vote share | ✓ | ✓ | ✓ | ✓ |
| County, quarter, hazard×band FE | ✓ | ✓ | ✓ | ✓ |

*Notes.* This table reports two-stage least squares estimates with the outcomes, samples and fixed effects of Panels B and C of Table 4. Columns 1 and 3 repeat its columns 1 and 3, in which chain size instruments interpretation and disaster coverage is a control. Columns 2 and 4 instrument interpretation and coverage jointly with chain size and audience reach. Panel A reports the county-level first stages per standard deviation of each instrument, in percentage points of the interpretation share and in log points of coverage. The coverage coefficient is in basis points per log point. In columns 2 and 4 the first-stage $F$ is the Kleibergen–Paap rank Wald $F$, the Anderson–Rubin test is of the joint zero, and the 90 percent set is its projection on interpretation. Standard errors, in parentheses, are clustered by county. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

# I Appendix to Section 7

TABLE I.1: COUNTY GREEN-ETF PORTFOLIO RESPONSE BY EXPOSURE BAND

| | *Dependent variable:* | | | |
|---|---|---|---|---|
| | Green-ETF share response (basis points) | | | |
| | Pre-event prices | | Current prices | |
| | (1) | (2) | (3) | (4) |
| Affected × Interpretation | | 37.27*** | | 37.93*** |
| | | (13.64) | | (13.56) |
| Band × Interpretation | 4.34 | −9.54 | 4.30 | −9.87 |
| | (6.52) | (7.38) | (6.57) | (7.53) |
| Affected × Attention | | −3.51 | | −4.70 |
| | | (14.10) | | (13.99) |
| Band × Attention | −10.13* | −9.47 | −9.31* | −8.09 |
| | (5.55) | (7.81) | (5.27) | (7.41) |
| Total Interpretation effect among affected and nearby counties | 4.35 | | 4.32 | |
| | (6.51) | | (6.57) | |
| Total Interpretation effect among nearby counties | | −9.52 | | −9.84 |
| | | (7.27) | | (7.45) |
| Total Interpretation effect among affected counties | | 27.74** | | 28.08** |
| | | (11.88) | | (11.80) |
| Observations | 105,662 | 105,662 | 105,662 | 105,662 |
| $R^2$ | .101 | .102 | .106 | .106 |
| Affected county-episodes | 244 | 244 | 244 | 244 |
| Nearby county-episodes | 569 | 569 | 569 | 569 |
| Affected and its interactions | | ✓ | | ✓ |
| Episode and quarter FE | ✓ | ✓ | ✓ | ✓ |
| Band × hazard × year FE | ✓ | ✓ | ✓ | ✓ |
| County FE | ✓ | ✓ | ✓ | ✓ |
| Controls and their band interactions | ✓ | ✓ | ✓ | ✓ |

*Notes.* See notes to Table 2. Band is one for counties in the episode footprint or nearby, within 50 miles of it; distant counties form the comparison group. Columns 1 and 3 include the band indicator and its interactions with Interpretation and Attention only, so that the band rows compare affected and nearby counties together with distant counties. Columns 2 and 4 add Affected and its interactions, so that the Band rows compare nearby with distant counties and the Affected × Interpretation coefficient is the difference between the total Interpretation effects among affected and among nearby counties. Each regression also includes Interpretation, Attention, the band indicator and, in columns 2 and 4, Affected; their coefficients are not reported. Band × hazard × year fixed effects replace the Affected × hazard × year fixed effects; the controls of Table 2, column 5, enter with their band interactions, together with county fixed effects. The total Interpretation effect among a group is the sum of the Interpretation coefficient and its interactions for that group, with standard errors from their joint covariance. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

TABLE I.2: PERSISTENCE OF THE COUNTY GREEN-ETF PORTFOLIO RESPONSE

| | *Dependent variable:* Green-ETF share response (basis points) | | | | | | |
|---|---|---|---|---|---|---|---|
| Quarter | $h = 0$ | $h = 1$ | $h = 2$ | $h = 3$ | $h = 4$ | $h = 5$ | $h = 6$ |
| *Panel A. Pre-event prices* | | | | | | | |
| Affected × Interpretation | 19.64** | 29.56** | 16.30 | 48.33** | 52.37* | 140.54*** | 162.69** |
| | (8.57) | (11.67) | (42.41) | (23.15) | (31.08) | (46.06) | (64.59) |
| Total Interpretation effect among affected counties | 20.29** | 30.64*** | 18.05 | 48.45** | 52.34* | 139.52*** | 159.43** |
| | (8.62) | (11.34) | (41.68) | (23.30) | (31.31) | (46.35) | (64.63) |
| Observations | 105,567 | 97,524 | 89,450 | 81,682 | 75,123 | 69,407 | 62,850 |
| Affected county-episodes | 244 | 224 | 195 | 179 | 151 | 139 | 121 |
| County clusters | 301 | 297 | 286 | 276 | 267 | 262 | 248 |
| *Panel B. Current prices* | | | | | | | |
| Affected × Interpretation | 19.39** | 28.35** | 15.20 | 48.79** | 50.62* | 132.18*** | 145.47** |
| | (8.68) | (11.55) | (42.81) | (23.56) | (28.88) | (43.58) | (57.78) |
| Total Interpretation effect among affected counties | 20.04** | 29.29*** | 16.77 | 48.61** | 50.19* | 130.60*** | 141.67** |
| | (8.73) | (11.24) | (42.14) | (23.68) | (29.05) | (43.87) | (57.91) |
| Observations | 105,567 | 97,524 | 89,450 | 81,682 | 75,123 | 69,407 | 62,850 |
| Affected county-episodes | 244 | 224 | 195 | 179 | 151 | 139 | 121 |
| County clusters | 301 | 297 | 286 | 276 | 267 | 262 | 248 |

*Notes.* See notes to Table E.1. This table reports the response at each quarter after the episode, one regression per column. The outcome is the county green-ETF share at the end of quarter $h$ minus its mean over quarters $-2$ and $-1$, in basis points; $h = 0$ is the disaster quarter. Panel A values holdings at prices and green classifications from the quarter before the episode; Panel B uses current prices and classifications. Both panels use the same observations. The sample covers high-salience flood, hurricane and coastal episodes, 2015–2024, and excludes counties with another sample episode in quarters $-2$ and $-1$ or after the episode through quarter $h$, so it varies across horizons. Interpretation and Attention, controls and fixed effects follow Table E.1, columns (4) and (9). Entries are responses to a one-standard-deviation increase in Interpretation (15.7 percentage points), conditional on Attention. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

TABLE I.3: CLIMATE BELIEFS AND GREEN ETF HOLDINGS

| | Dependent variable: Green share of ETF holdings (basis points) | | | | | |
|---|---|---|---|---|---|---|
| | Worried | U.S. harm now/within 10 years | U.S. harm already happening | Warming happening | Mostly human caused | Support $CO_2$ regulation |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| Belief share | 1.19*** | 1.09** | 1.69** | 1.32** | 0.86 | 0.94 |
| | (0.41) | (0.44) | (0.74) | (0.54) | (0.64) | (0.84) |
| DMA-years | 128 | 128 | 128 | 135 | 135 | 126 |
| Media markets | 20 | 20 | 20 | 21 | 21 | 19 |
| Advisers | 1,227 | 1,227 | 1,227 | 1,251 | 1,251 | 1,210 |
| Survey respondents | 7,047 | 7,017 | 7,017 | 7,547 | 7,542 | 6,851 |
| Mean green share (bp) | 5.61 | 5.61 | 5.61 | 5.32 | 5.32 | 5.66 |
| Year fixed effects | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

*Notes.* This table reports annual DMA-level regressions, 2009–2019. Each column relates the green share of advisers' ETF holdings to the indicated binary belief share. Coefficients are basis points per 10-percentage-point higher belief share. Advisers are those that predominantly serve households and whose known U.S. offices all lie within one DMA. Belief shares require at least 30 valid responses per DMA-year. Questions and coding are given in Figure 7. Standard errors clustered by DMA are in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

TABLE I.4: CLIMATE BELIEFS AFTER DISASTERS

| | Dependent variable: Binary survey answer (percentage points) | | | | | |
|---|---|---|---|---|---|---|
| | Worried | U.S. harm now/within 10 years | U.S. harm already happening | Warming happening | Mostly human caused | Support $CO_2$ regulation |
| | (1) | (2) | (3) | (4) | (5) | (6) |
| *Panel A: Average response* | | | | | | |
| Post-disaster | 6.83 | 13.44** | 11.41** | 10.42** | 0.69 | 6.09 |
| | (5.99) | (5.19) | (4.78) | (4.32) | (5.87) | (4.15) |
| Respondents | 1,172 | 1,157 | 1,157 | 1,204 | 1,203 | 1,131 |
| Media markets | 62 | 61 | 61 | 63 | 63 | 60 |
| *Panel B: Responses by ideology* | | | | | | |
| Post × Liberal | 4.82 | 6.17 | 4.04 | 7.43 | -7.18 | 1.48 |
| | (6.24) | (8.32) | (7.90) | (5.26) | (6.84) | (5.49) |
| Post × Moderate | 6.23 | 16.43*** | 12.72** | 13.34** | 3.57 | 13.23** |
| | (7.66) | (6.15) | (5.87) | (5.60) | (7.90) | (5.01) |
| Post × Conservative | 8.85 | 15.39** | 13.51** | 5.68 | 2.38 | 3.36 |
| | (7.53) | (7.33) | (6.20) | (5.07) | (6.12) | (6.23) |
| Respondents | 1,145 | 1,131 | 1,131 | 1,177 | 1,176 | 1,112 |
| Media markets | 62 | 61 | 61 | 63 | 63 | 60 |
| Liberal respondents | 319 | 318 | 318 | 328 | 328 | 310 |
| Moderate respondents | 434 | 427 | 427 | 441 | 441 | 422 |
| Conservative respondents | 392 | 386 | 386 | 408 | 407 | 380 |
| DMA × survey-year FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Interview year-month FE | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Event fixed effects | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Demographic/wording controls | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |

*Notes.* This table reports the change in climate beliefs after floods among Climate Change in the American Mind respondents in affected DMAs, interviewed from 28 days before through 27 days after the start of the flood, 2009–2019. Coefficients are percentage-point changes in the indicated binary answer. Panel B includes ideology indicators and separate post-disaster coefficients for liberals, moderates and conservatives. Controls are age, college education, gender, income and question wording. Questions and coding are given in Figure 7. Standard errors clustered by DMA are in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

TABLE I.5: GREEN SHARES, FLOODS AND CLIMATE BELIEFS

| | Green share | Answer | |
|---|---|---|---|
| | Belief | Post | Post × Interpretation |
| | (1) | (2) | (3) |
| Worry | .731* | .135 | .137* |
| | (.319) | (.089) | (.055) |
| | [.058] | [.398] | [.059] |
| Observations | 117 | 1,671 | 1,635 |
| Media markets | 117 | 76 | 71 |
| Harm sooner | 1.122*** | .294*** | .151** |
| | (.341) | (.082) | (.054) |
| | [.007] | [.003] | [.034] |
| Observations | 117 | 1,654 | 1,618 |
| Media markets | 117 | 76 | 71 |
| Happening | .912* | .181 | .061 |
| | (.384) | (.085) | (.088) |
| | [.058] | [.147] | [.493] |
| Observations | 117 | 1,316 | 1,280 |
| Media markets | 117 | 70 | 66 |
| Human causation | .841* | −.016 | .088 |
| | (.467) | (.104) | (.062) |
| | [.074] | [.881] | [.475] |
| Observations | 117 | 1,702 | 1,666 |
| Media markets | 117 | 77 | 73 |
| $CO_2$ regulation | .706** | .114 | .081 |
| | (.271) | (.081) | (.059) |
| | [.041] | [.398] | [.475] |
| Observations | 117 | 1,516 | 1,480 |
| Media markets | 117 | 68 | 63 |
| Media market × survey-year FE | | ✓ | ✓ |
| Interview year-month FE | | ✓ | ✓ |
| Episode FE | | ✓ | ✓ |
| Age, college, gender, income, wording | | ✓ | ✓ |

*Notes.* This table reports three regressions for each of five climate belief questions from the Yale–George Mason Climate Change in the American Mind surveys. Column (1) regresses the media market's average 2010–2017 green share of adviser holdings on its mean answer before 2018, one observation per market; the coefficient is in basis points of green share per between-market standard deviation of the mean answer. Columns (2) and (3) use respondents in affected media markets interviewed from 28 days before to 27 days after a flood starts, with answers standardized within survey wave. Post equals one on or after the start day. Column (3) interacts Post with the climate interpretation of local disaster coverage accumulated from the event's start through the interview day, the share of articles that link the event centrally to climate change within each outlet, averaged across the covering outlets with circulation weights; one standard deviation is 14.54 percentage points. The interaction is zero before the event starts, and post-event interviews with no coverage are dropped from column (3). Each sample keeps media markets with interviews both before and after the event in the same survey year and removes singleton fixed effects. Units and samples differ across columns. Regressions are unweighted. Standard errors, clustered by media market, are in parentheses. Holm-adjusted $p$-values are in brackets, computed separately over the five questions within each column; stars use these adjusted values. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

TABLE I.6: WILDFIRE AND FLOOD NEWS ATTRIBUTION AND FLOOD INSURANCE TAKE-UP: FIRST-LAG DIFFERENCE

| | *Dependent variable:* $CNew_{c,t_e+h} - CNew_{c,t_e-1}$ | | | |
|---|---|---|---|---|
| | Wildfire | | Flood | |
| | $h = 2$ | $h = 3$ | $h = 2$ | $h = 3$ |
| | (1) | (2) | (3) | (4) |
| Hazard-affected | 14.48** | 21.71** | 32.04*** | 40.88*** |
| | (7.15) | (10.48) | (8.06) | (10.35) |
| Total hazard-news volume | 6.32 | 10.02 | 4.38 | 5.21 |
| | (12.57) | (15.59) | (5.79) | (7.14) |
| Climate-attributed hazard-news volume | −8.49 | −13.97 | −1.58 | −1.13 |
| | (9.85) | (12.12) | (4.36) | (5.68) |
| Strict climate-only news volume | 4.76** | 6.07* | 3.35* | 3.39 |
| | (2.40) | (3.50) | (1.85) | (2.15) |
| Hazard-affected × Total hazard-news volume | −17.26* | −15.14 | −10.71 | −17.47 |
| | (9.79) | (13.61) | (11.11) | (13.62) |
| Hazard-affected × Climate-attributed hazard-news volume | 17.82** | 18.18* | −.15 | 2.11 |
| | (8.07) | (10.02) | (8.14) | (9.80) |
| Hazard-affected × Strict climate-only news volume | −1.27 | −8.78 | −2.36 | −2.70 |
| | (5.94) | (9.90) | (4.60) | (5.82) |
| Observations | 881 | 881 | 4,206 | 4,206 |
| Counties | 190 | 190 | 904 | 904 |
| Hazard episodes | 50 | 50 | 227 | 227 |
| $R^2$ | .846 | .835 | .662 | .676 |
| County FE | ✓ | ✓ | ✓ | ✓ |
| Hazard-episode FE | ✓ | ✓ | ✓ | ✓ |
| State × Year FE | ✓ | ✓ | ✓ | ✓ |

*Notes.* This table reports the response of first-time flood-insurance take-up to the volume of wildfire and flood news and to the share of it that links the hazard to climate change. The outcome is the first-lag difference $CNew_{c,t_e+h} - CNew_{c,t_e-1}$, so $t-1$ is the base period. The wildfire and flood columns are estimated separately. The unit is county by hazard episode and each sample begins in 2010. $CNew$ is cumulative county NFIP first-time voluntary policy issuance per 100,000 residents. Columns (1) and (3) report horizon $h = 2$; columns (2) and (4) report horizon $h = 3$. News volumes use county local-circulation weights and days 0–13 after the start of the hazard. Each news measure is standardized within its hazard-specific estimation sample after applying $\log(1 + x)$. Every column includes county, hazard-episode, and state-by-event-year fixed effects; standard errors are clustered by county. The wildfire sample excludes counties affected by a flood from $t - 6$ through $t + 3$. The flood sample applies the same exclusion from $t - 6$ through $t + 3$, since flood exposure directly affects policy issuance. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

TABLE I.7: PAST GREEN RETURNS AND THE ADVISER RESPONSE

| | Dependent variable: ETF green-share response, $h = 1$ (bp) | | |
|---|---|---|---|
| | Base | Prior green return added | Prior green return × interpretation |
| | (1) | (2) | (3) |
| Interpretation response | 19.34*** | 19.33*** | 20.70*** |
| | (7.14) | (7.10) | (7.21) |
| Disaster response × prior green return | | 0.23 | |
| | | (6.03) | |
| Interpretation response × prior green return | | | −1.45 |
| | | | (4.79) |
| Interpretation response, prior green return −1 SD | | | 22.15*** |
| | | | (8.38) |
| Interpretation response, prior green return +1 SD | | | 19.25** |
| | | | (8.92) |
| Observations | | 465,705 | |
| Affected adviser-episodes | | 1,158 | |

*Notes.* This table reports the adviser response to interpretation by the return of green ETFs before the event. Observations are adviser-episodes for flood, hurricane and coastal episodes, 2015Q1–2024Q4. The prior green return is the return of green ETFs minus all non-green ETFs over the 52 weeks before the event week, equal-weighted across funds, one value per episode, per standard deviation (6.1 percentage points). All columns include attention and its interaction with the disaster, affected-by-year terms, lagged log portfolio size, the county Democratic vote share, and episode and calendar-quarter fixed effects. Column (2) adds the prior green return interacted with the disaster. Column (3) interacts the prior green return with interpretation, the disaster and attention; its level is absorbed by the episode fixed effects, and the lower-order terms are not reported. Observations are weighted by the adviser's share of the county's ETF holdings at $t-1$. Standard errors, clustered by county, are in parentheses. $^{*}p < .10$, $^{**}p < .05$, $^{***}p < .01$.

# J Aggregation: Framework and Calibration

### 1. Aggregate ESG flows

The additional allocations $a^I_{i,e}$ derived in Appendix B generate demand for ESG funds and their underlying stocks. Consider the resulting purchase of $X$ dollars of the ESG portfolio, financed by selling $X$ dollars of the broad market portfolio, at initial prices.

First, I establish how much additional ESG investment this transaction generates. Let $K$ be the fixed capital of the investor population considered and $s$ its initial share in ESG funds. The other-fund portfolio in the individual problem is the non-ESG component of this broad portfolio. Because the broad market portfolio includes ESG funds with share $s$, selling it offsets $sX$ of the ESG purchase. The net dollar addition to ESG holdings is therefore $T = (1 - s)X$. Relative to initial ESG assets $G_0 = sK$, the corresponding proportional flow is

$$f = \frac{T}{G_0}. \tag{J.1}$$

### 2. Induced relative demand for green stocks

To trace the consequences for stock prices, I now look through the fund portfolios to the stocks they hold. The ESG purchase and broad-portfolio sale generate different stock demands because the two portfolios assign different weights to each stock.

Following van der Beck (2026), I define green stocks as those overweighted by ESG-labelled funds relative to the aggregate fund portfolio. In particular, the representative ESG portfolio pools the holdings of ESG-labelled funds: $w_n^{\text{ESG}}$ equals their total dollar holdings of stock $n$ divided by their aggregate assets. The benchmark weight $w_n^{\text{all}}$ is constructed in the same way across all funds, including ESG funds. Defining $\tau_n = w_n^{\text{ESG}} - w_n^{\text{all}}$, green stocks ($\tau_n > 0$) form group $G$ and the remaining stocks form group $N$.

I assume that the reallocation is implemented through purchases and sales of the underlying stocks in these portfolio proportions. Buying $X$ dollars of the ESG portfolio therefore means demanding $Xw_n^{\text{ESG}}$ of each stock $n$, while financing the purchase by selling the broad market portfolio generates sales of $Xw_n^{\text{all}}$. Writing $w_n^O$ for the stock weight in the other-fund portfolio gives $w_n^{\text{all}} = sw_n^{\text{ESG}} + (1 - s)w_n^O$. The individual-model reallocation of $T$ dollars from other funds into ESG funds therefore induces the same stock demand as the $X = T/(1 - s)$ broad-market-financed purchase:

$$T\left(w_n^{\text{ESG}} - w_n^O\right) = X\left(w_n^{\text{ESG}} - w_n^{\text{all}}\right) = X\tau_n. \tag{J.2}$$

The reallocation thus generates net purchases of green stocks and net sales of underweighted stocks.

Let $\tau_G$ and $\tau_N$ denote the sums of the weight differences within each group. Aggregating the stock demands gives total net demand for green and nongreen stocks of $X\tau_G$ and $X\tau_N$, respectively. Dividing each by its initial total market capitalization, $V_G$ or $V_N$, gives the corresponding proportional flow. Their difference is the relative flow into green versus nongreen stocks induced by the initial reallocation of $X$ dollars from the broad market portfolio into the ESG portfolio:

$$f^{\text{rel}} = \frac{X\tau_G}{V_G} - \frac{X\tau_N}{V_N}. \tag{J.3}$$

### 3. Valuation and price response

Finally, the relative prices of green and nongreen stocks adjust as other investors accommodate the reallocation. Van der Beck's (2026) *ESG flow multiplier*, denoted here by $\kappa_{\text{ESG}}$, measures the equilibrium difference in green and nongreen stocks' dollar valuation changes per dollar reallocated from the broad market portfolio into the ESG portfolio. It combines the weight differences $\tau_n$ defined above with holders' willingness to trade as prices change, captured by ownership-weighted demand elasticities estimated using other institutions' dividend-induced trading as an instrument for price changes.[75]

Let $\Delta V_G$ and $\Delta V_N$ denote the resulting changes in the total market values of green and nongreen stocks. Holding fundamentals and other demand drivers fixed, the calibrated first-order valuation response to this $X$-dollar reallocation, accounting for other investors' accommodation, is

$$\Delta V_G - \Delta V_N = \kappa_{\text{ESG}} X. \tag{J.4}$$

To express the same response in terms of relative returns, I define the *relative-return multiplier*, $M_{\text{ESG}}$, as the change in the return difference between value-weighted green and nongreen portfolios, $R_G - R_N$, per unit of relative flow $f^{\text{rel}}$. It is obtained from the same stock-price responses by dividing each portfolio's dollar valuation change by its initial market capitalization and the resulting return difference by $f^{\text{rel}}$:

$$\Delta(R_G - R_N) = \frac{\Delta V_G}{V_G} - \frac{\Delta V_N}{V_N} = M_{\text{ESG}} f^{\text{rel}}. \tag{J.5}$$

For the calibration, I use van der Beck's (2026) approach with his December 2022 replication inputs, including the portfolio weights used above to define green and nongreen stocks and their induced demand. These inputs give $\kappa_{\text{ESG}} \simeq 0.72$ and imply $M_{\text{ESG}} \simeq 1.12$.

### 4. From the local allocation response to aggregate ESG demand

The calculation in Parts 1–3 takes the aggregate reallocation $X$ as given. I now use the local allocation estimates to determine the additional reallocation generated by stronger climate interpretation, holding disaster attention fixed. The comparison keeps other demand drivers unchanged, so baseline allocations may differ across investors while interpretation contributes an additional component to aggregate demand.

**4.1. The additional allocation** The event-induced demand shift derived in (B.7) is $u_{i,e} = \eta_e \gamma_{i,e}$. Allow the portfolio parameters to differ across investors, and denote investor $i$'s responsiveness to log odds at initial prices by $\chi_i(P^0)$. The same solution gives $\gamma_{i,e} = \gamma_i(A_{i,e}, I_{i,e}, Z_e) = \chi_i(P^0)\ell_i(A_{i,e}, I_{i,e}, Z_e)$. Superscripts 0 and 1 denote the baseline and counterfactual information configurations. Relative to $(A^0_{i,e}, I^0_{i,e}, Z^0_e)$, the additional desired share at $(A, I, Z)$ is

$$\begin{aligned} a_{i,e}(A, I, Z) &\equiv u_{i,e}(A, I, Z) - u_{i,e}(A^0_{i,e}, I^0_{i,e}, Z^0_e) \\ &= \eta_e \left[\gamma_i(A, I, Z) - \gamma_i(A^0_{i,e}, I^0_{i,e}, Z^0_e)\right]. \end{aligned} \tag{J.6}$$

[75] Van der Beck (2026, Sections 5.A, 5.C and 6.A, and Appendix A.3).

The calibration changes interpretation while holding attention and the wider information environment fixed. Its individual allocation increment is therefore

$$\begin{aligned} a^I_{i,e} &= u_{i,e}(A_{i,e}, I^1_{i,e}, Z_e) - u_{i,e}(A_{i,e}, I^0_{i,e}, Z_e) \\ &= \eta_e \chi_i(P^0) \left[\ell_i(A_{i,e}, I^1_{i,e}, Z_e) - \ell_i(A_{i,e}, I^0_{i,e}, Z_e)\right]. \end{aligned} \tag{J.7}$$

This is the individual counterpart of the estimated additional allocation $\widehat{a}^I$ in (B.17).

**4.2. Aggregation** Let $K_i$ be investor $i$'s initial eligible capital, with total capital $K = \sum_i K_i$. Denote ESG-fund units by $Q_i$. At initial prices, a portfolio share $q_i$ purchases $Q_i = K_i q_i / P^0$ units. Write $Q^0_{i,e}$ and $Q^1_{i,e}$ for baseline and counterfactual units, and $a_{i,e}$ for the corresponding share increment. They satisfy

$$P^0\big(Q^1_{i,e} - Q^0_{i,e}\big) = K_i a_{i,e}. \tag{J.8}$$

This converts the derived share increment into desired dollar demand at initial prices. Summing across investors gives

$$T = \sum_i P^0\big(Q^1_{i,e} - Q^0_{i,e}\big) = \sum_i K_i \, a_{i,e}(A^1_{i,e}, I^1_{i,e}, Z^1_e). \tag{J.9}$$

Additional ESG purchases are financed by reducing other holdings, keeping each investor's capital fixed.

For the interpretation calibration, let $\mathcal{A}$ contain the investors activated by the counterfactual. Their combined capital represents a share $\alpha$ of total eligible capital:

$$\alpha = \frac{\sum_{i \in \mathcal{A}} K_i}{K}. \tag{J.10}$$

The central assumption is that investors "activated" in my counterfactual analyses would respond to stronger climate interpretation as the disaster-exposed investors in my sample do, at the specified level of attention and information environment. I use $\widehat{a}^I$ to estimate their capital-weighted average additional allocation $a^I$ per dollar of initial capital, while allowing individual responses, initial portfolios and baseline responses to differ. I apply this increment once to each investor's initial capital, treating it as the cumulative allocation adjustment over the event quarter and the following quarter.

The resulting additional ESG investment is

$$T^I = \sum_{i \in \mathcal{A}} K_i a^I_{i,e} = \alpha K a^I. \tag{J.11}$$

Using the accounting in Part 1, its proportional ESG flow and equivalent broad-market-to-ESG reallocation are

$$f^I = \frac{T^I}{G_0}, \qquad X^I = \frac{T^I}{1-s}, \qquad G_0 = sK, \tag{J.12}$$

where $s$ is the target population's initial ESG share.

Setting $\alpha = 1$ extends the estimated interpretation increment to all eligible capital, at the specified attention and information environment. Interpreting this calculation as the complete response to a national change additionally requires that any demand generated by a change in $Z_e$ be negligible or separately included.

To obtain the associated price response, I assume that the reallocation translates fully into underlying stock demand in the portfolio proportions defined above. **Assumption: market accommodation.** I hold

the estimated stock-demand elasticities and initial ownership weights fixed as the set of activated investors changes. Activation shifts desired allocations, while both activated and nonactivated investors retain their price responsiveness. These are the stock-quantity elasticities entering van der Beck's multiplier. Substituting $X^I$ for $X$ in Parts 2 and 3 gives the additional relative stock flow, denoted $f^{\text{rel},I}$, and the associated valuation changes and returns.

**4.3. The national information environment** In the calibration, I abstract from changes in the national information environment, holding $Z_e$ fixed as I extend the estimated interpretation response to the selected population.

A nationally covered event can also change $Z_e$. Public coverage and framing can shape expectations about other investors' responses (Morris and Shin, 2002). Investors might consequently buy green assets to profit from the price increases they expect others' purchases to generate (Schmidt-Engelbertz and Vasudevan, 2025).[76] This speculative motive lies outside the climate-belief channel modeled here. Within the model, $Z_e$ changes $\gamma_i(A, I, Z)$ through its effect on the climate-belief update $\ell_i(A, I, Z)$.

Let $T^{\text{fixed } Z}$ be the sum of investors' demand changes when attention and interpretation take their counterfactual values but the national information environment remains at $Z_e^0$. For the interpretation-only exercise, this is the calibrated $T^I$. National publicity can additionally change the climate-belief update through $Z_e$.

Denote by $T^Z$ the additional aggregate demand generated through climate-belief updating when that environment changes to $Z_e^1$, holding the same attention and interpretation profiles fixed:

$$\begin{aligned} T^Z &= \sum_i K_i \left[\gamma_i(A_{i,e}^1, I_{i,e}^1, Z_e^1) - \gamma_i(A_{i,e}^1, I_{i,e}^1, Z_e^0)\right] \eta_e, \\ T^{\text{national}} &= T^{\text{fixed } Z} + T^Z. \end{aligned} \tag{J.13}$$

The term $T^Z$, which is not included in my calibration, captures what the change in national coverage and framing adds beyond the specified changes in individual attention and interpretation. It can include both a response shared across investors and changes in how strongly they respond to attention or interpretation. When $T^Z > 0$, the national climate-belief response exceeds the sum of local responses evaluated at the baseline national information environment.

## 5. Calibration exercise

For this calibration exercise, I measure attention and interpretation using newspaper coverage of the event, assuming that coverage captures its broader local salience and interpretation captures the local discussion and narratives attributing it to climate change. I control for other important determinants of attention and interpretation, such as whether the county was affected and local political views, which can also influence interpretation and thus portfolio choices. By holding these factors at their average levels in the calibration sample, I focus on the additional event-specific portfolio response induced by variation in interpretation captured by local climate narratives.

**5.1. Applying the estimated allocation response** The numerical calibration uses the same interpretation contrast as (B.17), evaluated among affected counties. Its outcome is the cumulative change in the green-ETF

[76] See Schmidt-Engelbertz and Vasudevan (2025). Using investor surveys, they document that beliefs about others' beliefs can motivate positions that conflict with investors' own market valuations.

share from the pre-event quarter through the following quarter, valuing holdings at pre-event prices. I apply the price-response assumption of Appendix B to this measured share and horizon. The impact response in that section and the following-quarter response used here are separate estimates of the corresponding allocation adjustments.

Let $\widehat{\beta}_I^{\text{cal}} + \widehat{\delta}_I^{\text{cal}}$ be the estimated interpretation effect among affected counties for this calibration outcome, reported in basis points per standard deviation. Holding coverage and the other covariates fixed, the additional allocation in decimal share units is

$$\widehat{a}^I = \frac{\widehat{\beta}_I^{\text{cal}} + \widehat{\delta}_I^{\text{cal}}}{10{,}000} \frac{I^1 - I^0}{\sigma_I}. \tag{J.14}$$

This is the allocation increment used in $T^I = \alpha K \widehat{a}^I$.

For coverage, let $\widehat{\beta}_A^{\text{cal}} + \widehat{\delta}_A^{\text{cal}}$ be the affected-county response in basis points per standard deviation of $\log(1 + A)$, and $\sigma_A$ that standard deviation. For a joint change in coverage and interpretation, the estimated additional share $\widehat{a}$ is

$$\widehat{a} = \frac{1}{10{,}000} \left[ (\widehat{\beta}_A^{\text{cal}} + \widehat{\delta}_A^{\text{cal}}) \frac{\log(1 + A^1) - \log(1 + A^0)}{\sigma_A} + (\widehat{\beta}_I^{\text{cal}} + \widehat{\delta}_I^{\text{cal}}) \frac{I^1 - I^0}{\sigma_I} \right], \tag{J.15}$$

Here $A^0, A^1$ and $I^0, I^1$ are the baseline and counterfactual news measures.

**5.2. A stronger climate interpretation** Consider an increase in climate interpretation from its 25th to its 75th percentile, holding coverage at its median. In the calibration sample, this raises the share of coverage attributing the event to climate change from 8.07% to 27.51%, a change of approximately 1.24 standard deviations.

The estimated interpretation response is 26 allocation basis points per standard deviation. The implied additional reallocation is

$$\widehat{a}^I = \frac{25.5665 \times 1.23782}{10{,}000} \simeq 0.003165. \tag{J.16}$$

Thus, stronger interpretation increases the ESG allocation share by 32 basis points, or approximately 32 cents per 100 dollars of initial capital, over the event quarter and the following quarter. The 95% confidence interval is 7–56 basis points.

**5.3. Aggregate demand and the stock-price response** I consider three scenarios, depending on which investors are activated: ETF investors, mutual-fund and ETF investors together, and institutional equity investors. The institutional scenario is the main counterfactual. Table J.1 collects the calculations using the formulas in Parts 1–4, with full activation ($\alpha = 1$) within each investor base and the additional reallocation applied once to each investor's initial capital.

Table J.1: Additional ESG demand and stock-price responses by activated investor base

| | ETFs | Mutual funds + ETFs | Institutional equity investors |
|---|---|---|---|
| Additional ESG investment $T^I$ (\$bn) | 19.91 | 61.51 | — |
| Proportional ESG flow $f^I$ (%) | 30.47 | 26.95 | — |
| Reallocation $X^I$ (\$bn) | 20.12 | 62.24 | 79.21 |
| Relative stock flow $f^{\text{rel},I}$ (bp) | 7.73 | 23.90 | 30.41 |
| Relative stock return (bp) | 8.62 | 26.65 | 33.92 |
| Relative valuation change (\$bn) | 14.43 | 44.64 | 56.81 |

*Notes.* This table reports the additional ESG demand and the stock-price responses implied by the calibration for each activated investor base. All scenarios use the December 2022 multipliers from Part 3. The fund-flow measures $T^I$ and $f^I$ apply to the first two scenarios; stock flows, returns and valuation changes compare green with nongreen stocks. For the institutional scenario, the 95% confidence intervals are 7–60 basis points for the return difference and \$12.5–\$101.1 billion for the valuation difference. These reflect uncertainty in the estimated local response, holding capital and calibration assumptions fixed.

In November 2024, US domestic-equity ETFs managed approximately $K = \$6.29$ trillion, including $G_0 = \$65.35$ billion in ESG ETFs, corresponding to an initial ESG share of $s \simeq 1.039\%$. The first scenario applies the estimated interpretation increment to this capital.

The evidence that mutual-fund flows respond to climate interpretation supports extending the calculation from ETFs to mutual funds and ETFs together. I assume that investors in both types of fund make the same additional ESG allocation per dollar of capital as the investors represented in the local estimate.

In November 2024, US domestic-equity mutual funds and ETFs managed approximately $K = \$19.44$ trillion, including $G_0 = \$228.20$ billion in ESG funds, corresponding to an initial ESG share of $s \simeq 1.174\%$.

The evidence on mutual-fund managers' rebalancing in response to interpretation motivates extending the exercise to institutional investors' stock demand. In this main counterfactual, I assume that institutional investors make the same additional ESG-versus-benchmark reallocation per dollar of capital. The fund-sector calculation implies a reallocation rate of

$$\frac{X^I}{K} = \frac{\$62.24 \text{ billion}}{\$19.44 \text{ trillion}} \simeq 0.320\%. \tag{J.17}$$

In December 2022, institutions held $K_{\text{inst}} = \$24.74$ trillion of the stocks covered by the calibration, or 73.8% of their total market value. Applying the same reallocation rate gives

$$X^I_{\text{inst}} = K_{\text{inst}} \frac{X^I}{K} \simeq \$79.21 \text{ billion}. \tag{J.18}$$

Institutional holdings include stocks held by fund managers on behalf of their investors, so the underlying equity capital is counted once. The price multipliers retain accommodation by all investors, with the additional demand generated by institutions.

As a broader counterfactual, I extend the same reallocation rate to all investors in the covered equity market. Applying it to the \$33.53 trillion of equity capital covered by the December 2022 calibration, $K_{\text{all}} = V_G + V_N$, gives

$$X^I_{\text{all}} = K_{\text{all}} \frac{X^I}{K} \simeq \$107.36 \text{ billion}. \tag{J.19}$$

This implies 46 basis points of green-minus-nongreen appreciation and \$76.99 billion in relative valuation

change. This broader scenario includes institutional and non-13F owners, counting the underlying equity capital once.

**5.4. Comparison with annual ESG returns** Consider a year in which stronger climate interpretation induces additional reallocations equivalent to every institutional investor making the estimated adjustment once, at unchanged attention. These adjustments are completed during the year, holding the national information environment and other conditions unchanged.[77]

In dollar terms, this is the difference in annual ESG-versus-benchmark demand between the stronger- and weaker-interpretation scenarios:

$$\underbrace{X_{\text{year}}^{\text{stronger } I} - X_{\text{year}}^{\text{weaker } I}}_{X_{\text{inst}}^{I}} \simeq \$79.21 \text{ billion}. \tag{J.20}$$

Applying the calibrated price response gives 34 basis points more green-minus-nongreen appreciation in the stronger-interpretation scenario.

In van der Beck's (2026) decomposition, the average annual return spread is 223 basis points, with 186 basis points attributed to institutional ESG demand. Table J.2 expresses each calibrated effect as a percentage of these benchmarks.

TABLE J.2: CALIBRATED APPRECIATION RELATIVE TO ANNUAL ESG-RETURN BENCHMARKS

| Activated investor base | Percent of observed annual ESG return spread | Percent of annual institutional ESG-demand contribution |
|---|---|---|
| ETFs | 3.9 | 4.6 |
| Mutual funds + ETFs | 11.9 | 14.3 |
| Institutional equity investors | 15.2 | 18.2 |

*Notes.* This table expresses the calibrated green-minus-nongreen appreciation of each scenario as a percentage of the average annual ESG return spread in van der Beck (2026) and of the part of it attributed to institutional ESG demand.

This additional appreciation is equivalent in magnitude to 18.2% of the annual contribution attributed to institutional ESG demand and 15.2% of the observed annual return spread.[78]

For the broader all-equity counterfactual, the corresponding ESG-demand benchmark includes both institutional and non-13F demand and equals 155 basis points annually. The calibrated 46-basis-point effect is equivalent to 29.7% of that contribution and 20.6% of the observed annual spread.

[77] I measure the allocation adjustment from the quarter preceding the disaster through the following quarter, allowing for events occurring late in the quarter and for delayed portfolio adjustment. In the annual counterfactual, each investor's additional adjustment is counted once.

[78] These comparisons measure the size of the calibrated effect. Estimating interpretation's contribution to historical returns would require actual activation and interpretation changes over that sample. The calibration uses December 2022 portfolio weights and price responses; the historical returns reflect changing portfolios and market conditions and include dividends.